

A-Level Maths Supplementary Materials

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Abstract

These notes accompany my A-level mathematics tutoring and are designed to support students in developing clear, structured mathematical thinking. They are informed by common difficulties raised during tutoring sessions and focus on building confidence through worked examples and guided practice.

The worked solutions emphasise method selection and reasoning, rather than calculation alone. Students are encouraged to attempt each example before consulting the solutions. Make use of the table of contents to navigate the document via internal hyperlinks. This might require downloading the file and opening it for this to work properly.

I am a mathematics graduate with a Master of Advanced Study (Part III) in Pure Mathematics from the University of Cambridge, and I work professionally in data engineering and analytics. Alongside tutoring, I focus on explaining complex ideas clearly and accessibly.

If you spot any errors or have questions, feel free to get in touch:

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1 Functions

1.1 Compositions and Inverses

1.1.1 Compositions

Definition 1.1. If f and g are two functions, we can define a new function fg by

$$fg(x) = f(g(x))$$

Note that this relies on f being defined at the point $g(x)$. If f has a *singularity* at a point a (as in $f(a)$ divides by zero somewhere in the calculation) then if there is an x such that $g(x) = a$, $fg(x) = f(a)$, which would not be allowed.

We can think of this similar to bracket rules - you do the “innermost” one first, and work outwards. In the case of Definition 1.1 we compute $g(x)$ first, and then $f(x)$

This can keep going! If we have three functions f, g, h , then $fgh(x) = f(g(h(x)))$, and we evaluate the functions inside-out (h acts on x first, then g , then f).

1.1.2 Compositions Examples and Exercises

Example 1. Let f be a function defined by

$$f(x) = \frac{2x+1}{5}$$

Find $ff(x)$, and write your answer in the form

$$ff(x) = \frac{ax+b}{25}$$

Where a, b are numbers.

Solution. First write:

$$f(x) = \frac{2x+1}{5}$$

So

$$ff(x) = f(f(x)) = \frac{2f(x)+1}{5}$$

Substituting $f(x)$ into this equation gives

$$ff(x) = \frac{2\frac{2x+1}{5}+1}{5}$$

Here we have a fraction with fractions in it. Fractions are annoying, so we want to reduce the number of fractions. Notice that if we times the inner fraction by 5, it would remove the fraction. We can use the fact that any number times 1 is itself to multiply the fraction by 5/5:

$$\begin{aligned} ff(x) &= \frac{5}{5} \times \frac{2\frac{2x+1}{5}+1}{5} && \text{(substitute in } f(x)) \\ &= \frac{2 \times \cancel{5} \times \frac{2x+1}{\cancel{5}} + 5}{5 \times 5} && \text{(expand top and bottom)} \\ &= \frac{4x+2+5}{25} \\ &= \frac{4x+7}{25} \end{aligned}$$

Now set

$$\frac{4x+7}{25} = \frac{ax+b}{25}$$

Which finishes the question, finding $a = 4$, $b = 7$. □

Example 2. Let f be a function defined by

$$f(x) = \frac{2x+1}{4x}$$

Where $x \neq 0$. *This is required to not divide by 0.*

Find $ff(x)$, and write your answer in the most simplified form of

$$ff(x) = \frac{ax+b}{cx+d}$$

Where a, b, c, d are numbers. Make sure to state where $ff(x)$ is defined (what can x not be).

Solution. First write (whenever $x \neq 0$):

$$f(x) = \frac{2x+1}{4x}$$

So

$$ff(x) = f(f(x)) = \frac{2f(x)+1}{4f(x)}$$

Substituting $f(x)$ into this equation gives

$$ff(x) = \frac{2\frac{2x+1}{4x} + 1}{4\frac{2x+1}{4x}}$$

Here we have a fraction with fractions in it. Fractions are annoying, so we want to reduce the number of fractions. Notice that if we times the inner fractions by $4x$, it would remove the fraction. We can use the fact that any number times 1 is itself to multiply the fraction by $4x/4x$, which is allowed because $x \neq 0$:

$$\begin{aligned} ff(x) &= \frac{4x}{4x} \times \frac{2\frac{2x+1}{4x} + 1}{4\frac{2x+1}{4x}} && \text{(substitute in } f(x)) \\ &= \frac{2 \times \cancel{4x} \times \frac{2x+1}{\cancel{4x}} + 4x}{4 \times \cancel{4x} \times \frac{2x+1}{\cancel{4x}}} && \text{(expand top and bottom)} \\ &= \frac{4x + 2 + 4x}{8x + 4} \\ &= \frac{8x + 2}{8x + 4} \\ &= \frac{2}{2} \times \frac{4x + 1}{4x + 2} && \text{(factorise top and bottom)} \\ &= \frac{4x + 1}{4x + 2} \end{aligned}$$

Since we cannot simplify the equation any further, we now set

$$\frac{4x+1}{4x+2} = \frac{ax+b}{cx+d}$$

Which finds $a = c = 4$, $b = 1$, $d = 2$ by comparison. $ff(x)$ cannot divide by zero, so we know $4x+2 \neq 0$, and get that therefore $x \neq -\frac{1}{2}$. But remember f itself requires $x \neq 0$, so x cannot be 0 or $-\frac{1}{2}$. Which finishes the question. □

Example 3. Let f and g be functions defined by

$$f(x) = \sin(x) + \cos(3x)$$

$$g(x) = \frac{x+1}{3}$$

Find $fg(x)$, simplifying your answer.

Solution. First write:

$$f(x) = \sin(x) + \cos(3x)$$

$$g(x) = \frac{x+1}{3}$$

So

$$fg(x) = f(g(x)) = \sin(\textcolor{red}{g(x)}) + \cos(3\textcolor{red}{g(x)})$$

Substituting $g(x)$ into this equation gives

$$\begin{aligned} fg(x) &= \sin\left(\frac{x+1}{3}\right) + \cos\left(3 \times \frac{x+1}{3}\right) \\ &= \sin\left(\frac{x+1}{3}\right) + \cos(x+1) \end{aligned}$$

Which finishes the question. □

Exercise 1. *Try it yourself!*

Let f , g , h , be functions defined by

$$f(x) = \frac{1}{2}x^2$$

$$g(x) = \sin(x) - \cos(x)$$

$$h(x) = \cos(x) + \sin(x)$$

Show that

$$fg(x) + fh(x) = 1$$

Hint: you may use (or recall) without proof that $\sin(x)^2 + \cos(x)^2 = 1$.

Solution. We will approach this question by calculating fg and fh separately, and then plugging them in. First write:

$$f(x) = \frac{1}{2}x^2$$

$$g(x) = \sin(x) - \cos(x)$$

$$h(x) = \cos(x) + \sin(x)$$

So

$$fg(x) = f(g(x)) = \frac{1}{2}\textcolor{red}{g(x)}^2$$

Substituting $g(x)$ into this equation gives

$$fg(x) = \frac{1}{2}(\sin(x) - \cos(x))^2$$

Similarly,

$$fh(x) = f(h(x)) = \frac{1}{2}\textcolor{red}{h(x)}^2$$

Substituting $h(x)$ into this equation gives

$$fh(x) = \frac{1}{2}(\cos(x) + \sin(x))^2$$

Then we can calculate:

$$\begin{aligned}
 fg(x) + fh(x) &= \frac{1}{2}(\sin(x) - \cos(x))^2 + \frac{1}{2}(\cos(x) + \sin(x))^2 && \text{(plugging in the equations)} \\
 &= \frac{1}{2}((\sin(x) - \cos(x))^2 + (\cos(x) + \sin(x))^2) && \text{(factoring)} \\
 &= \frac{1}{2}(\sin(x)^2 - \cancel{2\sin(x)\cos(x)} + \cos(x)^2 + \cos(x)^2 + \cancel{2\sin(x)\cos(x)} + \sin(x)^2) && \text{(expanding)} \\
 &= \frac{1}{2} \times \cancel{2}(\sin(x)^2 + \cos(x)^2) && \text{(factoring)} \\
 &= \sin(x)^2 + \cos(x)^2 \\
 &= 1 && \text{(trig rules (the hint))}
 \end{aligned}$$

Which finishes the question. $\square$

1.1.3 Inverses

Definition 1.2. If f is a function, we define the inverse of f , denoted f^{-1} by the following equation:

$$ff^{-1}(x) = x$$

Or equivalently,

$$f^{-1}f(x) = x$$

The most common way to solve for f^{-1} is by:

1. Write $y = f(x)$
2. Rearrange so that we have x in terms of y
3. Swap x and y to find $f^{-1}(x)$

You can show that this satisfies the definition of the inverse of f . Intuitively, Step 2 above finds “what do you need to do to y to find x ”? This is because we are writing x as a bunch of operations applied to y , but y was chosen to be $f(x)$ in step 1, therefore it must “undo” the operations we did in Step 1. This argument can be made rigorous.

Here we have used the notation for function composition in Definition 1.1.

1.1.4 Inverses Examples and Exercises

Example 4. Example 1 continued.

Let f be a function defined by

$$f(x) = \frac{2x + 1}{5}$$

Find $f^{-1}(x)$, and write your answer in the form

$$f^{-1}(x) = \frac{ax + b}{2}$$

Where a, b are numbers.

Solution. First write

$$y = \frac{2x + 1}{5}$$

Then rearrange for x :

$$\begin{aligned} 5y &= 2x + 1 && \text{(multiply by 5)} \\ 5y - 1 &= 2x && \text{(minus 1)} \\ x &= \frac{5y - 1}{2} && \text{(divide by 2)} \end{aligned}$$

Therefore,

$$f^{-1}(x) = \frac{5x - 1}{2}$$

And setting it equal to $\frac{ax+b}{2}$, we find that $a = 5$ and $b = -1$ by comparison. □

Example 5. Example 4 continued.

Let f and g be functions defined by

$$\begin{aligned} f(x) &= \frac{2x + 1}{5} \\ g(x) &= \frac{5x - 1}{2} \end{aligned}$$

By computing $fg(x)$, verify that g is the inverse function of f , verifying the calculation in the previous question.

Solution. First write

$$\begin{aligned} f(x) &= \frac{2x + 1}{5} \\ g(x) &= \frac{5x - 1}{2} \end{aligned}$$

Then calculate

$$\begin{aligned} f(g(x)) &= \frac{2g(x) + 1}{5} \\ &= \frac{2 \frac{5x-1}{2} + 1}{5} \\ &= \frac{\cancel{5x-1} + 1}{5} \\ &= \frac{\cancel{5}x}{\cancel{5}} \\ &= x \end{aligned}$$

Therefore, $fg(x) = x$, which verifies g is the inverse function of f . □

Example 6. Example 2 continued.

Let f be a function defined by

$$f(x) = \frac{2x + 1}{4x}$$

Where $x \neq 0$. *This is required to not divide by 0.*

Find $f^{-1}(x)$, and write your answer in the form of

$$f^{-1}(x) = \frac{1}{ax + b}$$

Where a, b are numbers. Make sure to state where $f^{-1}(x)$ is defined (what can x not be).

Solution. Let $x \neq 0$. First write

$$y = \frac{2x+1}{4x}$$

Then rearrange for x :

$$\begin{aligned} 4xy &= 2x+1 && \text{(multiply by } 4x) \\ 4xy - 2x &= 1 && \text{(minus } 2x \text{ to get } x \text{ on one side)} \\ x(4y-2) &= 1 && \text{(factorise } x) \\ x &= \frac{1}{4y-2} && \text{(divide by } 4y-2) \end{aligned}$$

Therefore,

$$f^{-1}(x) = \frac{1}{4x-2}$$

And setting it equal to $\frac{1}{ax+b}$, we find that $a=4$ and $b=-2$ by comparison. This function is valid (not dividing by zero) whenever $4x-2 \neq 0$, therefore it is valid whenever $x \neq \frac{1}{2}$. $\square$

Exercise 2. Example 6 continued. *Try it yourself!*

Let f and g be functions defined by

$$\begin{aligned} f(x) &= \frac{2x+1}{4x}, \quad x \neq 0 \\ g(x) &= \frac{1}{4x-2}, \quad x \neq \frac{1}{2} \end{aligned}$$

By computing $fg(x)$, verify that g is the inverse function of f , verifying the calculation in the previous question.

Solution. First write

$$\begin{aligned} f(x) &= \frac{2x+1}{4x} \\ g(x) &= \frac{1}{4x-2} = \frac{1}{2} \times \frac{1}{2x-1} \end{aligned}$$

Then set $x \neq \frac{1}{2}$ and calculate

$$\begin{aligned} f(g(x)) &= \frac{2g(x)+1}{4g(x)} \\ &= \frac{\cancel{2} \times \frac{1}{\cancel{2}} \times \frac{1}{2x-1} + 1}{4 \times \frac{1}{2} \times \frac{1}{2x-1}} \\ &= \frac{\frac{1}{2x-1} + 1}{2 \times \frac{1}{2x-1}} \end{aligned}$$

Now we use my favourite trick of getting-rid-of-the-annoying-fractions-by-multiplying-by-the-inner-denominator, which is allowed because $x \neq \frac{1}{2}$:

$$\begin{aligned} f(g(x)) &= \frac{2x-1}{2x-1} \times \frac{\frac{1}{2x-1} + 1}{2 \times \frac{1}{2x-1}} && \text{(multiply by 1)} \\ &= \frac{(\cancel{2x-1}) \times \frac{1}{\cancel{2x-1}} + 2x-1}{2 \times (\cancel{2x-1}) \times \frac{1}{\cancel{2x-1}}} && \text{(simplify fractions)} \\ &= \frac{\cancel{x} + 2x - \cancel{x}}{2} \\ &= \frac{2x}{2} \\ &= x \end{aligned}$$

Therefore, $fg(x) = x$, which verifies g is the inverse function of f . $\square$

1.1.5 Additional Exercises

Exercise 3. The function f is defined by

$$f(x) = \frac{4x+3}{x-2}, \quad x \neq 2$$

1. Find f^{-1}
2. Show that

$$ff(x) = \frac{ax+b}{cx+d}$$

where a, b, c, d are integers to be found.

The point $P(3, 15)$ lies on the curve with equation $y = f(x)$.

3. Find the point to which P is mapped when $y = f(x)$ is transformed to the curve with equation $y = 2f(3x) + 8$.
4. (Bonus) Verify f^{-1} in the first part satisfies $ff^{-1}(x) = x$.

Solution. 1. Find f^{-1}

First write $y = f(x)$, where $x \neq 2$ i.e.

$$y = \frac{4x+3}{x-2}$$

Now rearrange for x , getting rid of any fractions.

$$\begin{aligned} y &= \frac{4x+3}{x-2} \\ y(x-2) &= 4x+3 && \text{(multiply by } (x-2)) \\ xy-2y &= 4x+3 && \text{(expand)} \\ xy-4x &= 2y+3 && \text{(collect } x \text{ terms to one side)} \\ x(y-4) &= 2y+3 && \text{(factorise LHS)} \\ x &= \frac{2y+3}{y-4} && \text{(assuming } y \neq 4) \end{aligned}$$

Hence, whenever $x \neq 4$,

$$f^{-1}(x) = \frac{2x+3}{x-4}$$

2. Show

$$ff(x) = \frac{ax+b}{cx+d}$$

Then calculate

$$\begin{aligned}
ff(x) &= f(f(x)) \\
&= \frac{4f(x) + 3}{f(x) - 2} && \text{(insert } f(x) \text{ wherever } x \text{ is)} \\
&= \frac{4 \times \frac{4x+3}{x-2} + 3}{\frac{4x+3}{x-2} - 2} && \text{(substitute } f(x)) \\
&= \frac{x-2}{x-2} \times \frac{4 \times \frac{4x+3}{x-2} + 3}{\frac{4x+3}{x-2} - 2} && \text{(multiply by 1 to remove the } x-2 \text{ denominators)} \\
&= \frac{(x-2)(4 \times \frac{4x+3}{x-2} + 3)}{(x-2)(\frac{4x+3}{x-2} - 2)} && \text{(multiply fractions)} \\
&= \frac{4(4x+3) + 3(x-2)}{4x+3 - 2(x-2)} && \text{(expand multiplication)} \\
&= \frac{16x+12+3x-6}{4x+3-2x+4} && \text{(expand multiplication)} \\
&= \frac{19x+6}{2x+7} && \text{(collect like terms)}
\end{aligned}$$

This is valid whenever $2x+7 \neq 0$, which is when $x \neq -\frac{7}{2}$

Hence we have found

$$ff(x) = \frac{19x+6}{2x+7} \quad x \neq 2, x \neq -\frac{7}{2}$$

3. We will reword the question slightly. Define g by

$$g(x) = 2f(3x) + 8$$

Find $g(3)$.

First calculate $f(3x)$.

$$f(3x) = \frac{4 \times 3x + 3}{3x - 2} = \frac{12x+3}{3x-2}$$

Therefore

$$\begin{aligned}
g(x) &= 2f(3x) + 8 \\
&= 2 \frac{12x+3}{3x-2} + 8 && \text{(substitute } f(3x)) \\
&= \frac{24x+6}{3x-2} + 8 && \text{(multiply fractions, noting } 2 = \frac{2}{1})
\end{aligned}$$

Then we can substitute 3 into the above equation:

$$\begin{aligned}
g(3) &= \frac{24 \times 3 + 6}{3 \times 3 - 2} + 8 \\
&= \frac{78}{7} + 8 \\
&= \frac{78 + 56}{7} \\
&= \frac{134}{7}
\end{aligned}$$

Therefore the point P is mapped to $(3, g(3))$ which is the point $(3, \frac{134}{7})$.

4. Show $ff^{-1}(x) = x$. This is an exercise in function composition, and we will follow the same style of approach as in e.g. Example 3.

First write

$$ff^{-1}(x) = f(f^{-1}(x))$$

Remember that

$$f(x) = \frac{4x+3}{x-2}$$

And that we found

$$f^{-1}(x) = \frac{2x+3}{x-4}$$

Therefore

$$\begin{aligned} f(f^{-1}(x)) &= \frac{4f^{-1}(x) + 3}{f^{-1}(x) - 2} \\ &= \frac{4\frac{2x+3}{x-4} + 3}{\frac{2x+3}{x-4} - 2} \\ &= \frac{x-4}{x-4} \times \frac{4 \times \frac{2x+3}{x-4} + 3}{\frac{2x+3}{x-4} - 2} && \text{(multiply by 1 to remove nested fractions)} \\ &= \frac{(x-4) \left(4 \times \frac{2x+3}{x-4} + 3 \right)}{(x-4) \left(\frac{2x+3}{x-4} - 2 \right)} && \text{(multiply fractions)} \\ &= \frac{4(3+2x) + 3(x-4)}{3+2x-2(x-4)} && \text{(expand)} \\ &= \frac{12+8x+3x-12}{3+2x-2x+8} && \text{(expand more)} \\ &= \frac{11x}{11} && \text{(collect terms)} \\ &= x \end{aligned}$$

As required, therefore $ff^{-1}(x) = x$.

□

1.2 Domain and Range

My school teacher said that he remembered which is range and what is domain by thinking of a lion's domain, and somehow he linked this to the inputs for a function. I never understood quite what he meant, but it stuck with me.

The domain of a function is the set of all input values for a function, and the range is the set of all possible outputs of that function, over the domain.

For example, consider the following function.

$$f(x) = \frac{1}{x^2}$$

This function is defined everywhere except for $x = 0$, where we would be dividing by 0. Therefore its (maximal) domain is every number except for 0. On this domain, we can see that it is always positive since the denominator is always positive, and it gets closer to 0 as x gets larger (or more negative). We can also observe that as x gets close to 0, the function outputs larger and larger numbers. Because of this, you might realise¹ that it can reach every number more than 0. Therefore, we have found the domain and range of $f(x)$.

The domain is $x < 0$ or $x > 0$, and the range is $0 < f(x)$.

¹You can prove this by setting $f(x) = a$ where $a > 0$, and showing you can solve for x .

Example 7. Consider $f(x) = 2x$, for $-2 \leq x < 1$. Find the range for this function, making sure to only allow the x range in the question.

Solution. We know that this function is a straight line that is increasing, so therefore the range has to be between whatever the leftmost point and the rightmost point is. The place to be careful is that we are told that $-2 \leq x < 1$, therefore it does not quite reach the rightmost point.

We can find the range therefore by plugging in:

$$\begin{aligned}f(-2) &= 2 \times (-2) = -4 \\f(1) &= 2 \times (1) = 2\end{aligned}$$

Therefore, when $-2 \leq x < 1$, we have that $-4 \leq f(x) < 2$ is the range. This completes the question. $\square$

1.2.1 The Modulus Function

The modulus function is defined in two cases:

1. If $x \geq 0$, then $|x| = x$
2. If $x < 0$, then $|x| = -x$

This has the effect of “removing-the-minus-sign” from the number.

Example 8.

$$\begin{aligned}|3| &= 3 \\|-3| &= 3 \\|-20| &= 20 \\|0| &= 0 \\|5| &= 5\end{aligned}$$

We are interested in how this can interact with a function $f(x)$, seen in Figure 1.

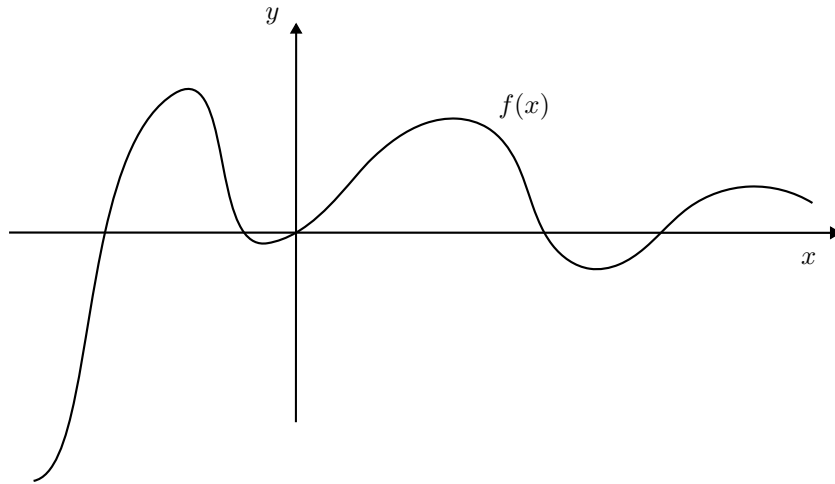


Figure 1: An example function

What would $|f(x)|$ look like? Since we know that the modulus function makes negative numbers positive, we care about when $f(x)$ is negative or positive.

Since $f(x)$ is on the y axis, we know whether it is negative or positive based on whether it is below or above the x axis. To plot the modulus of the function, observe that this is the same as reflecting all parts below the x axis to be above it, and leaving the parts above the x axis the same.

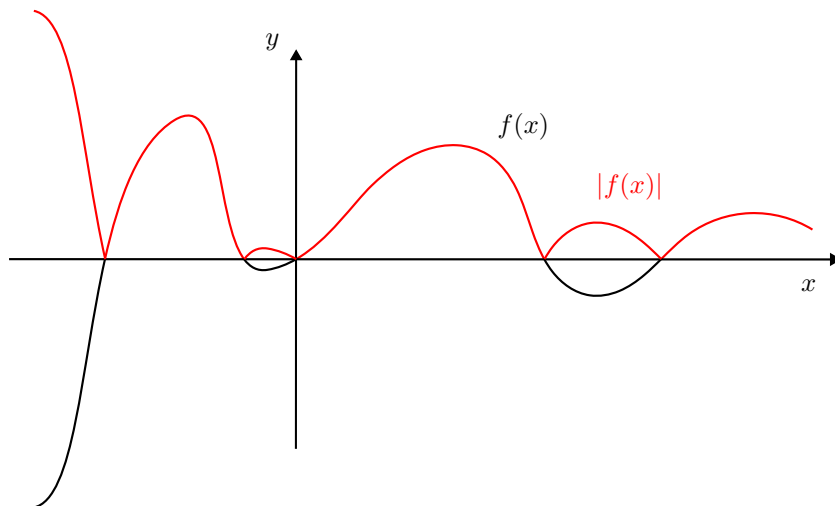


Figure 2: An example function, with the modulus of the function in red.

Now what might $f(|x|)$ look like instead? Since we know that the modulus function makes negative numbers positive, we care about when x is negative or positive.

Since x is on the x axis (of course it is!), we know whether it is negative or positive based on

whether it is to the left or the right of the y axis. When it is on the right of the y axis, x is positive and therefore $f(|x|)$ is unchanged. When it is on the left of the y axis, the x point that we plug into $f(x)$ is shifted to be the positive x point. This means that the y position of $f(|x|)$ is given by whatever the y position is by going an equal amount to the right of the y axis than you are on the left of it. This is the same as simply reflecting the right side of the y axis to the left side. This is shown in Figure 3.

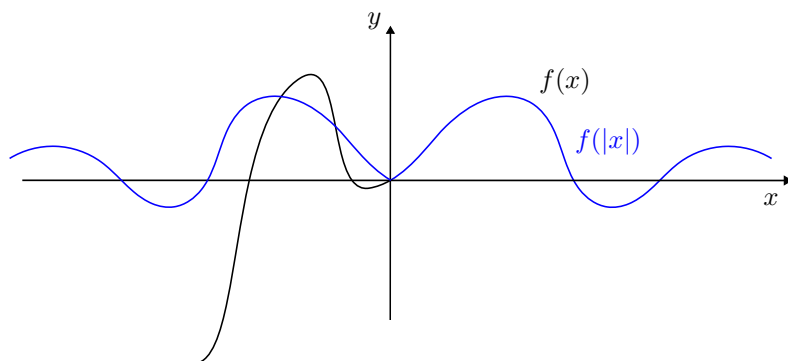


Figure 3: An example function, with the function using the modulus of the input in blue.

Now what about a combination of the above? What would $-|f(|x|)|$ look like? We can split this up into steps, as in when inputting an x , what steps happen to x ? This works inside-out in order:

$$\begin{array}{ll}
 x \mapsto |x| & \text{(taking the modulus)} \\
 \mapsto f(|x|) & \text{(applying } f) \\
 \mapsto |f(|x|)| & \text{(taking the modulus)} \\
 \mapsto -|f(|x|)| & \text{(taking the minus sign)}
 \end{array}$$

Now we can sketch each one in order, since each step individually we have done above. For example, we can start with $f(|x|)$, which we sketched in Figure 3. From here, the next step is to take the modulus. From our discussion leading up to Figure 2, we know that this amounts to reflecting everything below the x axis to above the x axis. We will do the same thing, but to $f(|x|)$. This is sketched in Figure 4.

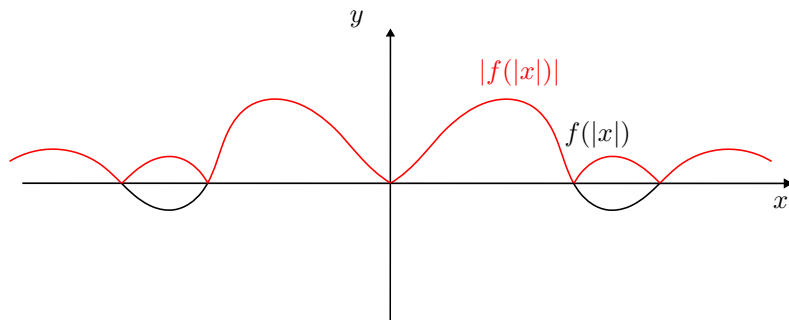


Figure 4: An example function, with the function using the modulus of the input, with the modulus applied in red

And for the last step, we can take the minus sign of the red line sketched for $|f(|x|)|$. This reflects every number about the x axis. This gives us Figure 5.

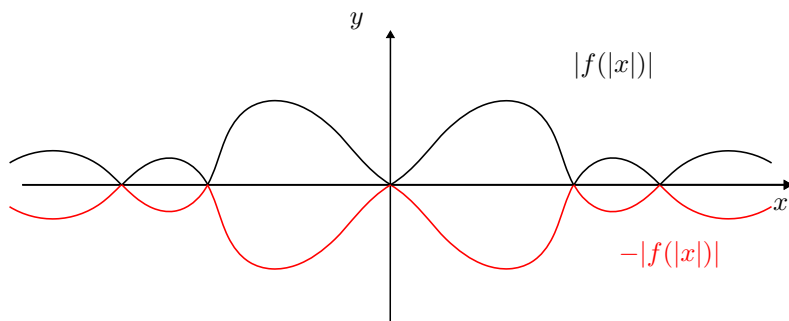


Figure 5: An example function, with the function using the modulus of the input, with the modulus applied, with a negative sign in red

The final result would have been hard to spot directly from Figure 1 without breaking it down step-by-step.

We are often interested in how the range of a function is affected by the modulus.

Example 9. Given that $f(x)$ has range $-16 < f(x) \leq 5$ and reaches every point in between, what

range does $|f(x)|$ have?

Solution. Since the range means that the y values are between -16 and 5 (but not equal to -16 in this case), and we know the modulus turns negative values into positive values, we therefore know that -16 will transform into 16 . Since this is more than 5 , this is the new top-end of the range. For the lower end, we are told that f reaches every point in between. Therefore the lowest point the modulus can reach is 0 .

This gives us the range $0 \leq |f(x)| < 16$, since the original function did not quite reach -16 , the modulus can't reach 16 either. This completes the question. $\square$

The key point here is that the modulus does not “remove” values; it makes them positive. One way to think about the previous question is that -16 is “further negative” than 5 is positive, the new upper end of the range must go to the one that is further out, in this case $|-16|$.

Example 10. Let $f(x) = 2x - 1$, for $-2 \leq x \leq 1$.

1. What is the range of $f(x)$?
2. What is the range of $|f(x)|$?
3. What is the range of $f(|x|)$ if we are given that f is only defined for inputs between -2 and 1 ? What is the range if we allowed $-2 \leq x \leq 1$ for $f(|x|) = 2|x| - 1$?²

Solution. 1. We know that this function is a straight line that is increasing, so therefore the range has to be between whatever the leftmost point and the rightmost point is. The place to be careful is that we are told that $-2 \leq x \leq 1$.

We can find the range therefore by plugging in:

$$\begin{aligned}f(-2) &= 2 \times (-2) - 1 = -4 - 1 = -5 \\f(1) &= 2 \times (1) - 1 = 2 - 1 = 1\end{aligned}$$

Therefore, when $-2 \leq x \leq 1$, we have that $-5 \leq f(x) \leq 1$ is the range. This completes the first part of the question.

2. For this part, it is useful to sketch the graph, Figure 6

²The difference here is whether we impose that f itself is only defined on inputs between -2 and 1 , or if we consider $f(|x|)$ on all inputs between -2 and 1 , which actually implies that we are allowed to use f between the points 0 and 2 , since we are taking the modulus first before plugging into f . This is an important clarification since 2 is outside our original allowed range of x values.

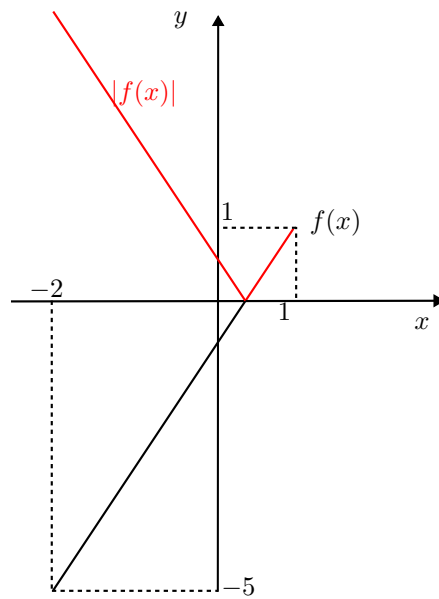


Figure 6: (Sketch) The function $2x - 1$, with the modulus of the function in red, for $-2 \leq x \leq 1$.

From here we can see that the new minimum of the function is 0, and the new maximum is the old minimum, which in this case is $|-5| = 5$.

This gives us the range $0 \leq |f(x)| \leq 5$, which completes this part of the question.

3. For this part, it is again useful to sketch the graph, Figure 7

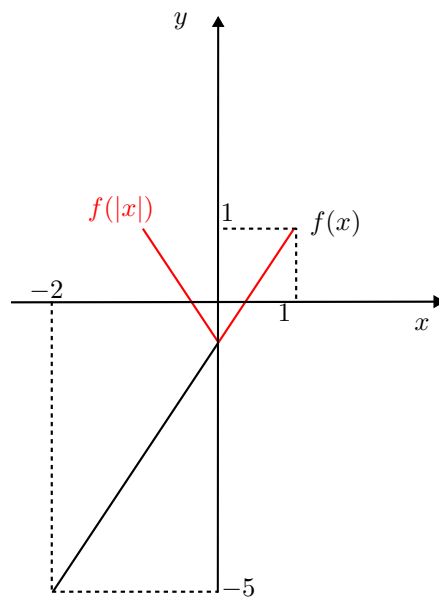


Figure 7: (Sketch) The function $2x - 1$, with the function using the modulus of the input in red, for $-2 \leq x \leq 1$.

Notice that the function cuts off on the left side. This is because we are only allowed to use $f(x)$ when x is between -2 and 1 from the question. But if x is less than -1 , then $|x|$ is more than 1 , which is not allowed to be input into the function.

From here we can see that the new minimum of the function is where the old function

intersected the y axis, at $x = 0$, and the new maximum is the maximum of the right side of $f(x)$, which in this case is at $x = 1$ or $x = -1$.

At $x = 0$,

$$f(|0|) = 2 \times 0 - 1 = -1$$

This gives the minimum as -1 . At $x = 1$,

$$f(|1|) = 2 \times 1 - 1 = 1$$

This gives us the range $-1 \leq f(|x|) \leq 1$, which completes this part of the question.

If we allow $-2 \leq x \leq 1$ as in the second half of this question, the red line would extend backwards to $x = -2$, where $f(|x|) = 2 \times |-2| - 1 = 3$, and therefore the range in this case is $-1 \leq f(|x|) \leq 3$.

This completes the full question.

□

2 Components of Forces or Vectors

It is useful to revisit what sin and cos are.

Let's draw a circle with radius r on an x, y axis, and label a point $\vec{a} = \begin{bmatrix} a \\ b \end{bmatrix}$ on the circle. Note that the equation for a circle of radius r is $x^2 + y^2 = r^2$.

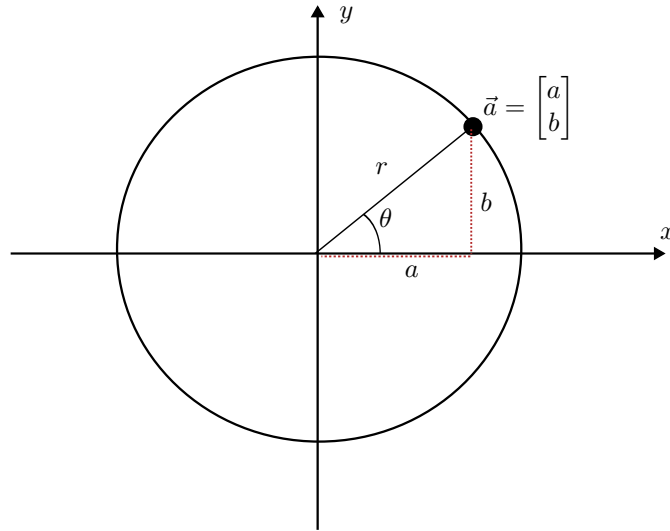


Figure 8: A labelled circle

Then $a = r \cos \theta$ and $b = r \sin \theta$ – you might remember this from GCSE via SOH-CAH-TOA. If we interpret r as the magnitude of a force pointed in the direction of $\vec{a}$, we can interpret this setup as “ a is the component of the force in the x direction, and b is the component of the force in the y direction.”

We always consider our force as the hypotenuse of the triangle when using these trigonometry rules. From Pythagoras, we get that $r^2 = a^2 + b^2$, which is the same as our circle equation!

And because we can write a and b in terms of the magnitude r and the angle θ , we can write:

$$\vec{a} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}$$

Which explains what we mean when we talk about components – given the angle and magnitude of a force from an axis, we can figure out how much of that force is in the direction of each axis.

In other words, if you are a 2-dimensional person standing at the origin $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and a force of magnitude r is pushing you in the direction of $\vec{a}$ (i.e. at an angle of θ from the x -axis), we can write that the amount the force is pushing you to the right (the x -direction) is $r \cos \theta$ and the amount the force is pushing you up (the y -direction) is $r \sin \theta$.

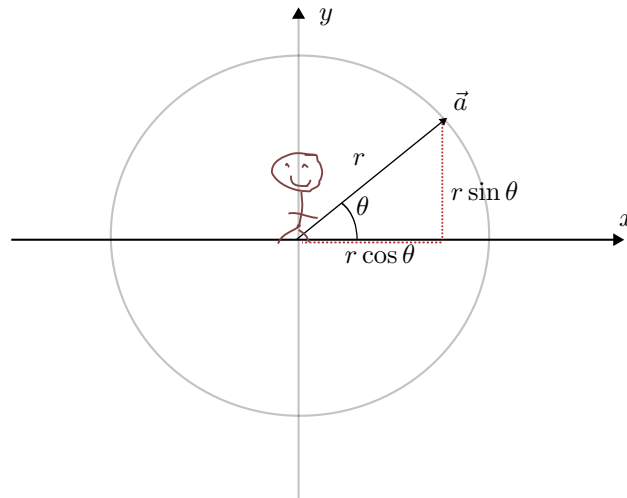


Figure 9: A labelled circle interpreted as forces

Now what happens when we have a physics problem which involves inclines? Well, depending on which direction we care about, we can turn our head and think about it like the circles above!

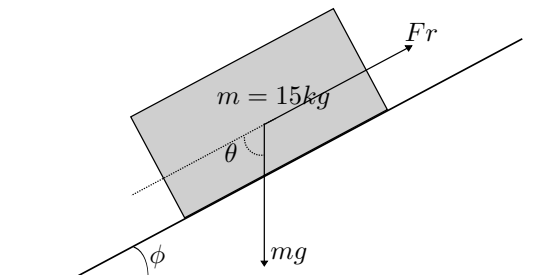


Figure 10: A block on an incline

In this type of question where a force (e.g. friction Fr) is pointing up or down the slope, and another force (e.g. due to gravity) is pointing in another direction, we often care about the direction of the plane that the block is on. We can “turn our head” like this:

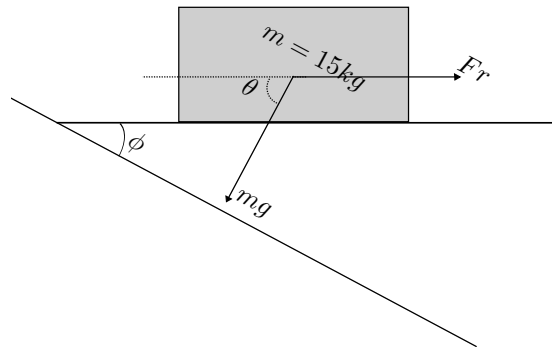


Figure 11: A block on an incline, with your head turned

Then to find, for example, the amount of force in the direction of going down the inclined plane, we apply our knowledge of sin or cos to this problem! Remembering to **make the force itself the hypotenuse**, we can draw our triangle:

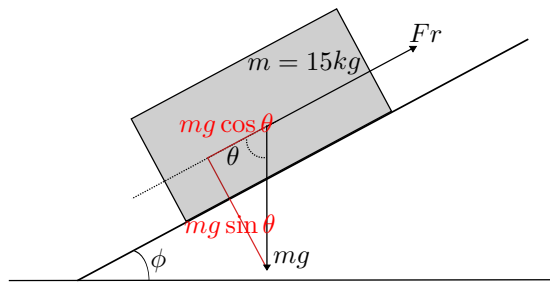


Figure 12: A block on an incline, with a triangle

The force in the direction of going down the slope would therefore be:

$$F = mg \cos \theta - Fr$$

We took away Fr because it is going in the opposite direction to down the slope.

But here we conveniently have θ ! If we were only given ϕ , the angle of the incline, we need to do some geometry.

Exercise 4. Show using geometry that, in the slope problem above, $\theta = \frac{\pi}{2} - \phi$ in radians (90 degrees minus ϕ). Hint: look at the diagram below. Why have I labelled the purple ϕ there?

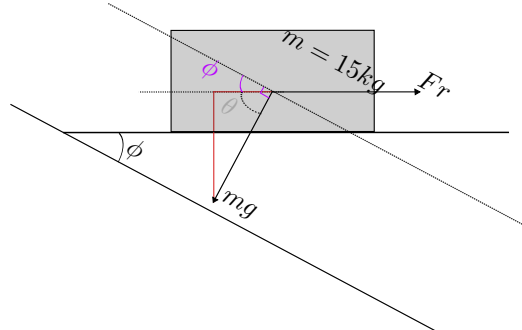


Figure 13: A block on an incline, with a triangle, with your head turned

Now using the above and sin and cos rules, we know that:

$$\begin{aligned}\cos(\theta) &= \cos\left(\frac{\pi}{2} - \phi\right) \\ &= \cos\left(\phi - \frac{\pi}{2}\right) \\ &= \sin(\phi)\end{aligned}$$

This gives us an alternative formula (actually, three alternative formulae), which is secretly the same formula for the amount of force pointing down the slope in terms of the incline of the slope ϕ :

$$F = mg \sin(\phi) - Fr$$

But despite using sin here, we got to this point by using all of our component ideas by drawing the triangle!

The point is: figure out the direction you care about, draw the triangle you want, and then use sin or cos depending on where your force is pointing, and the direction of the force is always considered the hypotenuse when using trig rules.

Example 11. A block of mass 20 kg is on a smooth slope of incline 20° , as in Figure 14

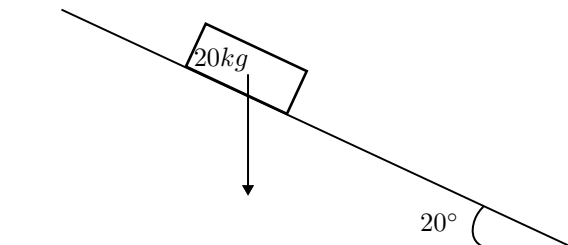


Figure 14: A block on an incline

1. What forces are acting on the block? Draw a labelled diagram of the forces.
2. How fast is the block accelerating down the slope?
3. If you introduce a new force acting on the block pointing up the slope, how much force would need to be applied to stop the block from accelerating down the slope?
4. How would you introduce friction? What coefficient of friction would require the block to not accelerate down the slope if no other forces are applied?

Solution. 1. Since the block is not falling through the plane, there must be a force R perpendicular to the plane acting on the block to balance the force of gravity pushing into the plane. The force due to gravity is mg pointing downwards. Therefore we get the diagram Figure 15. From the analysis above, we know that the angle between the perpendicular and the force due to gravity is 20° .

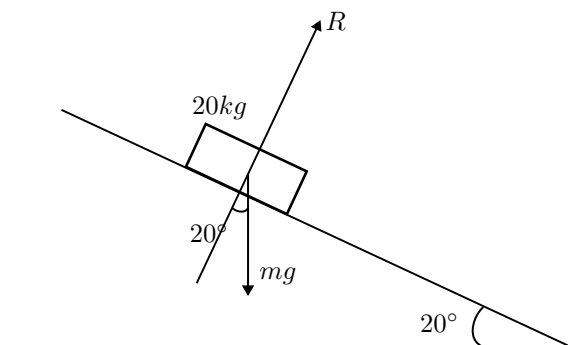


Figure 15: A block on an incline, labelled

What is R ? To find R , we must find the component of gravity in the axis perpendicular to R . Then we can use trigonometry to find the magnitude of the component of gravity in the direction of the perpendicular to the plane is $mg \cos 20^\circ$.

Since the block is not falling through the plane, we know that the total forces perpendicular to the plane must sum to 0. Since gravity is pointing more into the plane than out of the plane, we know it must have the opposite sign to R itself.

Let's choose our positive direction as in the direction of R . Then we have that:

$$\begin{aligned}\text{sum of forces in direction perpendicular to plane} &= 0 \\ \therefore R - mg \cos 20^\circ &= 0 \\ \therefore R &= mg \cos 20^\circ\end{aligned}$$

What would have happened if we chose the opposite direction as our positive direction? Then we would have R is negative and the force due to gravity in the axis perpendicular to the plane is positive:

$$\begin{aligned}\text{sum of forces in direction perpendicular to plane} &= 0 \\ \therefore -R + mg \cos 20^\circ &= 0 \\ \therefore R &= mg \cos 20^\circ\end{aligned}$$

We get the same answer, as expected, because we are consistent with our choice of positive direction.

Using $m = 20$ and $g = 9.8$, we have that $R = 20 \times 9.8 \times \cos 20^\circ$.

2. For the acceleration down the slope, we must find the component of all forces going down the slope. Using trigonometry, we know that the component of gravity in the axis of the slope is $mg \sin 20^\circ$. Let's choose our positive direction as going down the slope. Since gravity is pointing more down the slope than up the slope, then our component of gravity in the axis of the slope has a positive sign.

For demonstration purposes, we will also resolve the force R in the direction of the slope. In this case, we have the component of R in the axis of the slope is $R \cos 90^\circ$, but this is 0 since $\cos 90^\circ = 0$. This is actually true in general; forces do not contribute to the perpendicular direction they act in.

Therefore, the total force down the slope is $mg \sin 20^\circ$ (+0 if you also resolved R). Using $F = ma$ we can rearrange to find $a = \frac{F}{m}$, we have that the acceleration down the slope is

$$\begin{aligned}a &= \frac{F}{m} \\ &= \frac{mg \sin 20^\circ}{m} \\ &= g \sin 20^\circ \\ &= 9.8 \sin 20^\circ\end{aligned}$$

Therefore, $a = 9.8 \sin 20^\circ \text{ ms}^{-2}$.

3. We are now asked to introduce a new force F pointing up the slope. We can see the additional force sketched in Figure 16, which is the force we would need to apply up the slope to stop the block from accelerating down the slope.

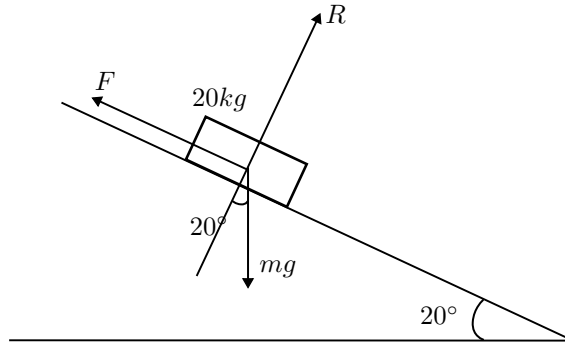


Figure 16: A block on an incline, labelled with additional force

We are asked to stop the block from accelerating down the slope, which means we want the total force down the slope to be 0.

Using the previous question, we found that the force down the slope (without F) is $mg \sin 20^\circ$. Keeping with the same positive direction as pointing down the slope, since F is pointing up the slope, it is working against the force down the slope, so we need to take it away from the force down the slope to get the total force.

$$\text{sum of forces in direction of the slope} = mg \sin 20^\circ - F$$

To stop the block accelerating, we need this total to be 0:

$$mg \sin 20^\circ - F = 0$$

Which we can rearrange to find

$$F = mg \sin 20^\circ$$

4. We are now asked to introduce friction. Recall that the force of friction is given by $F_{max} = \mu R$ where μ is the coefficient of friction and R is the normal reaction force we calculated in the first part.

Ignoring F from the previous part, we now have that the total force down the slope is

$$\begin{aligned} \text{sum of forces in direction of the slope} &= mg \sin 20^\circ - F_{max} \\ &= mg \sin 20^\circ - \mu R \\ &= mg \sin 20^\circ - \mu mg \cos 20^\circ \end{aligned}$$

To stop the block accelerating, we need this total to be 0:

$$\begin{aligned} mg \sin 20^\circ - \mu mg \cos 20^\circ &= 0 \\ \therefore \mu mg \cos 20^\circ &= mg \sin 20^\circ \\ \therefore \mu &= \frac{mg \sin 20^\circ}{mg \cos 20^\circ} = \tan 20^\circ \end{aligned}$$

□

It is important to choose a positive direction and stick to it when resolving forces, and use the nearest direction to determine what sign a force has compared to your chosen positive direction.

The final part of the previous question is interesting since we found that the coefficient of friction required to stop the block from accelerating down the slope only uses the slope's angle. In fact, it is possible to show that the coefficient of friction required to stop the block from accelerating down the slope is $\tan \phi$ for any slope of incline ϕ .

Example 12. A block of mass m is on a rough slope of incline ϕ . What coefficient of friction μ would be required to stop the block from accelerating down the slope if no other forces are applied?

What happens if the slope is 0 degrees? What happens near to 90 degrees?

Solution. We can use the same analysis as in the previous question, but leaving m and ϕ as unknowns.

Resolving in the direction of the slope, using down the slope as our positive direction we find that we must have that:

$$\begin{aligned} 0 &= mg \sin \phi - \mu mg \cos \phi \\ \therefore \mu mg \cos \phi &= mg \sin \phi \\ \therefore \mu &= \frac{mg \sin \phi}{mg \cos \phi} = \tan \phi \end{aligned}$$

Therefore the coefficient of friction required to stop the block from accelerating down the slope is $\tan \phi$.

If the slope is 0 degrees, then $\tan 0^\circ = 0$, so we would need a coefficient of friction of 0 to stop the block from accelerating down the slope. This makes sense since if the slope is flat, there is no force down the slope, so we do not need any friction to stop the block from accelerating.

If the slope is near to 90 degrees, then $\tan \phi$ shoots off to infinity, so we would need a very large coefficient of friction to stop the block from accelerating down the slope. This also makes sense since if the slope is nearly vertical, there is a large force down the slope, so we would need a large amount of friction to stop the block from accelerating.

□

2.1 Additional Exercises

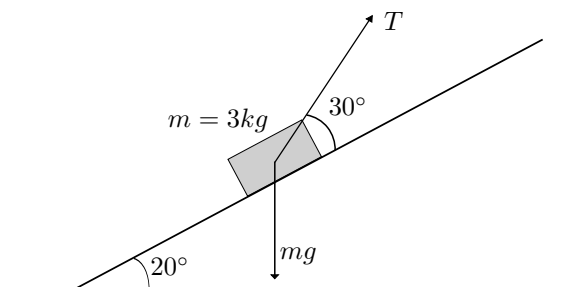


Figure 17: A block on an incline, pulled by a rope

Exercise 5. A small box of mass 3 kg moves on a rough plane which is inclined at an angle of 20° to the horizontal. The box is pulled up a line of greatest slope of the plane using a rope which is attached to the box. The rope makes an angle of 30° with the plane, as shown in figure 17. The rope lies in the vertical plane which contains a line of greatest slope of the plane. The coefficient of friction between the box and the plane is 0.2. The tension in the rope is 25 N.

The box is modelled as a particle, the rope is modelled as a light inextensible string and air resistance is ignored.

Using the model, find the acceleration of the box.

Solution. First write the relevant equations: $F = ma$, $F_{max} = \mu R$. Then draw a diagram with the forces labelled, and with the triangles drawn using geometric rules as in Figure 18.

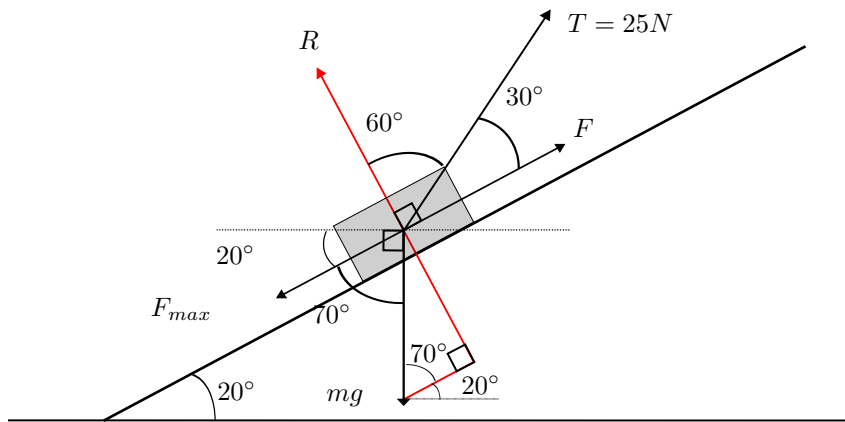


Figure 18: A block on an incline, pulled by a rope, annotated

There are four forces:

1. Friction F_{max} , pointing down the slope.
2. Gravity, pointing downwards.
3. Tension T , pointing in the direction of the rope being pulled.
4. The resultant normal force R from the plane the block is on, causing it to not sink into the plane.

To find the acceleration up the slope, we need the total force going up the slope.

For this, we find the components of each of the forces using the labelled diagram and the appropriate use of sin and cos, e.g. via SOH-CAH-TOA.

Reminder: the original force direction is always considered the hypotenuse.

To find R , we need all forces that act in the direction of perpendicular to the slope. We can then find R by writing down the necessary equation that all the forces in that direction must sum to zero – the block is in equilibrium in the direction of the normal to the plane. We have some

component of tension T from the rope pointing in that direction, gravity is pointing down, away from R , giving gravity a negative sign.

$$R + T(\nearrow) - F_{gravity}(\searrow) = 0$$

And therefore

$$R = F_{gravity}(\searrow) - T(\nearrow)$$

Let's first find the component of the resultant force of gravity pointing in the direction perpendicular to the slope:

$$F_{gravity}(\searrow) = mg \sin(70^\circ)$$

Now find the component of tension pointing in the direction perpendicular to the slope

$$T(\nearrow) = 25 \cos(60^\circ)$$

Then

$$R = mg \sin(70^\circ) - 25 \cos(60^\circ)$$

So

$$F_{max} = 0.2R = 0.2(mg \sin(70^\circ) - 25 \cos(60^\circ))$$

Now we have found the friction force, we need to also find the remaining two forces which contribute in the direction of the slope, gravity and tension.

For gravity, we can find its component going up the slope via

$$F_{gravity}(\swarrow) = mg \cos(70^\circ)$$

Note this is pointing down the slope because the arrow is closer to being down the slope (acute angle) than going up the slope (obtuse angle).

For tension, we can again find its component going up the slope via

$$T(\nearrow) = 25 \cos(30^\circ)$$

Then, combining it all together, we can find the total force up the slope is:

$$\begin{array}{lll} F = T(\nearrow) & -F_{gravity}(\swarrow) & -F_{max} \\ = 25 \cos(30^\circ) & -mg \cos(70^\circ) & -0.2(mg \sin(70^\circ) - 25 \cos(60^\circ)) \end{array}$$

We take away forces going in the opposite direction because they act against our "positive" direction, up the slope.

Then we can compute acceleration by rearranging $F = ma$ to $a = F/m$, which gives us that the acceleration of the block up the slope is

$$a = \frac{25 \cos(30^\circ) - mg \cos(70^\circ) - 0.2(mg \sin(70^\circ) - 25 \cos(60^\circ))}{m} \text{ m/s}^2$$

If you are confused about the correct signs of each of them, look at the force diagram again. We want to take the tension direction up the slope as positive, and the gravity and friction directions to work against the rope down the slope as negative. $\square$

2.2 The reusable approach to mechanics questions

As long as we are consistent with what direction our forces are relative to the direction we chose, the maths works out. The key thing is to be sure what direction the component of the force you are looking at faces, like we did in Exercise 11 with gravity pointing more away from R than towards it.

Most physics questions become exactly the following pattern:

- Find which direction(s) we have an equilibrium
- Decide a positive direction for that direction
- Resolve your forces relative to that positive direction
- Write your force equation, usually $F(\text{in some direction}) = 0$

The exception is when we have a non-equal force being applied, e.g. when there is acceleration. The only difference is that the total force is no longer equal to 0, or the acceleration is not equal to zero. Then you can use

$$F = ma$$

to find that

$$a = \frac{F}{m},$$

and use

$$\frac{F(\text{in some direction})}{m} = a$$

instead of $F(\text{in some direction}) = 0$. Still, the approach is the same.

2.3 Friction and R

Friction is found via the formula $F_{max} = \mu R$ where F_{max} is the magnitude of force friction applies opposing the movement of the object (the direction it is acting in the most), μ is the coefficient of friction, and R is the normal reaction force due to the block being in equilibrium in the direction normal to the plane, i.e. the force pointing perpendicular to the surface the block is on which is preventing the block from sinking into the ground. For example, see Figure 18.

You can reason your way to R needing to exist by considering the following. When any object is in equilibrium in a direction (not moving or in constant speed), the forces applied to it in that direction must overall become zero. If you had a block on a horizontally flat plane with gravity being the only force on the block, we would have the force downwards not be balanced by upward force, resulting in it accelerating downwards due to $F = ma$. Explicitly, $a = F/m$, and because in this thought experiment we have only one force (gravity) acting downwards and a non-zero mass, a must be a positive number. Therefore, if our block is not falling through the plane, there *must* be an additional force being applied to it so that $F = 0$ in the direction perpendicular to the plane.

In fact, most physics problems can be reduced to just looking at the equation $F = ma$ in a smart way, using that the forces in certain directions must be 0, etc.

To find R , we calculate the magnitude of force in that direction the same way we always calculate forces in a direction: we find each force that is acting on the object, and find the component of each force in the same direction of R via trigonometry, where we are considering the direction of R to be our positive direction. We then use the fact that the block is not falling through the plane to know that $F = 0$ in the direction perpendicular to the plane. Then we can rearrange the resulting equation for R .

Note that this construction will always result in R being equal the components of all the forces pointed in its opposite direction.

Note also that most of the time we compute R to calculate F_{max} . If a question does not include friction, we often do not need to compute R .

In summary, R is the additional force which acts on the block to prevent it from sinking into the plane it is on, and can be calculated because therefore $F = 0$ in the perpendicular direction to the plane.

A keen-eyed reader would notice this is mostly an application of Newton's Third Law.

Let's see some examples for calculating R , independently from the context of friction.

Example 13. A 5 kg block is on a 20° inclined plane, and an angry man does not want the block to slide onto his property. To counteract this, he blows on the block with a force of 10 N at an angle of 30° to the inclined plane, above the inclined plane, in an attempt to blow the block away.

What is the magnitude of R , the normal reaction force to the inclined plane?

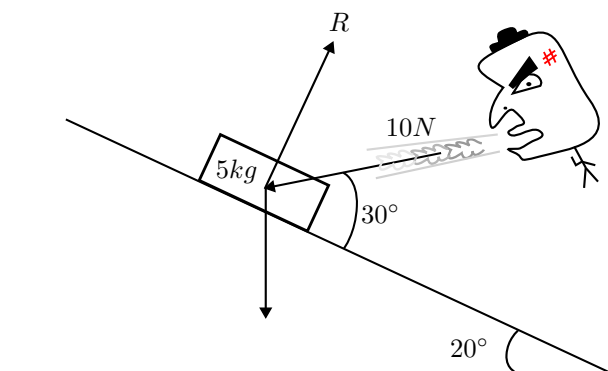


Figure 19: A block on an incline, being blown away by an angry man

Solution. First draw a diagram with the forces labelled, and with the triangles drawn using geometric rules, Figure 20.

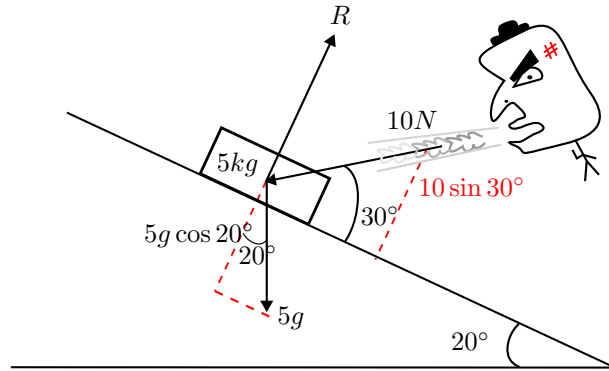


Figure 20: A block on an incline, being blown away by an angry man, annotated

There are three relevant forces:

1. The man blowing, pointing in the direction of 30° to the slope. We will call this force B .
2. Gravity, pointing downwards.
3. The normal reaction force R of the plane on the block, preventing the block from falling through the plane.
4. (Ignored) Friction, which we are ignoring in our calculation of R , and it was not mentioned in this question anyway!

To find R , we need to find the component of each force going in the direction of R , and then use the fact that $F = 0$ in the direction perpendicular to the plane since the block is not falling through the plane. We are taking the direction of R to be our positive direction. This means that since the man blowing and gravity are both more away from R than toward it, we are going to take care to include negative signs.

$$R - F_{gravity}(\swarrow) - B(\swarrow) = 0$$

Therefore

$$R = F_{gravity}(\swarrow) + B(\swarrow)$$

Using our labelled force diagram, we can calculate:

$$\begin{aligned} F_{gravity}(\swarrow) &= 5g \cos 20^\circ \\ B(\swarrow) &= 10 \sin 30^\circ \end{aligned}$$

Therefore, $R = 5g \cos 20^\circ + 10 \sin 30^\circ$.

□

Example 14. A 5 kg block is on a 20° inclined plane, and an angrier mole-man does not want the block near his property. To counteract this, he blows on the block with a force of 10 N at an angle of 30° to the inclined plane, below the inclined plane (he is a mole-man that can blow through the ground), in an attempt to blow the block away.

What is the magnitude of R , the normal reaction force to the inclined plane?

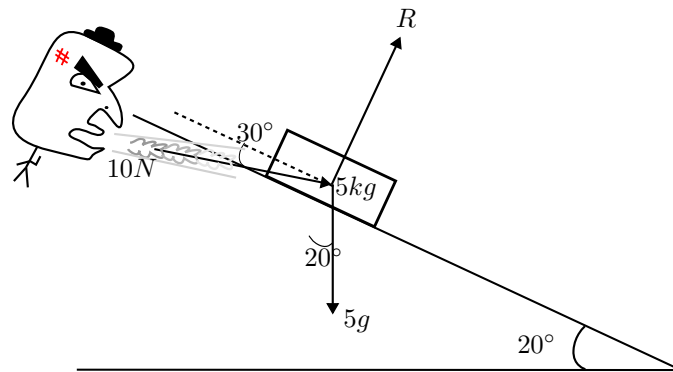


Figure 21: A block on an incline, being blown away by an angry mole-man

Solution. First draw a diagram with the forces labelled, and with the triangles drawn using geometric rules, Figure 22.

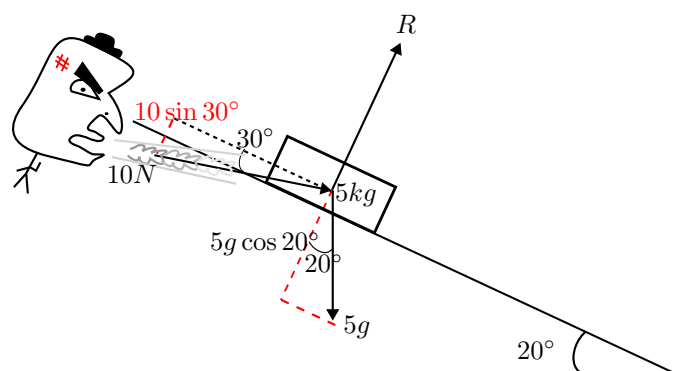


Figure 22: A block on an incline, being blown away by an angry mole-man, annotated

There are three relevant forces:

1. The mole-man blowing, pointing in the direction of 30° to the slope from below. We will call this force B .
2. Gravity, pointing downwards.
3. The normal reaction force R of the plane on the block, preventing the block from falling through the plane.
4. (Ignored) Friction, which we are ignoring in our calculation of R , and it was not mentioned in this question anyway!

To find R , we need to find the component of each force going in the direction of R , and then use the fact that $F = 0$ in the direction perpendicular to the plane since the block is not falling through the plane. We are taking the direction of R to be our positive direction. This means that since gravity is both more away from R than towards it, and since the mole-man blowing is more in the direction of R than away from it, we are going to take care to include negative signs where appropriate.

$$R - F_{gravity}(\swarrow) + B(\nearrow) = 0$$

Therefore

$$R = F_{gravity}(\swarrow) - B(\nearrow)$$

Using our labelled force diagram, we can calculate:

$$\begin{aligned} F_{gravity}(\swarrow) &= 5g \cos 20^\circ \\ B(\nearrow) &= 10 \sin 30^\circ \end{aligned}$$

Therefore, $R = 5g \cos 20^\circ - 10 \sin 30^\circ$. □

Example 15. A 10 kg block is on a rough flat plane, and a force P is applied to the block at an angle of 45° to the plane. The coefficient of friction is 0.1. Given that the block is accelerating along the plane at a rate of 0.3 ms^{-2} and moving in the direction of acceleration, what is P ?

Solution. First draw a diagram with the forces labelled, and with the triangles drawn using geometric rules, Figure 23.

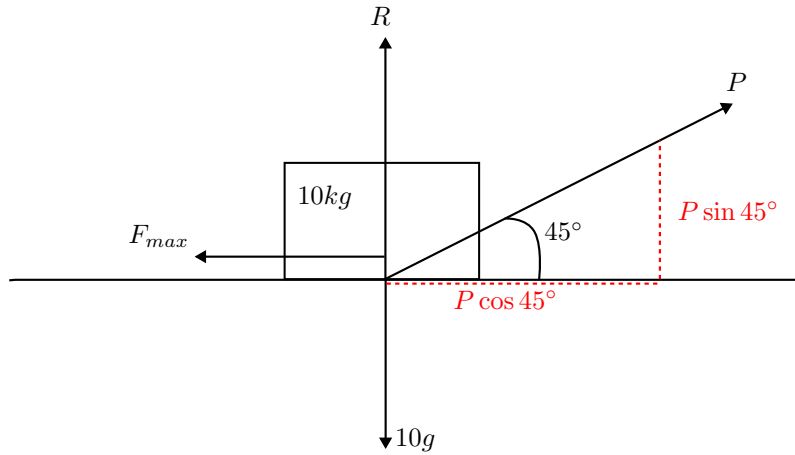


Figure 23: A block on a plane, with an unknown force

Write the relevant equations $F = ma$ and $F_{max} = \mu R$.

There are four forces:

1. The unknown force P , at an angle of 45° to the horizontal.
2. Gravity, pointing downwards.
3. The normal reaction force R of the plane on the block, preventing the block from falling through the plane.
4. Friction, pointing away from the direction of movement.

We will calculate the force F in the direction of the block moving (to the right in our diagram).

$$F = P(\rightarrow) - F_{max}$$

To find R , we need to find the component of each force going in the direction of R , and then use the fact that $F = 0$ in the direction perpendicular to the plane since the block is not falling through the plane. We are taking the direction of R to be our positive direction. This means that since gravity is both more away from R than towards it, and since P is more in the direction of R than away from it, we are going to take care to include negative signs where appropriate.

$$R - F_{gravity} + P(\uparrow) = 0$$

Therefore

$$R = F_{gravity} - P(\uparrow)$$

Using our labelled force diagram, we can calculate:

$$\begin{aligned} F_{gravity} &= 10g \\ P(\uparrow) &= P \sin 45^\circ \end{aligned}$$

Therefore, because gravity is pointing more away from R than towards it, and we have chosen R to be our positive direction,

$$R = 10g - P \sin 45^\circ$$

Hence, using $F_{max} = \mu R$, we have found that

$$F_{max} = 0.1(10g - P \sin 45^\circ)$$

Then, use the fact that

$$P(\rightarrow) = P \cos 45^\circ$$

Which lets us write

$$F = P \cos 45^\circ - 0.1(10g - P \sin 45^\circ)$$

Using that $a = 0.3$, and that $a = F/m$, we have that

$$0.3 = \frac{P \cos 45^\circ - 0.1(10g - P \sin 45^\circ)}{10}$$

$$\begin{aligned} 0.3 &= -\frac{1}{10}g + \frac{P}{10} \cos 45^\circ + \frac{P}{100} \sin 45^\circ && \text{(expanding)} \\ &= -\frac{1}{10}g + P\left(\frac{1}{10} \cos 45^\circ + \frac{1}{100} \sin 45^\circ\right) && \text{(collecting } P \text{ terms)} \end{aligned}$$

We can rearrange for P

$$\begin{aligned} 0.3 + \frac{1}{10}g &= P\left(\frac{1}{10} \cos 45^\circ + \frac{1}{100} \sin 45^\circ\right) \\ \frac{0.3 + \frac{1}{10}g}{\frac{1}{10} \cos 45^\circ + \frac{1}{100} \sin 45^\circ} &= P \end{aligned}$$

and calculate that $P = 16.5$, rounded to 1 decimal place. □

2.4 Moments

Moment about a point is *Perpendicular Force* times *Perpendicular Distance* to the point, and acts rotationally to the point, in the orientation that the perpendicular force is pointing towards. You can choose which point to look at as your base point (or pivot)! If you are careful the maths will always work out.

Moment of force = Perpendicular force $\times$ Distance from the base point

As with all force questions, you need to account for all of the forces!

Why would you choose one point as your base point over another? Looking at the equation, we have that any forces at the point itself gives zero contribution to the moment of force at that point.

This is because the distance is 0, and the right-hand side of the equation cancels. This can be used to cancel unknown forces, or to remove a point that has many forces acting on it from giving us a headache.

As with other force equations, you *choose* which direction is your positive direction. Except with moments, you choose which *orientation* your positive direction is – clockwise or anticlockwise – around your base point or pivot. See Figure 25.

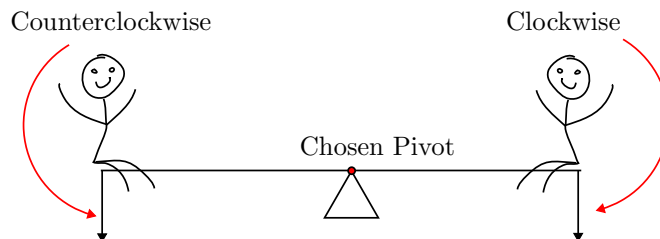


Figure 24: Two happy friends on a see-saw

As long as you *make sure clockwise and counter-clockwise have opposite signs* – just like how we make sure “up” and “down” have opposite signs in the usual force questions – everything will work out!

Our approach will be the following:

1. Draw and label all the forces affecting the object in question, giving them unknown variables if necessary.
2. Label all the distances to the forces.
3. Resolve forces to be perpendicular to the moment if necessary.
4. Choose a base point smartly to remove unknown variables.
5. Choose a positive direction and use equilibria (my favourite) to get force equations and eliminate unknowns.
6. Repeat using equilibria until you get your answer.

What do we mean by equilibria? Just like with force questions before, if our object is not accelerating (or moving) in a direction then the total force in that direction must be zero. Similarly, if our object is not *rotating*, then the *total moment force around a pivot must be zero*.

Pretty much all A-level physics questions become $F = 0$ in a smart way.

Ignoring everything above, let’s start with an example where we do not have an equilibrium.

Example 16. Consider a weightless rod of length 10 m, with a pivot half way along the rod. One happy 60 kg person is sat 2 m from the end of the rod. What is the total moment force on the rod around the pivot? See Figure 25.

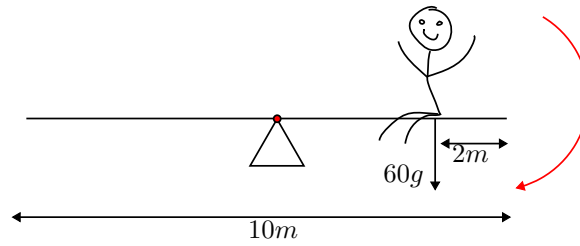


Figure 25: One happy friend on a see-saw

Solution. To begin with, let's draw and label *all* the forces affecting the rod. Since the rod is weightless, the only force is from the person. We need to verify we are looking at the perpendicular component of the force to the rod, which we are. Therefore there are no additional forces to label or compute.

For the distances, we can see that the distance from the half-way point at 5 m in and the person is 3 m, labelled in Figure 26.

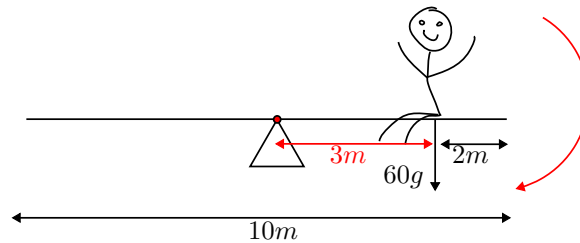


Figure 26: One happy friend on a see-saw, labelled

For lack of a better choice, we can choose clockwise to be our positive direction. Since there are no other pieces of information to use, we can calculate that our total moment force M around the pivot is

$$M = 3 \times 60g = 180g$$

In this case, there is an unbalanced amount of force, meaning this see-saw would start rotating. $\square$

Example 17. Consider a uniform horizontal 4 kg rod of length 5 m, with a pivot 2 m from the left side, and a block on the left side of the rod. The rod is in equilibrium. What is the mass of the block? See Figure 27. For convenience, I will mark the additional calculated forces and distances in red.

Solution. Let's choose our base point to be at the pivot.

To begin with, let's draw and label *all* the forces affecting the rod. The rod is uniform, meaning its weight is acting from the centre. The centre is half way into the rod, which is 2.5 m. Therefore force due to gravity is acting downwards as $4g$ Newtons.

If we label the mass of the block as x , we can write that the force due to gravity at the block is xg Newtons. There are no other (relevant) forces acting on the rod – we will revisit this.

The distances between the centre of the rod and the pivot, and between the pivot and the block can be calculated from taking the difference of 2 and half of 5.

We need to verify we are looking at the perpendicular component of the forces to the rod, which we are.

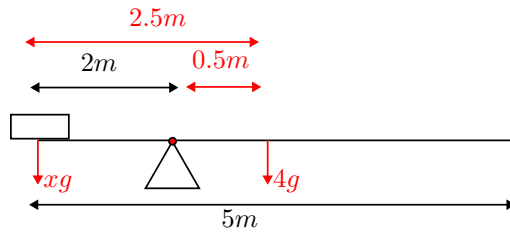


Figure 27: A block on an uneven see-saw, labelled

We can choose clockwise to be our positive direction. We are told the rod is in equilibrium. Therefore all of the moments of force around the pivot (or from any other base point) must cancel.

$$F(\odot) - F(\ominus) = 0$$

Let's calculate the moment of force going clockwise, which we will label $F(\odot)$. The only force acting in this direction is $4g$, and therefore by the moment equation, the moment of force due to the rod from the pivot is

$$F(\odot) = 0.5 \times 4g$$

Similarly for the anticlockwise moment of force on the pivot due to the block is the following.

$$F(\ominus) = 2 \times xg$$

Since there are no other pieces of information to use, we can substitute into our equilibrium equation:

$$\begin{aligned} 0 &= F(\odot) - F(\ominus) \\ &= 0.5 \times 4g - 2 \times xg \end{aligned}$$

Therefore

$$0 = 0.5 \times 4g - 2 \times xg$$

Which we can rearrange for x :

$$0 = 0.5 \times 4g - 2 \times xg$$

$$0 = 2 - 2x \quad (\text{canceling } g)$$

$$0 = 1 - x \quad (\text{dividing by 2})$$

$$x = 1 \quad (\text{rearranging for } x)$$

Therefore, the block is 1 kg. □

Why did we choose our base point as the pivot? We chose this to cancel any forces at the pivot itself. Because the rod is in equilibrium vertically too, we know there must be a reaction force from the pivot onto the rod R which prevents the rod from falling into the pivot. Since we do not know what R is, we chose our base point to be at the pivot itself, cancelling R in our moment equation! If we chose another point, we would need to include R acting upwards from the pivot!

The point: choose the point of rotation smartly: the forces at the point itself can be ignored!

Example 18. Consider a uniform horizontal 20 kg rod of length 7 m, held in equilibrium by two ropes at right angles to the rod as in Figure 28. Find the tension in both ropes.

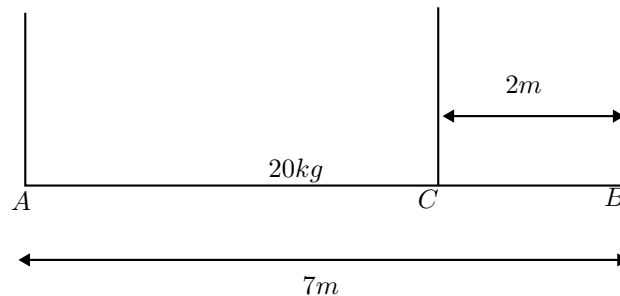


Figure 28: A rod with two ropes attached

Solution. To begin with, let's draw and label *all* the forces affecting the rod. The rod is uniform, meaning its weight is acting from the centre. The centre is half way into the rod, which is 3.5 m. Therefore force due to gravity is acting downwards as $20g$ Newtons.

The other forces are tension in the ropes, pointing upwards. We can label them with T_A and T_C . There are no other forces acting on the rod in this system. See Figure 29.

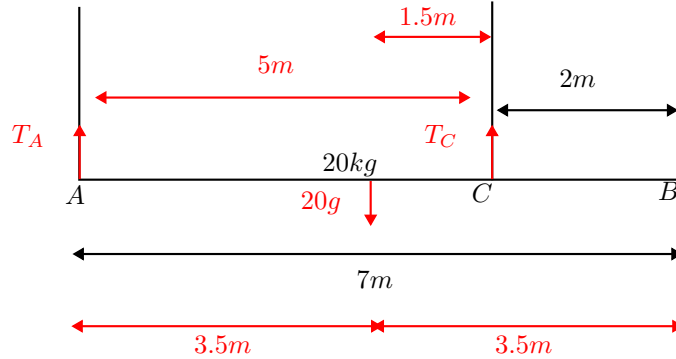


Figure 29: A rod with two ropes attached, labelled

We need to verify we are looking at the perpendicular component of the forces to the rod, which we are.

Since there are two unknowns, a smart choice of base point is at either A or C . We will continue as setting C to be our base point, cancelling T_C from our equations.

We can choose clockwise to be our positive direction.

We are told the rod is in equilibrium. Therefore all of the moments of force around the pivot (or from any other base point) must cancel.

$$F(\odot) - F(\oslash) = 0$$

Let's calculate the moment of force going clockwise, which we will label $F(\odot)$. The only force acting in this direction is T_A , and therefore by the moment equation, the moment of force due to T_A on C is

$$F(\odot) = 5 \times T_A$$

Similarly for the anticlockwise moment of force on the pivot due to gravity is the following.

$$F(\oslash) = 1.5 \times 20g$$

Since there are no other pieces of information to use, we can substitute into our equilibrium equation:

$$\begin{aligned} 0 &= F(\odot) - F(\oslash) \\ &= 5 \times T_A - 1.5 \times 20g \end{aligned}$$

Which we can rearrange for T_A :

$$\begin{aligned}
0 &= 5 \times T_A - 1.5 \times 20g \\
1.5 \times 20g &= 5T_A \\
\frac{30g}{5} &= T_A \\
T_A &= 6g
\end{aligned}$$

Now we have found T_A , we still need to find T_C . There are a few approaches. My favourite is to use another kind of equilibrium: since the rod is not rising or falling, the vertical forces must total to zero. Therefore, we can write:

$$T_A + T_C = 20g$$

And rearrange

$$\begin{aligned}
T_C &= 20g - T_A \\
T_C &= 20g - 6g && \text{(substituting } T_A) \\
T_C &= 14g
\end{aligned}$$

Alternatively, we could write the moment equation from C , and repeat the rearranging from above:

$$5T_C - 3.5 \times 20g = 0$$

Or even you could set the base point as somewhere entirely different and substitute T_A and rearrange. For example, setting the base point to be at B gives the following equation:

$$2T_C - 3.5 \times 20g + 7T_A = 0$$

Each of the three approaches result in the correct answer, so choosing the simplest approach is best. I like the vertical equilibrium option, but the moment at C is perfectly acceptable – just don't do the third option of the moment at B ! $\square$

Summary: look at *all* equilibria! $F = 0$ in that direction.

So far we have only seen questions where the forces are nicely already perpendicular. Let's see what happens when they are not.

Example 19. A 4 m 20 kg uniform ladder is resting in equilibrium against a smooth wall at an angle of 60° to the horizontal. The base of the ladder A is on a rough floor and is at the point of slipping, with the coefficient of friction being 0.4. An 80 kg man is at a point C on the ladder. What is the length AC ? See Figure 30.

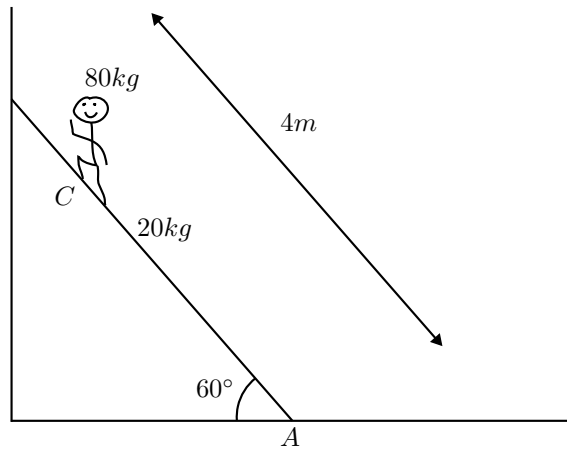


Figure 30: A man on a ladder

Solution. To begin with, let's draw and label *all* the forces affecting the rod.

The rod is uniform, meaning its weight is acting from the centre. The centre is half way into the rod, which is 2 m. Therefore force due to gravity on the ladder is acting downwards as $20g$ Newtons. There is another force due to gravity on the man, acting downwards as $80g$ Newtons at point C .

Since the ladder is slipping at the base, friction F_{max} is acting towards the wall, opposing the slipping away from the wall.

The other forces are resultant forces due to the contact between the ladder and the floor and wall. These forces must exist since we know the ladder is not falling into the floor or the wall, and they act perpendicular to the floor and wall each. We will label the floor resultant force R and the wall resultant force S .

See Figure 31.

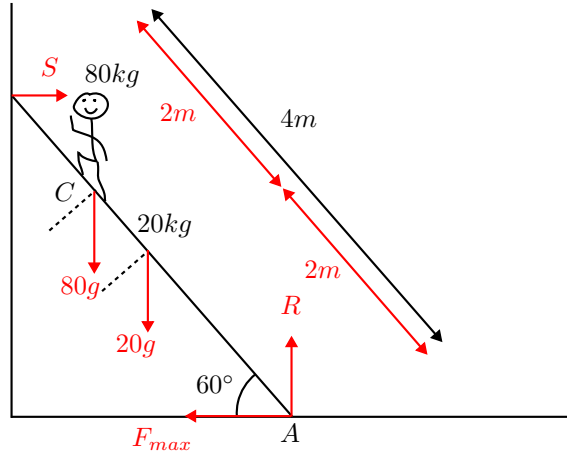


Figure 31: A man on a ladder, labelled

We need to verify we are looking at the perpendicular component of the forces to the rod, which we are not (!). To rectify this, we will need to do some geometry with 60° . This was covered in Section 2 to recap. Here I will simply write the correct usage of sin or cos, and leave it to the reader to verify.

Before starting with moment equations, we can gather information using equilibrium equations.

Since the ladder is in equilibrium horizontally, the horizontal forces must total zero. Choosing S to be our positive direction, we find that

$$S - F_{max} = 0$$

Therefore

$$S = F_{max}$$

Vertically, choosing R to be our positive direction, we find

$$R - 20g - 80g = 0$$

Therefore

$$R = 100g$$

Using $F_{max} = \mu R$, we can find that

$$\begin{aligned} F_{max} &= 0.4R & (\mu = 0.4) \\ F_{max} &= 0.4 \times 100g & (R = 100g) \\ S &= 0.4 \times 100g & (S = F_{max}) \end{aligned}$$

Now we have deduced as much as possible using equilibria, we can tackle the moments part of the problem. Let us set our base point at A to not deal with both F_{max} and R . We are free to do this

since there are no more unknown forces in the system to account for. Let's label the distance AC as x for convenience.

We can choose clockwise to be our positive direction. The clockwise force is from S and the anticlockwise forces are from $80g$ and $20g$, but they are not pointing perpendicular to the ladder. Resolving the forces in the perpendicular direction to the ladder, and using that the total moment of force must be zero, working from left to right, we find

$$4S \sin 60^\circ - 80gx \cos 60^\circ - 2 \times 20g \cos 60^\circ = 0$$

Which is our equilibrium moment equation from A .

We can then finish the question by substituting what S is from above, and rearranging for x , left as an exercise. $\square$

3 Differentiation

Why? Differentiation is the study of rates of change. For example, if you were to model how an epidemic moves through a population, you might want to know how quickly people are infected before and after a vaccination has been introduced. “How quickly” implies a rate of change, and therefore it would be useful to set up a framework where we can find the rate of change of enough functions where we can model even extremely complicated situations.

We have the concept of differentiation from first principles. This is where the definition of “rate of change” is used to explicitly calculate a derivative. This involves something called a limit, and is very tedious and difficult to do in even slightly complex scenarios.

Instead, we have come up with a great framework, where we have a set of standard functions,

$$ax, x^n, \sin x, \cos x, e^x, \ln x$$

Which are varied enough that we can combine them to make any function we could want.

By “combine” we mean three main things:

1. Adding or subtracting functions, e.g. $ax - e^x$
2. Multiplying or dividing functions, e.g. $\frac{x \sin x}{\ln x}$
3. Composing functions as in Definition 1.1, e.g. $\sin(x^3)$

It turns out this is enough to (approximate) *any* function!

As such, we would like to be able to study the rate of change of these combinations of standard functions. This is done in three ways too:

1. For adding or subtracting, we can use that you can differentiate each additive term separately, and add together the answers
2. For multiplying or dividing, we can use the product rule
3. For composition of functions, we can use the chain rule

And combining these three methods allows us to differentiate anything we could want.

3.1 First Principles

The derivative of a function is the slope of the function at a point. You can approximate the slope of a function by calculating the difference in x values and the difference in y values between two points, and divide the difference in y by the difference in x . This is the formula for the slope of a straight line you find in GCSE.

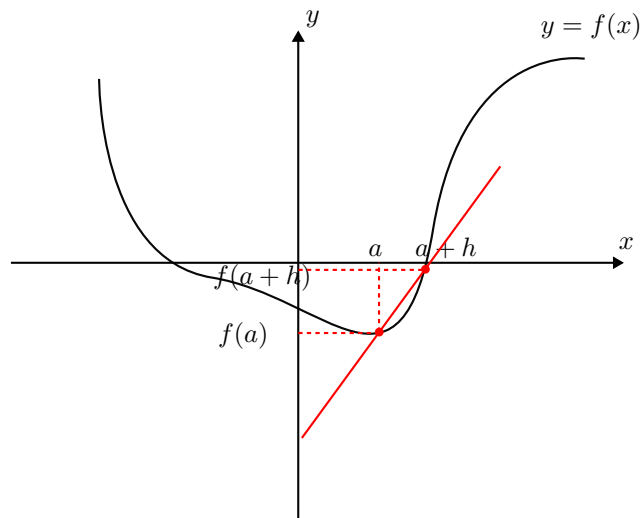


Figure 32: Slope between two points on a function

The slope of the red line in Figure 32 is therefore:

$$\frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h}$$

This approximates the slope of the function $f(x)$ at the point a , but not very well. We can make it better by moving the second point closer and closer to the first, by making h become smaller and smaller.

Taking this to the limit, where h is sent all the way to zero, gives us the derivative of f at a , denoted $f'(a)$:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

This is taking the derivative of f from first principles.

Example 20. Let $f(x) = x^2$. Using first principles, find the derivative of f .

Solution. Let's substitute x^2 into the definition of the derivative.

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h} && \text{(substituting into the formula)} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{a^2} + 2ah + h^2 - \cancel{a^2}}{h} && \text{(expanding numerator)} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(2a+h)}{\cancel{h}} && \text{(factorise } h \text{ from numerator)} \\ &= \lim_{h \rightarrow 0} 2a + h && \text{(cancel } h \text{ from top and bottom)} \\ &= 2a && \text{(now set } h = 0) \end{aligned}$$

Since this is true at every a , we can write $f'(x) = 2x$ □

Because the derivative of a function is the slope of a function, what happens to constants when you differentiate them?

If we plot $f(x) = c$, where c is a constant, we get the graph in Figure 33.

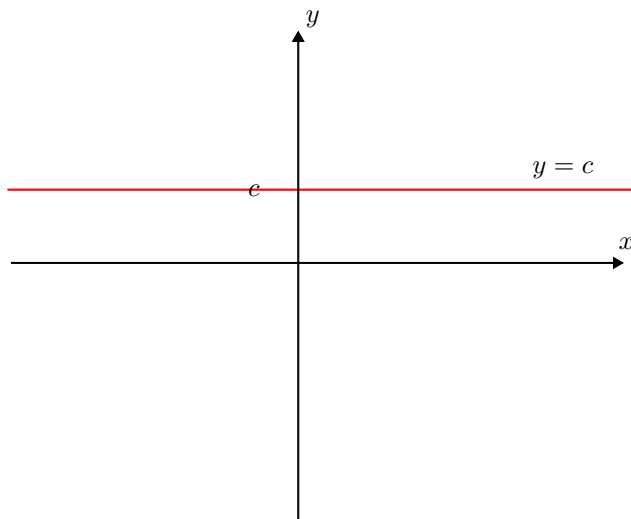


Figure 33: A constant function

What is the slope of this graph? Well, the difference in y between any two points is 0 because they are always at the same height. Therefore the difference in y divided by the difference in x is always 0 since the numerator is always 0.

Therefore if $f(x) = c$ is a constant function, then $f'(x) = 0$.

What about more complex differentiation by substitution?

Example 21. From first principles, find the derivative of $\cos x$.

You may assume the following.

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$$

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\lim(a + b) = \lim(a) + \lim(b)$$

You may also assume that if $f(x)$ is a function of only x , and $g(h)$ is a function of only h , then

$$\lim_{h \rightarrow 0} (f(x)g(h)) = f(x) \lim_{h \rightarrow 0} g(h)$$

Then deduce the derivative of $\sin x$.

Solution. Let's substitute into the first principles formula.

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} && \text{(definition of derivative)} \\
&= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} && \text{(using the third hint)} \\
&= \lim_{h \rightarrow 0} \left(\frac{\cos x \cos h - \cos x}{h} - \frac{\sin x \sin h}{h} \right) && \text{(separate the fraction)} \\
&= \cos x \lim_{h \rightarrow 0} \left(\frac{\cos h - 1}{h} \right) - \sin x \lim_{h \rightarrow 0} \left(\frac{\sin h}{h} \right) && \text{(fourth and fifth hint)} \\
&= \cos x \times 0 - \sin x \times 1 && \text{(first and second hint)} \\
&= -\sin x
\end{aligned}$$

Therefore the derivative of $\cos(x)$ is $-\sin(x)$.

To deduce the derivative of $\sin(x)$, first recall that $\sin(x - \frac{\pi}{2}) = -\cos(x)$ and $\cos(x - \frac{\pi}{2}) = \sin x$. Therefore, using the chain rule with $y = x - \frac{\pi}{2}$ (or by using first principles on $f(x+a)$ for some constant a),

$$\begin{aligned}
\sin'(x) &= \cos'(x - \frac{\pi}{2}) && \text{(using } \cos(x - \frac{\pi}{2}) = \sin(x) \text{)} \\
&= -\sin(x - \frac{\pi}{2}) && \text{(using the chain rule)} \\
&= -(-\cos(x)) && \text{(using } \sin(x - \frac{\pi}{2}) = -\cos(x) \text{)} \\
&= \cos(x)
\end{aligned}$$

Therefore the derivative of $\sin(x)$ is $\cos(x)$. □

To prove the limit of trigonometry identities used as hints, you can use the Taylor expansion of $\sin$ and $\cos$ – try it!

3.2 Product rule

The Product Rule tells us that if we have two functions $u = u(x)$ and $v = v(x)$, then

$$(uv)' = uv' + u'v$$

Example 22. Find the derivative of the function $x \sin(x)$.

Solution. Let $u = x$ and $v = \sin(x)$. So $u' = 1$ and $v' = \cos(x)$. Then by the product rule,

$$(x \sin(x))' = x \cos(x) + 1 \cdot \sin(x)$$

□

Example 23. Find the derivative of the function $\cos x \ln x + x \sin x$.

Solution. Let's write this as $f(x) + g(x)$, where $f(x) = \cos x \ln x$ and $g(x) = x \sin x$.

Then since we can differentiate each additive term separately and add the results, we want to know $f'(x)$ and $g'(x)$.

For $f(x)$, we can let $u = \cos x$ and $v = \ln x$ so that $u' = -\sin x$ and $v' = \frac{1}{x}$. Then by the product rule,

$$\begin{aligned}
(uv)' &= uv' + u'v \\
&= \cos x \cdot \frac{1}{x} + (-\sin x) \cdot \ln x \\
&= \frac{\cos x}{x} - \ln x \sin x
\end{aligned}$$

Which gives that $f'(x) = \frac{\cos x}{x} - \ln x \sin x$. For $g'(x)$, we can reuse the work from the previous question:

$$(x \sin(x))' = x \cos(x) + 1 \cdot \sin(x)$$

Therefore

$$\begin{aligned}
(f(x) + g(x))' &= f'(x) + g'(x) \\
&= \frac{\cos x}{x} - \ln x \sin x + x \cos(x) + 1 \cdot \sin(x)
\end{aligned}$$

Which completes the question. □

3.3 Implicit Differentiation

When you are taking the derivative of a function with respect to x , you are actually applying a function $\frac{d}{dx}$ to both sides of the equation:

$$\begin{aligned}
y &= f(x) \\
\frac{dy}{dx} &= \frac{d}{dx}(f(x))
\end{aligned}$$

The notation on the right hand side is what you are used to doing when you differentiate an equation – you apply all of your differentiation rules to wherever you see an x .

If you had a more complex equality such as

$$y + xy = x^2$$

You can still differentiate this by differentiating *both sides*, and applying any differentiation rules (product rule, chain rule, etc.) necessary.

$$\begin{aligned}
\frac{d}{dx}(y + xy) &= \frac{d}{dx}(x^2) && \text{(differentiating both sides)} \\
\frac{dy}{dx} + \frac{d}{dx}(xy) &= 2x && \text{(expanding LHS and evaluating RHS)} \\
\frac{dy}{dx} + x \frac{d}{dx}(y) + y \frac{d}{dx}(x) &= 2x && \text{(product rule)} \\
\frac{dy}{dx} + x \frac{dy}{dx} + y &= 2x && \text{(using } \frac{dx}{dx} = 1)
\end{aligned}$$

In this way, our equation *implicitly* contains $\frac{dy}{dx}$, but we would need to solve for it. If we are asked to find the value of $\frac{dy}{dx}$ at a single point, e.g. at $x = 1$, $y = \frac{1}{2}$, we can substitute it into our final equation:

$$\frac{dy}{dx} + 1 \frac{dy}{dx} + \frac{1}{2} = 2$$

We can rearrange this to find that, at $x = 1$, $y = \frac{1}{2}$, the gradient $\frac{dy}{dx}$ is

$$\frac{dy}{dx} = 1 - \frac{1}{4} = \frac{3}{4}$$

Remember that this gradient is specifically at the point we substituted in! The gradient at any point would need to be calculated by solving

$$\frac{dy}{dx} + x \frac{dy}{dx} + y = 2x$$

by rearranging for $\frac{dy}{dx}$.

Example 24. Find the gradient of the normal to the curve defined by the equation

$$2x - 2y^2 + 4x^2y = 4x^2 + 8$$

at the point $x = 5$, $y = 1$.

Solution. To approach this question, we can first recall that the normal is the negative reciprocal of the gradient. Therefore, if we find $\frac{dy}{dx}$ at the point $x = 5$, $y = 1$, we can simply take the negative reciprocal of it to find the gradient of the normal.

Hence, let us find the derivative implicitly by taking the derivative of both sides.

$$\begin{aligned} \frac{d}{dx}(2x - 2y^2 + 4x^2y) &= \frac{d}{dx}(4x^2 + 8) \\ 2 - 4y \frac{dy}{dx} + 8xy + 4x^2 \frac{dy}{dx} &= 8x \end{aligned} \quad \text{(using the chain and product rule)}$$

Hence now we can find $\frac{dy}{dx}$ at $x = 5$, $y = 1$ by substituting $x = 5$, $y = 1$ into the above equation and rearranging.

$$\begin{aligned} 2 - 4 \frac{dy}{dx} + 8 \times 5 + 4 \times 5^2 \frac{dy}{dx} &= 8 \times 5 && \text{(substituting } x = 5, y = 1) \\ 2 - 4 \frac{dy}{dx} + 100 \frac{dy}{dx} &= 0 && \text{(simplifying)} \\ \frac{dy}{dx} &= -\frac{2}{96} = -\frac{1}{48} && \text{(rearranging)} \end{aligned}$$

Hence, at $x = 5$, $y = 1$, the gradient is $-\frac{1}{48}$, which means that the normal has gradient 48.

We could find the equation for the normal at this point by using the equation for a line with slope 48 that passes through a point $x = 5$, $y = 1$. $\square$

3.4 The Chain Rule

The chain rule is an essential tool for differentiating, and allows you to differentiate any function as long as you can write it in terms of functions you know how to differentiate (sometimes you might need the product rule too).

It goes as follows.

Theorem 3.1. (The Chain Rule)

If $y = g(u)$ for some function g , and $u = f(x)$ for some function f , then y can be written as a function of x via

$$y = g(f(x))$$

And we have that

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

For more details on function compositions, see Definition 1.1.

What this means is that we can differentiate any composition of the following standard functions:

ax , x^n , $\sin x$, $\cos x$, e^x , $\ln x$.

The easiest way to remember the chain rule is by pretending that they are actual fractions, and then seeing that the right-hand side would cancel to give the left-hand side in the equation.

This is the easiest to see with some examples.

Example 25. Using the chain rule, differentiate $(2x)^2$, using $u = 2x$.

Solution. Obviously, we could write this as $4x^2$ and use our knowledge of differentiation to solve. This is not what the question is asking.

Let's write $y = (2x)^2$, $u = 2x$ so that $y = u^2$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned} \frac{dy}{du} &= 2u && \text{(using the derivative of } u^2) \\ \frac{du}{dx} &= 2 && \text{(using the derivative of } 2x) \end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= 2u \times 2 \\ &= 4u \\ &= 8x && \text{(using } u = 2x) \end{aligned}$$

Which gives us our answer. □

Example 26. Using the chain rule, differentiate $(\ln x)^2$.

Solution. The question is: if we were to replace any part of $(\ln x)^2$ with u , how can we turn it into one of the functions that we know how to differentiate:

au , u^n , $\sin u$, $\cos u$, e^u , $\ln u$?

Let's write $y = (\ln x)^2$, $u = \ln x$ so that $y = u^2$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned} \frac{dy}{du} &= 2u && \text{(using the derivative of } u^2) \\ \frac{du}{dx} &= \frac{1}{x} && \text{(using the derivative of } \ln x) \end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\
&= \frac{2u}{x} && \text{(substituting our workings)} \\
&= \frac{2 \ln x}{x} && \text{(using } u = \ln x)
\end{aligned}$$

Which gives us our answer. □

Example 27. Using the chain rule, differentiate $e^{2 \sin x}$.

Solution. The question is: if we were to replace any part of $e^{2 \sin x}$ with u , how can we turn it into one of the functions that we know how to differentiate:

au , u^n , $\sin u$, $\cos u$, e^u , $\ln u$?

Let's write $y = e^{2 \sin x}$, $u = 2 \sin x$ so that $y = e^u$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned}
\frac{dy}{du} &= e^u && \text{(using the derivative of } e^u) \\
\frac{du}{dx} &= 2 \cos x && \text{(using the derivative of } 2 \sin x)
\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\
&= e^u \times 2 \cos x && \text{(substituting our workings)} \\
&= 2e^{2 \sin x} \cos x && \text{(using } u = 2 \sin x)
\end{aligned}$$

Which gives us our answer. □

Example 28. (*Hard*)

Using the chain rule, differentiate $\sin(\cos(\ln x))$.

Solution. The question is: if we were to replace any part of $\sin(\cos(\ln x))$ with u , how can we turn it into one of the functions that we know how to differentiate:

au , u^n , $\sin u$, $\cos u$, e^u , $\ln u$?

Let's write $y = \sin(\cos(\ln x))$, $u = \cos(\ln x)$ so that $y = \sin u$.

We need the derivative of y with respect to u and the derivative of u with respect to x . We can find the first:

$$\begin{aligned}
\frac{dy}{du} &= \cos u && \text{(using the derivative of } \sin u) \\
\frac{du}{dx} &= (\cos(\ln x))' && \text{(leave for later)}
\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\
&= \cos u (\cos(\ln x))' && \text{(substituting our workings)} \\
&= \cos(\cos(\ln x)) (\cos(\ln x))' && \text{(using } u = \cos(\ln x))
\end{aligned}$$

Which almost gives us our answer, except we still need to differentiate $\cos(\ln x)$. To do this, we can use the chain rule again!

I will use a new symbol v for book-keeping: write $u = \cos(\ln x)$, $v = \ln x$ so that $u = \cos v$.

The chain rule will then be

$$\frac{du}{dx} = \frac{du}{dv} \frac{dv}{dx}$$

We need the derivative of u with respect to v and the derivative of v with respect to x .

$$\begin{aligned} \frac{du}{dv} &= -\sin v && \text{(using the derivative of } \cos v) \\ \frac{dv}{dx} &= \frac{1}{x} && \text{(using the derivative of } \ln x) \end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned} \frac{du}{dx} &= \frac{du}{dv} \frac{dv}{dx} \\ &= -\sin v \times \frac{1}{x} && \text{(substituting our workings)} \\ &= -\frac{\sin(\ln x)}{x} && \text{(using } v = \ln x) \end{aligned}$$

This gives us the derivative of $z = \cos(\ln x)$ with respect to x

$$\frac{du}{dx} = -\frac{\sin(\ln x)}{x}$$

so

$$(\cos(\ln x))' = -\frac{\sin(\ln x)}{x}$$

Which we can substitute into our workings for the $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx} &= \cos(\cos(\ln x))(\cos(\ln x))' && \text{(from before)} \\ &= \cos(\cos(\ln x)) \times -\frac{\sin(\ln x)}{x} \\ &= -\frac{\sin(\ln x) \cos(\cos(\ln x))}{x} \end{aligned}$$

Which gives us our funny looking answer. Note that if we have n function compositions, we will need n chain rules. $\square$

Note that in the above, we worked through finding

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

But since we did not know how to work with the final term, we then did the chain rule again to find

$$\frac{du}{dx} = \frac{du}{dv} \frac{dv}{dx}$$

Substituting this in gives

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx}$$

Which then gives us a hint about how the chain rule generalises. If we were to pretend that these were genuine fractions, the right-hand side would cancel to give the left-hand side. This is true in general, that we can keep inserting additional multiples of derivatives as long as they “cancel” if you consider them as actual fractions.

Therefore, we can use the chain rule to compute the derivative of any composition of functions that we know how to differentiate.

What this means in practice is that we can use a combination of the chain rule and the product rule to differentiate any composition and multiple of the following functions:

ax , x^n , $\sin x$, $\cos x$, e^x , $\ln x$.

3.4.1 Generalising Standard Derivatives

We can now use our knowledge of standard derivatives to get generalised results almost for free.

In this section, we will use that $f(x)$ is a function of x .

Example 29. Using the chain rule, differentiate $(f(x))^n$ with respect to x . Give your answer in terms of $f(x)$ and $f'(x)$, where n is a real number.

Solution. The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let's write $y = (f(x))^n$, $u = f(x)$ so that $y = u^n$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned} \frac{dy}{du} &= nu^{n-1} && \text{(using the derivative of } u^n) \\ \frac{du}{dx} &= f'(x) && \text{(in terms of } f'(x)) \end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= nu^{n-1} \times f'(x) && \text{(substituting our workings)} \\ &= nf(x)^{n-1} f'(x) && \text{(using } u = f(x)) \end{aligned}$$

Which gives us our answer. □

Example 30. Using the chain rule, differentiate $e^{f(x)}$ with respect to x . Give your answer in terms of $f(x)$ and $f'(x)$.

Solution. The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let's write $y = e^{f(x)}$, $u = f(x)$ so that $y = e^u$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned}\frac{dy}{du} &= e^u && \text{(using the derivative of } e^u\text{)} \\ \frac{du}{dx} &= f'(x) && \text{(in terms of } f'(x)\text{)}\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= e^u \times f'(x) && \text{(substituting our workings)} \\ &= f'(x)e^{f(x)} && \text{(using } u = f(x)\text{)}\end{aligned}$$

Which gives us our answer. □

Example 31. Using the chain rule, differentiate $\ln(f(x))$ with respect to x . Give your answer in terms of $f(x)$ and $f'(x)$.

Solution. The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let's write $y = \ln(f(x))$, $u = f(x)$ so that $y = \ln u$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned}\frac{dy}{du} &= \frac{1}{u} && \text{(using the derivative of } \ln u\text{)} \\ \frac{du}{dx} &= f'(x) && \text{(in terms of } f'(x)\text{)}\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{u} \times f'(x) && \text{(substituting our workings)} \\ &= \frac{f'(x)}{f(x)} && \text{(using } u = f(x)\text{)}\end{aligned}$$

Which gives us our answer. □

Example 32. Using the chain rule, differentiate $\sin(f(x))$ with respect to x . Give your answer in terms of $f(x)$ and $f'(x)$.

Solution. The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let's write $y = \sin(f(x))$, $u = f(x)$ so that $y = \sin u$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned}\frac{dy}{du} &= \cos u && \text{(using the derivative of } \sin u\text{)} \\ \frac{du}{dx} &= f'(x) && \text{(in terms of } f'(x)\text{)}\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= \cos u \times f'(x) && \text{(substituting our workings)} \\ &= f'(x) \cos(f(x)) && \text{(using } u = f(x)\text{)}\end{aligned}$$

Which gives us our answer. $\square$

Example 33. Using the chain rule, differentiate $\cos(f(x))$ with respect to x . Give your answer in terms of $f(x)$ and $f'(x)$.

Solution. The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let's write $y = \cos(f(x))$, $u = f(x)$ so that $y = \cos u$.

We need the derivative of y with respect to u and the derivative of u with respect to x .

$$\begin{aligned}\frac{dy}{du} &= -\sin u && \text{(using the derivative of } \cos u\text{)} \\ \frac{du}{dx} &= f'(x) && \text{(in terms of } f'(x)\text{)}\end{aligned}$$

Therefore, using the chain rule, we have that

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= -\sin u \times f'(x) && \text{(substituting our workings)} \\ &= -f'(x) \sin(f(x)) && \text{(using } u = f(x)\text{)}\end{aligned}$$

Which gives us our answer. $\square$

3.5 Mixed Exercises

Example 34. Differentiate $\sin(x) \cos(\sin(x))$

Solution. Let's first define two functions $f(x) = \sin(x)$ and $g(x) = \cos(\sin(x))$ so that the thing we are trying to differentiate becomes

$$\sin(x) \cos(\sin(x)) = f(x) \cdot g(x)$$

Then we can use the product rule:

$$(f(x) \cdot g(x))' = f'(x)g(x) + f(x)g'(x)$$

Now we can find $f'(x)$, since this is just the derivative of $f(x) = \sin(x)$, which is $\cos(x)$, but what about the derivative of $g(x) = \cos(\sin(x))$?

For this we need the chain rule.

How can we define u so that $g(x)$ becomes something we know how to differentiate? If we set $u = \sin(x)$, and write $y = g(x)$, then $y = \cos u$. The chain rule then gives us that

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Since $u = \sin(x)$, we find that $\frac{du}{dx} = \cos x$, and since $y = \cos u$, we find that $\frac{dy}{du} = -\sin u$.

Substituting this in gives

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= -\sin u \cos x \\ &= -\sin(\sin(x)) \cos x \quad (\text{using } u = \sin(x)) \end{aligned}$$

But this was all to find what $g'(x)$, i.e. the derivative of $\cos(\sin(x))$, which we have found to be

$$-\sin(\sin(x)) \cos x$$

Now we can substitute this into our product rule, recalling that $f(x) = \sin(x)$ and $g(x) = \cos(\sin(x))$, and $f'(x) = \cos x$ to find:

$$\begin{aligned} (f(x) \cdot g(x))' &= f'(x)g(x) + f(x)g'(x) \\ &= \cos x g(x) + f(x)g'(x) && (\text{substituting } f'(x)) \\ &= \cos x \cos(\sin(x)) + f(x)g'(x) && (\text{substituting } g(x)) \\ &= \cos x \cos(\sin(x)) + \sin(x)g'(x) && (\text{substituting } f(x)) \\ &= \cos x \cos(\sin(x)) + \sin(x) \cdot (-\sin(\sin(x)) \cos x) && (\text{substituting } g'(x)) \\ &= \cos x \cos(\sin(x)) - \sin(\sin(x)) \cos x \sin x \end{aligned}$$

Therefore, we have found the derivative

$$(\sin(x) \cos(\sin(x)))' = \cos x \cos(\sin(x)) - \sin(\sin(x)) \cos x \sin x$$

□

Example 35. Consider $e^{2 \sin x} + \cos(x^2)$. First find the derivative, and then find the derivative of the derivative.

Solution. Since we know that you can differentiate each additive term separately and then add them together, we can consider them one by one.

Let $f(x) = e^{2 \sin x}$ and $g(x) = \cos(x^2)$ so that we are trying to differentiate

$$f(x) + g(x) = e^{2 \sin x} + \cos(x^2)$$

Let's start with $f(x)$.

The question we ask ourselves is: if we could replace any part of the function with u , can we turn it into one of our standard functions?

Let $y = e^{2 \sin x}$, $u = 2 \sin x$ so that $y = e^u$.

Then we can compute our derivatives:

$$\begin{aligned} \frac{du}{dx} &= 2 \cos x \\ \frac{dy}{du} &= e^u \end{aligned}$$

So that, by the chain rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= e^u \cdot 2 \cos x && \text{(substituting)} \\ &= e^{2 \sin x} \cdot 2 \cos x && \text{(using } u = 2 \sin x \text{)}\end{aligned}$$

Therefore

$$f'(x) = 2 \cos x e^{2 \sin x}$$

For $g(x)$:

Let $y = \cos(x^2)$, $u = x^2$ so that $y = \cos u$.

Then we can compute our derivatives:

$$\begin{aligned}\frac{du}{dx} &= 2x \\ \frac{dy}{du} &= -\sin u\end{aligned}$$

So that, by the chain rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} \\ &= -\sin u \cdot 2x && \text{(substituting)} \\ &= -\sin(x^2) \cdot 2x && \text{(using } u = x^2 \text{)}\end{aligned}$$

Therefore

$$g'(x) = -2x \sin(x^2)$$

Which gives us our derivative:

$$(e^{2 \sin x} + \cos(x^2))' = f'(x) + g'(x) = 2 \cos x e^{2 \sin x} - 2x \sin(x^2)$$

This completes the first part of the question. We are then asked to differentiate this function again.

Let's again split it into two cases based on the additive terms:

$$2 \cos x e^{2 \sin x}$$

and

$$-2x \sin(x^2)$$

Now for the purposes of this solution, we will only walk through the second term. The first is left as an exercise.

Since $-2x \sin(x^2)$ is a multiple of two terms, we need to use the product rule. Let $u = -2x$ and $v = \sin(x^2)$. Then $u' = -2$, and v' must be calculated via the chain rule.

Let's use the chain rule with $y = \sin(x^2)$, $w = x^2$ so that $y = \sin w$. We need the derivatives:

$$\begin{aligned}\frac{dw}{dx} &= 2x \\ \frac{dy}{dw} &= \cos w\end{aligned}$$

Then the chain rule tells us that

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{dw} \frac{dw}{dx} \\ &= \cos w \cdot 2x && \text{(substituting)} \\ &= \cos(x^2) \cdot 2x\end{aligned}$$

Therefore $v' = 2x \cos(x^2)$.

Then the product rule tells us that

$$\begin{aligned}(uv)' &= uv' + u'v \\ &= -2x \cdot 2x \cos(x^2) - 2 \cdot \sin(x^2)\end{aligned}$$

And therefore, we have found that

$$(-2x \sin(x^2))' = -4x^2 \cos(x^2) - 2 \sin(x^2)$$

A similar process can be used to find

$$(2 \cos x e^{2 \sin x})' = 4 \cos^2 x e^{2 \sin x} - 2 \sin x e^{2 \sin x}$$

Therefore, the derivative we are looking for is:

$$\begin{aligned}(2 \cos x e^{2 \sin x} - 2x \sin(x^2))' &= 4 \cos^2 x e^{2 \sin x} - 2 \sin x e^{2 \sin x} \\ &\quad - 4x^2 \cos(x^2) - 2 \sin(x^2)\end{aligned}$$

Which finishes the question, finding the second derivative of $e^{2 \sin x} + \cos(x^2)$. □

4 Integration

4.1 Definite vs Indefinite Integration

An *indefinite integral* is an integral in which you do not evaluate at two points, and can be defined as the inverse of the derivative, up to a constant. That is,

$$\int f' dx = f + c$$

Where c is a constant. Where does the constant come from? We know that taking the derivative of a constant is always 0 (see Section 3.1), so we can calculate:

$$(f + c)' = f' + c' = f' + 0 = f'$$

Which regains f' .

Example 36. Find the indefinite integral of $\cos(x)$.

Solution. We know that if A is a constant, then $(A \sin(x))' = A(\sin(x))' = A \cos(x)$. Therefore, if we set A to be 1, we have found that $(\sin(x))' = \cos(x)$. Therefore,

$$\int \cos(x) dx = \sin(x) + c$$

Where c is a constant. □

Example 37. Find the indefinite integral of $\sin(x)$.

Solution. We know that if A is a constant, then $(A \cos(x))' = A(\cos(x))' = -A \sin(x)$. Therefore, if we set $-A = 1$, we find $A = -1$ (since we want to find the indefinite integral of $\sin(x)$) gives that $(-\cos(x))' = \sin(x)$. Therefore,

$$\int \sin(x) dx = -\cos(x) + c$$

Where c is a constant. □

Example 38. Find the indefinite integral of x^8 .

Solution. We know that if A is a constant, then $(Ax^9)' = A(x^9)' = 9Ax^8$. Therefore, if we set $9A = 1$, we find $A = \frac{1}{9}$ (since we want to find the indefinite integral of x^8) gives that $(\frac{1}{9}x^9)' = x^8$. Therefore,

$$\int x^8 dx = \frac{1}{9}x^9 + c$$

Where c is a constant. □

A *definite integral* is one where we evaluate at two points, and can be defined as the area under a function f between those two points³. In practice, we compute this by the following:

$$\int_a^b f dx = \left[\int f dx \right]_{x=a}^{x=b}$$

Where the notation on the RHS means that we calculate the indefinite integral at b and subtract the indefinite integral at a . Recall that the indefinite integral always has a constant added c . If we take two evaluations of the indefinite integral, the c will cancel. For example, let's use the example of when we are integrating a derivative.

³In fact, it is a signed area since if a function is negative, the integral is also negative.

$$\begin{aligned}
\int_a^b f' dx &= [f + c]_{x=a}^{x=b} && \text{(substitute indefinite integral of } f') \\
&= [f(b) + c - (f(a) + c)] && \text{(expand)} \\
&= f(b) - f(a) && \text{(cancel } c)
\end{aligned}$$

Example 39. Calculate the following integral:

$$\int_0^\pi \cos(x) dx$$

Solution. From Example 36, we know that $\int \cos(x) dx = \sin(x) + c$ where c is a constant. Therefore (noting c would cancel),

$$\begin{aligned}
\int_0^\pi \cos(x) dx &= [\sin(x)]_{x=0}^{x=\pi} \\
&= [\sin(\pi) - \sin(0)] \\
&= 0 - 0 \\
&= 0
\end{aligned}$$

□

Example 40. Calculate the following integral:

$$\int_0^\pi \sin(x) dx$$

Solution. From Example 37, we know that $\int \sin(x) dx = -\cos(x) + c$ where c is a constant. Therefore (noting c would cancel),

$$\begin{aligned}
\int_0^\pi \sin(x) dx &= [-\cos(x)]_{x=0}^{x=\pi} \\
&= [-\cos(\pi) - (-\cos(0))] \\
&= [-(-1) - (-1)] && (\cos(0) = 1, \cos(\pi) = -1) \\
&= 2
\end{aligned}$$

□

Example 41. Calculate the following integral:

$$\int_0^1 x^8 dx$$

Solution. From Example 38, we know that $\int x^8 dx = \frac{1}{9}x^9 + c$ where c is a constant. Therefore (noting c would cancel),

$$\begin{aligned}
\int_0^1 x^8 dx &= [\frac{1}{9}x^9]_{x=0}^{x=1} \\
&= [\frac{1}{9}1^9 - \frac{1}{9}0^9] \\
&= \frac{1}{9}
\end{aligned}$$

□

In summary, indefinite integral you don't evaluate at two points, and definite integral you do evaluate at two points.

You might ask why not just use the trapezium rule (Section 12.1) if we only care about the area? The trapezium rule gives an estimate, whereas it is useful to know the precise, exact answer. However, in the “real world” the estimate is almost always enough.

4.1.1 Additional Exercises

Example 42. Calculate the indefinite integral $\int \sin^2 x dx$.

Solution. The trick to higher powers of sin or cos is to use double angle formulae to reduce it to lower powers.

Recall

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x && \text{(double angle formula for cos)} \\ &= 1 - \sin^2 x - \sin^2 x && \text{(using } \sin^2 x + \cos^2 x = 1) \\ &= 1 - 2\sin^2 x && \text{(collecting terms)}\end{aligned}$$

Therefore, rearranging gives us

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

Then we can integrate

$$\begin{aligned}\int \sin^2 x dx &= \int \frac{1}{2}(1 - \cos 2x) dx && \text{(substituting the power reduction formula)} \\ &= \frac{1}{2} \int 1 dx - \frac{1}{2} \int \cos 2x dx && \text{(expanding and factorising)} \\ &= \frac{1}{2} x - \frac{1}{4} \sin 2x + c && \text{(integrating)}\end{aligned}$$

Where we deduced the integral of $\cos 2x$ by differentiating $\sin 2x$, and seeing that we get $2 \cos 2x$. Therefore we must divide by 2 to cancel the factor of 2 in front:

$$\int \cos 2x dx = \frac{1}{2} \sin 2x + c$$

Alternatively, you can calculate this via substitution $y = 2x$, which is covered in Section 4.3. $\square$

4.2 Integration by Parts

It is useful to know how to integrate two functions multiplied together.

To do so, let's look at two functions $f(x)$ and $g(x)$.

Consider the derivative of $f(x)g(x)$

$$\frac{d(f \cdot g)}{dx} = \frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

Now recall that the integral of a derivative is the function itself:

$$\begin{aligned} f \cdot g &= \int \frac{d(f \cdot g)}{dx} dx = \int \left(\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx} \right) dx \\ &= \int \frac{df}{dx} \cdot g dx + \int f \cdot \frac{dg}{dx} dx \end{aligned}$$

Then we can rearrange this formula to get the integration by parts formula.

$$\int f \cdot \frac{dg}{dx} dx = f \cdot g - \int \frac{df}{dx} \cdot g dx$$

In textbooks, this is often written with u, v instead, and using shorthand u', v' for their derivatives:

$$\int uv' dx = uv - \int u'v dx$$

Note that if we are integrating between $x = a_0$ and $x = a_1$, this will become:

$$\int_{a_0}^{a_1} uv' dx = [uv]_{x=a_0}^{x=a_1} - \int_{a_0}^{a_1} u'v dx$$

Which is equivalent to

$$\int_{a_0}^{a_1} uv' dx = u(a_1)v(a_1) - u(a_0)v(a_0) - \int_{a_0}^{a_1} u'v dx$$

The reason this is useful is because often u and v' can be chosen so that the integral on the right is simpler!

A handy mnemonic is LATE⁴: Logarithm (or $\ln$), Algebra (powers of x), Trig, Exponential – this is usually a good guess as your choice of u in order from left to right. This is mostly because these functions become simpler after differentiating – the u' on the right hand side will be easier to integrate.

You should only use integration by parts if you cannot integrate it another way – if you can see that your function is easy to integrate without this trick you should not over-complicate it.

Example 43. Find the indefinite integral of $x \cos(x)$. Then calculate

$$\int_0^\pi x \cos(x) dx$$

Solution. First we check if this looks like the derivative of a function we know. Since it doesn't, we continue with integration by parts.

Following LATE, we see that x is algebra and comes before $\cos(x)$ which is trig. Therefore choose $u = x$ and $v' = \cos(x)$. Now we know $\sin(x)$ differentiates to become $\cos(x)$ therefore $v = \sin(x)$. Also, $u' = 1$. I like to write this in a box:

$$\begin{array}{ll} u = x & v = \sin(x) \\ u' = 1 & v' = \cos(x) \end{array}$$

⁴Some use ILATE instead, where I is inverse trig. If you have not encountered differentiating inverse trig functions, ignore this.

Now we remind ourselves of integration by parts:

$$\int uv' dx = uv - \int u'v dx$$

Substituting, we get

$$\int x \cos x dx = x \sin(x) - \int 1 \cdot \sin(x) dx$$

Now we know $\cos(x)$ differentiates to become $-\sin(x)$. Therefore $\int \sin(x) dx = -\cos(x) + c'$, where c' is a constant. Therefore:

$$\begin{aligned} \int x \cos x dx &= x \sin(x) - \int \sin(x) dx \\ &= x \sin(x) - (-\cos(x) + c') \\ &= x \sin(x) + \cos(x) + c \end{aligned} \quad (\text{letting } c = -c')$$

This is the indefinite integral. To calculate the definite integral in the second part of the question, we need to include the bounds

$$\begin{aligned} \int_0^\pi x \cos x dx &= [x \sin(x)]_{x=0}^{x=\pi} - \int_0^\pi \sin(x) dx \\ &= \pi \cdot \sin(\pi) - 0 \cdot \sin(0) - [-\cos(x)]_{x=0}^{x=\pi} \\ &= 0 - 0 - [-\cos(\pi) + \cos(0)] \quad (\sin(\pi) = 0) \\ &= -[1 + 1] \\ &= -2 \end{aligned}$$

Which finishes the question. □

Example 44. Find the indefinite integral of $x \ln x$. Then calculate

$$\int_1^2 x \ln(x) dx$$

Solution. First we check if this looks like the derivative of a function we know. Since it doesn't, we continue with integration by parts.

Following LATE, we see that $\ln(x)$ is logarithm and comes before x which is algebra. Therefore choose $u = \ln(x)$ and $v' = x$. Now we know x^2 differentiates to become $2x$ therefore $v = \frac{1}{2}x^2$. Also, $u' = \frac{1}{x}$. I like to write this in a box:

$$\begin{array}{ll} u = \ln(x) & v = \frac{1}{2}x^2 \\ u' = \frac{1}{x} & v' = x \end{array}$$

Now we remind ourselves of integration by parts:

$$\int uv' dx = uv - \int u'v dx$$

Substituting, we get

$$\int x \ln x dx = \frac{1}{2}x^2 \ln(x) - \int \frac{1}{x} \cdot \frac{1}{2}x^2 dx$$

Therefore:

$$\begin{aligned}\int x \ln x dx &= \frac{1}{2}x^2 \ln(x) - \frac{1}{2} \int x dx \\ &= \frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 + c \\ &= \frac{1}{2}x^2 \left(\ln(x) - \frac{1}{2} \right) + c\end{aligned}$$

Where c is a constant. This is the indefinite integral. To calculate the definite integral in the second part of the question, we need to include the bounds

$$\begin{aligned}\int_1^2 x \ln x dx &= \left[\frac{1}{2}x^2 \ln(x) \right]_{x=1}^{x=2} - \frac{1}{2} \int_1^2 x dx \\ &= \dots\end{aligned}$$

Which we could continue like before. Alternatively, we can use the indefinite integral and calculate:

$$\begin{aligned}\int_1^2 x \ln x dx &= \left[\frac{1}{2}x^2 \left(\ln(x) - \frac{1}{2} \right) + c \right]_{x=1}^{x=2} \\ &= \frac{1}{2}2^2 \left(\ln(2) - \frac{1}{2} \right) - \frac{1}{2}1^2 \left(\ln(1) - \frac{1}{2} \right) \\ &= 2 \left(\ln(2) - \frac{1}{2} \right) + \frac{1}{4} \\ &= \ln(4) - \frac{3}{4}\end{aligned}$$

Which finishes the question. □

A quirky application of this is how to integrate $\ln(x)$.

Exercise 6. *Try it yourself!* By writing $\ln(x)$ as $1 \cdot \ln(x)$, find the indefinite integral of $\ln(x)$.

Solution. First we check if this looks like the derivative of a function we know. Since it doesn't, we continue with integration by parts.

Following LATE, we see that $\ln(x)$ is logarithm and comes before 1 which we consider as algebra here. We can construct our square as follows

$$\begin{array}{ll}u = \ln(x) & v = x \\ u' = \frac{1}{x} & v' = 1\end{array}$$

Now we remind ourselves of integration by parts:

$$\int uv' dx = uv - \int u'v dx$$

Substituting, we get

$$\begin{aligned}\int \ln(x) dx &= x \ln(x) - \int \frac{x}{x} dx \\ &= x \ln(x) - x\end{aligned}$$

As a bonus exercise, verify that this differentiates to become $\ln(x)$. □

4.3 Integration by Substitution

It is possible to change what variable is being integrated by. This is useful in two ways:

1. To convert an integral to one you know.
2. To simplify an integral.

How does it work?

Let's consider the following integral.

$$\int_a^b f(x)dx$$

For clarity, we will clarify that the a and b are integrating between $x = a$ and $x = b$ in the following way.

$$\int_{x=a}^{x=b} f(x)dx$$

Now what happens if we have a new variable in terms of x ? Let's write it as $y = g(x)$. We also require that we can write x as a function of y , i.e. that g^{-1} exists – see Section 1.1.3. Then $x = g^{-1}(y)$.

Now let's abuse notation⁵ a lot:

$$\begin{aligned}\frac{dx}{dy} &= \frac{dx}{dy} \\ dx &= \frac{dx}{dy} dy && \text{('multiply' by } dy \text{ (abuse of notation))} \\ dx &= \frac{dg^{-1}}{dy} dy && \text{(using } x = g^{-1}(y)\text{)}\end{aligned}$$

Now we have all the tools to replace all instances of x in our integral. We will do this one-by-one from left to right for clarity.

$$\begin{aligned}\int_{x=a}^{x=b} f(x)dx &= \int_{g^{-1}(y)=a}^{g^{-1}(y)=b} f(x)dx && \text{(using } x = g^{-1}(y)\text{)} \\ &= \int_{y=g(a)}^{y=g(b)} f(x)dx && \text{(rearranging the bounds)} \\ &= \int_{y=g(a)}^{y=g(b)} f(g^{-1}(y))dx && \text{(using } x = g^{-1}(y)\text{)} \\ &= \int_{y=g(a)}^{y=g(b)} f(g^{-1}(y)) \frac{dg^{-1}}{dy} dy && \text{(using } dx = \frac{dg^{-1}}{dy} dy\text{)}\end{aligned}$$

Since the final line is *completely* in terms of y , we can continue by computing the integral as usual, but in the y variable instead. Note that the final formula uses both g and g^{-1} , meaning that we need g^{-1} to exist for the above formula.

⁵This can be made rigorous by looking at differentials.

Summary: **exchange every instance of x with the equivalent y , and then integrate in the y variable.**

We have only dealt with definite integrals here, but for indefinite you do the same procedure!

Note that you could keep doing substitution if needed.

This has been very abstract, so let's see some examples.

Example 45. Compute the following integral.

$$\int_0^1 (x+1)^3 dx$$

Then state the indefinite integral of $(x+1)^3$.

Solution. Firstly, note that you *could* expand the brackets and then integrate each term. Instead, let's utilise substitution.

Define $x+1 = y$. We now want to rewrite everything in terms of y . Rearranging gives $x = y-1$, and taking the derivative of both sides gives $\frac{dy}{dx} = 1$, where we abuse notation to 'multiply by dx ' and write $dy = dx$. Let's collect our three equations:

$$\begin{aligned}y &= x+1 \\x &= y-1 \\dx &= dy\end{aligned}$$

Now let's substitute everything until it is in terms of y only. For clarity, we will do this from left to right.

$$\begin{aligned}\int_0^1 (x+1)^3 dx &= \int_{x=0}^{x=1} (x+1)^3 dx \\&= \int_{y-1=0}^{y-1=1} (x+1)^3 dx && \text{(using } x = y-1 \text{)} \\&= \int_{y=1}^{y=2} (x+1)^3 dx && \text{(rearranging integral bounds)} \\&= \int_{y=1}^{y=2} y^3 dx && \text{(using } y = x+1 \text{)} \\&= \int_{y=1}^{y=2} y^3 dy && \text{(using } dx = dy \text{)}\end{aligned}$$

Now we can integrate this in y as usual.

$$\begin{aligned}\int_{y=1}^{y=2} y^3 dy &= \left[\frac{1}{4} y^4 \right]_{y=1}^{y=2} \\&= \frac{1}{4} (2^4 - 1^4) \\&= \frac{15}{4}\end{aligned}$$

The indefinite integral can be written as just *the-bit-in-the-square-brackets-plus-c*:

$$\int (x+1)^3 dx = \frac{1}{4}y^4 + c$$

And now all we need to do is substitute back in what y was:

$$\int (x+1)^3 dx = \frac{1}{4}(x+1)^4 + c$$

□

Example 46. Compute the following integral.

$$\int_0^1 \cos(\pi x + \pi) dx$$

Then state the indefinite integral of $\cos(\pi x + \pi)$.

Solution. Notice that we know what the integral of $\cos(x)$ is, therefore it would be nice to make a substitution that gives us something we know.

Define $\pi x + \pi = y$. We now want to rewrite everything in terms of y . Rearranging gives $x = \frac{1}{\pi}(y - \pi)$, and taking the derivative of both sides gives $\frac{dy}{dx} = \pi$, where we abuse notation to ‘multiply by dx ’ and write $dy = \pi dx$. Further rearranging gives

$$dx = \frac{1}{\pi} dy$$

Let’s collect our three equations:

$$\begin{aligned} y &= \pi x + \pi \\ x &= \frac{1}{\pi}(y - \pi) \\ dx &= \frac{1}{\pi} dy \end{aligned}$$

Now let’s substitute everything until it is in terms of y only. For clarity, we will do this from left to right.

$$\begin{aligned} \int_0^1 \cos(\pi x + \pi) dx &= \int_{x=0}^{x=1} \cos(\pi x + \pi) dx \\ &= \int_{x=0}^{x=1} \cos(\pi x + \pi) dx \\ &= \int_{\frac{1}{\pi}(y-\pi)=0}^{\frac{1}{\pi}(y-\pi)=1} \cos(\pi x + \pi) dx && \text{(using } x = \frac{1}{\pi}(y - \pi) \text{)} \\ &= \int_{y=\pi}^{y=2\pi} \cos(\pi x + \pi) dx && \text{(rearranging integral bounds)} \\ &= \int_{y=\pi}^{y=2\pi} \cos(y) dx && \text{(using } y = \pi x + \pi \text{)} \\ &= \int_{y=\pi}^{y=2\pi} \cos(y) \frac{1}{\pi} dy && \text{(using } dx = \frac{1}{\pi} dy \text{)} \\ &= \frac{1}{\pi} \int_{y=\pi}^{y=2\pi} \cos(y) dy && \text{(rearranging)} \end{aligned}$$

Now we can integrate this in y as usual.

$$\begin{aligned}\frac{1}{\pi} \int_{y=\pi}^{y=2\pi} \cos(y) dy &= \frac{1}{\pi} [\sin(y)]_{y=\pi}^{y=2\pi} \\ &= \frac{1}{\pi} (\sin(2\pi) - \sin(\pi)) \\ &= \frac{1}{\pi} (0 - 0) \\ &= 0\end{aligned}$$

The indefinite integral can be written as just *the-bit-in-the-square-brackets-plus-c*, remembering the $\frac{1}{\pi}$ we pulled out:

$$\int \cos(\pi x + \pi) dx = \frac{1}{\pi} \sin(y) + c$$

And now all we need to do is substitute back in what y was:

$$\int \cos(\pi x + \pi) dx = \frac{1}{\pi} \sin(\pi x + \pi) + c$$

If the dividing by π confuses you, go through the steps above but without the integral bounds! $\square$

Example 47. Compute the following integral.

$$\int_0^1 \frac{1}{2x+1} dx$$

Then state the indefinite integral of $\frac{1}{2x+1}$.

Solution. Notice that we know what the integral of $\frac{1}{x}$ is, therefore it would be nice to make a substitution that gives us something closer to what we know.

Define $2x+1 = y$. We now want to rewrite everything in terms of y . Rearranging gives $x = \frac{1}{2}(y-1)$, and taking the derivative of both sides using the chain rule gives $\frac{dy}{dx} = 2$, where we abuse notation to ‘multiply by dx ’ and write $dy = 2dx$. Further rearranging gives

$$dx = \frac{1}{2} dy$$

Let’s collect our three equations:

$$\begin{aligned}y &= 2x + 1 \\ x &= \frac{1}{2}(y - 1) \\ dx &= \frac{1}{2} dy\end{aligned}$$

Now let’s substitute everything until it is in terms of y only. For clarity, we will do this from left to right.

$$\begin{aligned}
\int_0^1 \frac{1}{2x+1} dx &= \int_{x=0}^{x=1} \frac{1}{2x+1} dx \\
&= \int_{\frac{1}{2}(y-1)=0}^{\frac{1}{2}(y-1)=1} \frac{1}{2x+1} dx && \text{(using } x = \tfrac{1}{2}(y-1)\text{)} \\
&= \int_{y=1}^{y=3} \frac{1}{2x+1} dx && \text{(rearranging integral bounds)} \\
&= \int_{y=1}^{y=3} \frac{1}{y} dx && \text{(using } y = 2x+1\text{)} \\
&= \int_{y=1}^{y=3} \frac{1}{y} \frac{1}{2} dy && \text{(using } dx = \tfrac{1}{2} dy\text{)} \\
&= \frac{1}{2} \int_{y=1}^{y=3} \frac{1}{y} dy && \text{(rearranging)}
\end{aligned}$$

Now we can integrate this in y as usual.

$$\begin{aligned}
\frac{1}{2} \int_{y=1}^{y=3} \frac{1}{y} dy &= \frac{1}{2} [\ln(y)]_{y=1}^{y=3} \\
&= \frac{1}{2} (\ln(3) - \ln(1)) \\
&= \ln(3)
\end{aligned}$$

The indefinite integral can be written as just *the-bit-in-the-square-brackets-plus-c*, remembering the $\frac{1}{2}$ we pulled out:

$$\int \frac{1}{2x+1} dx = \frac{1}{2} \ln(y) + c$$

And now all we need to do is substitute back in what y was:

$$\int \frac{1}{2x+1} dx = \frac{1}{2} \ln(2x+1) + c$$

If the dividing by 2 confuses you, go through the steps above but without the integral bounds! $\square$

Example 48. Compute the following integral.

$$\int_0^{\pi/4} \tan(x) dx$$

Then state the indefinite integral of $\tan(x)$.

Solution. Firstly, recall that

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

And here we can make a smart substitution by noticing that $\cos(x)$ differentiates into $-\sin(x)$, therefore when we replace dx , there might be some canceling.

Define $\cos(x) = y$. We now want to rewrite everything in terms of y . Rearranging gives $x = \arccos(y)$, and taking the derivative of both sides gives $\frac{dy}{dx} = -\sin(x)$, where we abuse notation to ‘multiply by dx ’ and write $dy = -\sin(x)dx$. Further rearranging gives

$$dx = -\frac{1}{\sin(x)} dy$$

Let's collect our three equations:

$$\begin{aligned} y &= \cos(x) \\ x &= \arccos(y) \\ dx &= -\frac{1}{\sin(x)} dy \end{aligned}$$

Now let's substitute everything until it is in terms of y only. For clarity, we will do this from left to right.

$$\begin{aligned} \int_0^{\pi/4} \tan(x) dx &= \int_{x=0}^{x=\pi/4} \tan(x) dx \\ &= \int_{x=0}^{x=\pi/4} \frac{\sin(x)}{\cos(x)} dx \\ &= \int_{\arccos(y)=0}^{\arccos(y)=\pi/4} \frac{\sin(x)}{\cos(x)} dx && \text{(using } x = \arccos(y)\text{)} \\ &= \int_{y=\cos(0)}^{y=\cos(\pi/4)} \frac{\sin(x)}{\cos(x)} dx && \text{(rearranging integral bounds)} \\ &= \int_{y=1}^{y=\cos(\pi/4)} \frac{\sin(x)}{\cos(x)} dx && \text{(simplifying integral bounds)} \\ &= - \int_{y=\cos(\pi/4)}^{y=1} \frac{\sin(x)}{\cos(x)} dx && \text{(swapping integral bounds)} \\ &= - \int_{y=\cos(\pi/4)}^{y=1} \frac{\sin(x)}{y} dx && \text{(using } y = \cos(x)\text{)} \\ &= - \int_{y=\cos(\pi/4)}^{y=1} \frac{\sin(x)}{y} \left(-\frac{1}{\sin(x)}\right) dy && \text{(using } dx = -\frac{1}{\sin(x)} dy\text{)} \\ &= \int_{y=\cos(\pi/4)}^{y=1} \frac{1}{y} dy && \text{(simplifying)} \end{aligned}$$

Now we can integrate this in y as usual.

$$\begin{aligned} \int_{y=\cos(\pi/4)}^{y=1} \frac{1}{y} dy &= [\ln(y)]_{y=\cos(\pi/4)}^{y=1} \\ &= (\ln(1) - \ln(\cos(\pi/4))) \\ &= -\ln(\cos(\pi/4)) \end{aligned}$$

The indefinite integral can be written as just *the-bit-in-the-square-brackets-plus-c*, but since we swapped the integral bounds, we need to include a minus sign:

$$\int \tan(x) dx = -\ln(y) + c$$

And now all we need to do is substitute back in what y was:

$$\int \tan(x)dx = -\ln(\cos(x)) + c$$

If the minus sign confuses you, go through the steps above but without the integral bounds! $\square$

4.3.1 Trigonometric Substitutions

One thing that appears frequently is the use of less direct substitutions, where the substitution you require does not directly appear in the integral itself as a change of variable.

A good rule of thumb for when you might need these is if you can spot that the integral looks like an identity that we know if we were to swap out the variable for a different function.

For example, we know that $1 - \sin^2 \theta = \cos^2 \theta$, and $1 - \cos^2 \theta = \sin^2 \theta$. Therefore, if we spot a term that looks like $1 - x^2$ in an integral, we might think to use the substitution $x = \sin \theta$ or $x = \cos \theta$.

This is because after the substitution this term would simplify:

$$1 - x^2 = 1 - \sin^2 \theta = \cos^2 \theta$$

This is important because this often makes a complicated integral much simpler.

We can get a suite of identities from starting with $\sin^2 \theta + \cos^2 \theta = 1$:

$\sin^2 \theta + \cos^2 \theta = 1$	(main identity)
$\sin^2 \theta = 1 - \cos^2 \theta$	(rearranging main identity)
$\cos^2 \theta = 1 - \sin^2 \theta$	(rearranging main identity)
$\tan^2 \theta + 1 = \sec^2 \theta$	(dividing main identity by $\cos^2 \theta$)
$\sec^2 \theta - 1 = \tan^2 \theta$	(rearranging previous identity)
$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$	(dividing main identity by $\sin^2 \theta$)
$\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta$	(rearranging previous identity)

This means that we can always transform a term that looks like $1 \pm x^2$ or $x^2 \pm 1$ into a single trig term via any of the above identities. Which one to use then comes down to how the *rest* of the integral transforms (especially the dx part)!

Now the other key part is that we can multiply any of the above equations by any number and they are still true!

For example, we have that $a \cos^2 \theta = a - a \sin^2 \theta$. So what if we have something of the form $a - bx^2$? Let's work backwards from what we want to use, and figure out what x^2 must be. We will highlight the non- x part red to show what we are trying to match with the identity.

$$a - bx^2 \quad (\text{the starting point})$$

What is a relevant trig identity?

$$a \cos^2 \theta = a - a \sin^2 \theta \quad (\text{what we want to use})$$

$$= a - bx^2 \quad (\text{what we want to replace})$$

$$\therefore bx^2 = a \sin^2 \theta \quad (\text{what we need to use})$$

$$\therefore x = \sqrt{\frac{a}{b}} \sin \theta \quad (\text{the substitution we get})$$

You could go and work out all examples of possible substitutions for each combination of $a \pm bx^2$, but:

Using the technique above to multiply the trig identities by a constant to get the non- x part in the correct form, and deduce what the x part needs to be to achieve this is very powerful. This will allow you to deduce any trig substitution in the form $a \pm bx^2$.

The above is related to the fact that the equation of a circle at the origin is $x^2 + y^2 = r^2$, and we can rearrange to find $y = \sqrt{r^2 - x^2}$, and trigonometry usually appears when we are dealing with circles.

The thing to be careful of is how the bounds transform in these cases; for example, the integral:

$$\int_0^2 1 - x^2 \, dx$$

The upper bound of the integral is 2, so what would happen if we try to use $x = \sin \theta$? Substituting the upper bound as we did in the previous section gives:

$$\begin{aligned} x &= 2 \\ \therefore \sin \theta &= 2 \end{aligned}$$

But now we have a problem! $\sin \theta$ can never be 2 since $\sin \theta$ can only ever be between -1 and 1 .

Example 49. Using a trigonometric substitution, find the indefinite integral

$$\int \sqrt{1 - x^2} \, dx$$

Then state what integral bounds this is valid for when using that substitution.

Solution. Firstly, it is possible to integrate this directly in a few ways, using a substitution of $1 - x^2$ being the most straightforward, or even a difference of two squares method. Since the question is asking for trig, we will do that instead. The identities that look like $1 - x^2$ are $1 - \sin^2 \theta = \cos^2 \theta$, and $1 - \cos^2 \theta = \sin^2 \theta$. This should suggest that we use $\sin$ or $\cos$

Let's substitute the following $x = \sin \theta$. Then calculate

$$\frac{dx}{d\theta} = \cos \theta$$

We can 'pretend-to-be-able-to-multiply-by- θ ' to find $dx = \cos \theta \, d\theta$.

Then continue substituting every instance of x with the appropriate thing in terms of θ from left to right.

$$\begin{aligned}
\int \sqrt{1-x^2} dx &= \int \sqrt{1-\sin^2 \theta} dx && \text{(using } x = \sin \theta) \\
&= \int \sqrt{1-\sin^2 \theta} \cos \theta d\theta && \text{(using } dx = \cos \theta d\theta) \\
&= \int \sqrt{\cos^2 \theta} \cos \theta d\theta && \text{(trig identity)} \\
&= \int \cos \theta \cos \theta d\theta \\
&= \int \cos^2 \theta d\theta
\end{aligned}$$

Note that here we used that $\sqrt{\cos^2 \theta} = \cos \theta$, which implicitly means that we need to restrict to where $\cos \theta$ is positive, i.e. $0 \leq \theta \leq \pi$

From here, we must remember how to integrate a power of $\cos$ and $\sin$. To do this, we (repeatedly if necessary) apply

$$\begin{aligned}
\cos(2\theta) &= \cos^2 \theta - \sin^2 \theta && \text{(trig identity)} \\
\therefore \cos(2\theta) &= \cos^2 \theta - (1 - \cos^2 \theta) && \text{(trig identity)} \\
\therefore \cos(2\theta) &= 2\cos^2 \theta - 1 && \text{(simplifying)} \\
\therefore \cos^2 \theta &= \frac{\cos(2\theta) + 1}{2} && \text{(rearranging)}
\end{aligned}$$

This allows us to integrate directly:

$$\begin{aligned}
\int \sqrt{1-x^2} dx &= \int \cos^2 \theta d\theta && \text{(where we left off)} \\
&= \int \frac{\cos(2\theta) + 1}{2} \theta d\theta && \text{(using above identity)} \\
&= \frac{1}{2} \int \cos(2\theta) d\theta + \frac{1}{2} \int 1 d\theta && \text{(separating the integral)} \\
&= \frac{1}{2} \left(\frac{1}{2} \sin(2\theta) \right) + \frac{1}{2} \theta + c && \text{(evaluating the integrals)}
\end{aligned}$$

In the last line, we have used that $\frac{1}{2} \sin(2\theta)$ differentiates to become $\cos(2\theta)$, which you can verify using the chain rule. In fact, because of the chain rule Section 3.4, we could have worked this out by knowing that the function which differentiates to give $\cos(2\theta)$ must be of the form $A \sin(2\theta)$, and we can figure out what A must be by differentiating it and setting it equal to what we want:

$$\begin{aligned}
\cos(2\theta) &= (A \sin(2\theta))' \\
&= 2A \cos(2\theta) && \text{(using the chain rule)} \\
\therefore 2A &= 1 && \text{(since } \cos(2\theta) = 2A \cos(2\theta)) \\
\therefore A &= \frac{1}{2}
\end{aligned}$$

You always have time to do little calculations like this if you are not sure!

If you dislike this method, you can always do another substitution of $2\theta = u$ and run through the integration by substitution method.

To return to the question, we have found the indefinite integral in terms of θ via a trig substitution. We want to replace any θ with the equivalent x terms. To do this, we need to remember that

$x = \sin \theta$. Therefore, $\theta = \sin^{-1} x$, whenever we can invert it⁶. We can continue to simplify by also remembering that

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

and that you can write $\cos \theta = \sqrt{1 - \sin^2 \theta}$ to rearrange it in terms of $\sin$ or θ :

$$\begin{aligned} \int \sqrt{1-x^2} dx &= \frac{1}{2} \left(\frac{1}{2} \sin(2\theta) \right) + \frac{1}{2} \theta + c \\ &= \frac{1}{2} \left(\frac{1}{2} 2 \sin \theta \cos \theta \right) + \frac{1}{2} \theta + c && \text{(using } \sin 2\theta \text{ identity)} \\ &= \frac{1}{2} (\sin \theta \sqrt{1 - \sin^2 \theta}) + \frac{1}{2} \theta + c && \text{(writing } \cos \text{ in terms of } \sin) \\ &= \frac{1}{2} (x \sqrt{1 - x^2}) + \frac{1}{2} \theta + c && \text{(using } x = \sin \theta) \\ &= \frac{1}{2} (x \sqrt{1 - x^2}) + \frac{1}{2} \sin^{-1} x + c && \text{(using } \theta = \sin^{-1} x) \end{aligned}$$

This almost finishes the question, giving us that, subject to certain restrictions on x .

$$\int \sqrt{1-x^2} dx = \frac{1}{2} (x \sqrt{1-x^2}) + \frac{1}{2} \sin^{-1} x + c$$

But what are the restrictions? We used $x = \sin \theta$, and therefore x cannot be more than 1 or less than -1 since $\sin$ is between -1 and 1 . This means that this indefinite integral is only valid between $-1 \leq x \leq 1$.

This completes the question.

□

This question also shows a major downside using some trigonometric substitutions – the bounds are heavily restricted! For example, we cannot use the indefinite integral above⁷ to find

$$\int_0^2 \sqrt{1-x^2} dx$$

Since the upper bound is outside of -1 and 1 , whereas we *can* compute

$$\begin{aligned} \int_0^1 \sqrt{1-x^2} dx &= \left[\frac{1}{2} (x \sqrt{1-x^2}) + \frac{1}{2} \sin^{-1} x \right]_{x=0}^{x=1} \\ &= \frac{1}{2} (1 \sqrt{1-1^2}) + \frac{1}{2} \sin^{-1} 1 - \left(\frac{1}{2} (0 \sqrt{1-0^2}) + \frac{1}{2} \sin^{-1} 0 \right) \\ &= \frac{1}{2} \sin^{-1} 1 \\ &= \frac{\pi}{4} \end{aligned}$$

Now we look at examples where it indeed helps us, and consider definite integrals. When considering definite integrals, we need to be very careful with the bounds since we will be inverting trigonometric functions.

In each example, we will first try a substitution that *looks* plausible, and then discard it if it complicates the integrand. Then we will choose a trigonometric substitution that matches one of the identities

$$1 - \sin^2 \theta = \cos^2 \theta, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad \sec^2 \theta - 1 = \tan^2 \theta,$$

or a scaled version of them.

⁶Explicitly, this is when $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

⁷In fact, simply because we cannot allow $1 - x^2$ to be negative is sufficient to deduce this, but the point still stands.

Example 50. Using a trigonometric substitution, compute

$$\int_{1/2}^1 \frac{1}{x\sqrt{1-x^2}} dx.$$

Solution. First, it is tempting to try the “direct” substitution $u = 1 - x^2$. Then $du = -2x dx$, so $dx = -\frac{1}{2x} du$. Substituting this into the integral gives

$$\int_{1/2}^1 \frac{1}{x\sqrt{1-x^2}} dx = \int \frac{1}{x\sqrt{u}} \left(-\frac{1}{2x}\right) du = -\frac{1}{2} \int \frac{1}{x^2\sqrt{u}} du,$$

This is complicating the integral because x^2 is still present, which will incorporate another u due to $x^2 = u - 1$. So we discard this.

Since the integrand contains $\sqrt{1-x^2}$, we should aim to use $1 - \cos^2 \theta = \sin^2 \theta$ or $1 - \sin^2 \theta = \cos^2 \theta$. Both $x = \sin \theta$ and $x = \cos \theta$ are candidates.

We pick one and see if it helps. If not, we can try the other one.

Now proceed with $x = \cos \theta$. Since $1/2 \leq x \leq 1$, we take $0 \leq \theta \leq \frac{\pi}{2}$, where $\cos \theta \geq 0$ and $\sin \theta \geq 0$.

Compute the pieces:

$$\begin{aligned} x &= \cos \theta \\ dx &= -\sin \theta d\theta \\ \sqrt{1-x^2} &= \sqrt{1-\cos^2 \theta} = \sqrt{\sin^2 \theta} = \sin \theta \quad (\text{since } \sin \theta \geq 0) \end{aligned}$$

Convert the bounds:

$$\begin{aligned} x = \frac{1}{2} &\therefore \cos \theta = \frac{1}{2} \therefore \theta = \frac{\pi}{3} \\ x = 1 &\therefore \cos \theta = 1 \therefore \theta = 0 \end{aligned}$$

Substitute from left to right:

$$\begin{aligned} \int_{1/2}^1 \frac{1}{x\sqrt{1-x^2}} dx &= \int_{x=1/2}^{x=1} \frac{1}{x\sqrt{1-x^2}} dx \\ &= \int_{\theta=0}^{\theta=\pi/3} \frac{1}{x\sqrt{1-x^2}} dx \\ &= \int_{\theta=0}^{\theta=\pi/3} \frac{1}{\cos \theta \sqrt{1-\cos^2 \theta}} dx \\ &= \int_{\theta=0}^{\theta=\pi/3} \frac{1}{\cos \theta \sqrt{1-\cos^2 \theta}} (-\sin \theta d\theta) \\ &= \int_{\theta=\pi/3}^{\theta=0} \frac{1}{\cos \theta \cdot \sin \theta} (-\sin \theta d\theta) \\ &= \int_{\theta=\pi/3}^{\theta=0} -\sec \theta d\theta \\ &= \int_0^{\pi/3} \sec \theta d\theta. \end{aligned}$$

Now integrate using the formula book:

$$\int_0^{\pi/3} \sec \theta d\theta = [\ln |\sec \theta + \tan \theta|]_0^{\pi/3} = \ln \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - \ln(\sec 0 + \tan 0).$$

Evaluate the trig values:

$$\sec \frac{\pi}{3} = 2, \quad \tan \frac{\pi}{3} = \sqrt{3}, \quad \sec 0 = 1, \quad \tan 0 = 0.$$

Therefore

$$\int_{1/2}^1 \frac{1}{x\sqrt{1-x^2}} dx = \ln(2 + \sqrt{3}) - \ln(1) = \ln(2 + \sqrt{3}).$$

□

Example 51. Using a trigonometric substitution, compute

$$\int_1^2 \frac{1}{\sqrt{x^2-1}} dx$$

Solution. The expression $x^2 - 1$ matches the identity

$$\sec^2 \theta - 1 = \tan^2 \theta$$

This suggests the substitution

$$x = \sec \theta$$

We need to replace all instances of x with θ , including the bounds and dx , therefore we compute

$$\begin{aligned} x &= \sec \theta \\ dx &= \sec \theta \tan \theta d\theta \\ \sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} = \sqrt{\tan^2 \theta} = \tan \theta \end{aligned}$$

Convert the bounds

$$\begin{aligned} x = 1 &\therefore \sec \theta = 1 \therefore \theta = 0 \\ x = 2 &\therefore \sec \theta = 2 \therefore \theta = \frac{\pi}{3} \end{aligned}$$

Substitute from left to right

$$\begin{aligned} \int_1^2 \frac{1}{\sqrt{x^2-1}} dx &= \int_{x=1}^{x=2} \frac{1}{\sqrt{x^2-1}} dx \\ &= \int_{\theta=0}^{\theta=\pi/3} \frac{1}{\sqrt{\sec^2 \theta - 1}} dx \\ &= \int_{\theta=0}^{\theta=\pi/3} \frac{1}{\tan \theta} \sec \theta \tan \theta d\theta \\ &= \int_0^{\pi/3} \sec \theta d\theta \end{aligned}$$

Use the formula book

$$\int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c$$

Therefore

$$\begin{aligned} \int_0^{\pi/3} \sec \theta d\theta &= [\ln |\sec \theta + \tan \theta|]_0^{\pi/3} \\ &= \ln \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - \ln (\sec 0 + \tan 0) \end{aligned}$$

Evaluate

$$\begin{aligned}\sec \frac{\pi}{3} &= 2 & \tan \frac{\pi}{3} &= \sqrt{3} \\ \sec 0 &= 1 & \tan 0 &= 0\end{aligned}$$

Therefore

$$\int_1^2 \frac{1}{\sqrt{x^2-1}} dx = \ln(2 + \sqrt{3}) - \ln(1) = \ln(2 + \sqrt{3})$$

□

Example 52. Let $a > 0$ and $b > 0$. Using a trigonometric substitution, compute

$$\int_0^{\frac{b}{a}} \frac{1}{\sqrt{a^2x^2 + b^2}} dx$$

Solution. We want to get it into a form where we can spot a term like $1 \pm x^2$ or $x^2 \pm 1$.

The expression $a^2x^2 + b^2$ can be written as

$$a^2x^2 + b^2 = b^2 \left(1 + \left(\frac{ax}{b} \right)^2 \right)$$

and we recognise the identity

$$1 + \tan^2 \theta = \sec^2 \theta$$

This suggests the substitution

$$\frac{ax}{b} = \tan \theta$$

which we write as

$$x = \frac{b}{a} \tan \theta$$

We need to replace all instances of x with θ , including the bounds and dx , therefore we compute

$$\begin{aligned}x &= \frac{b}{a} \tan \theta \\ dx &= \frac{b}{a} \sec^2 \theta d\theta \\ \sqrt{a^2x^2 + b^2} &= \sqrt{a^2 \left(\frac{b}{a} \tan \theta \right)^2 + b^2} \\ &= \sqrt{b^2 \tan^2 \theta + b^2} \\ &= b\sqrt{1 + \tan^2 \theta} \\ &= b \sec \theta\end{aligned}$$

Convert the bounds

$$\begin{aligned}x = 0 &\therefore \tan \theta = 0 \therefore \theta = 0 \\ x = \frac{b}{a} &\therefore \tan \theta = 1 \therefore \theta = \frac{\pi}{4}\end{aligned}$$

Substitute from left to right

$$\begin{aligned}
 \int_0^{\frac{b}{a}} \frac{1}{\sqrt{a^2 x^2 + b^2}} dx &= \int_{x=0}^{x=\frac{b}{a}} \frac{1}{\sqrt{a^2 x^2 + b^2}} dx \\
 &= \int_{\theta=0}^{\theta=\pi/4} \frac{1}{\sqrt{a^2 \left(\frac{b}{a} \tan \theta\right)^2 + b^2}} dx \\
 &= \int_{\theta=0}^{\theta=\pi/4} \frac{1}{b \sec \theta} \frac{b}{a} \sec^2 \theta d\theta \\
 &= \frac{1}{a} \int_0^{\pi/4} \sec \theta d\theta
 \end{aligned}$$

Use the formula book

$$\int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c$$

Therefore

$$\begin{aligned}
 \frac{1}{a} \int_0^{\pi/4} \sec \theta d\theta &= \frac{1}{a} [\ln |\sec \theta + \tan \theta|]_0^{\pi/4} \\
 &= \frac{1}{a} \left(\ln \left(\sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right) - \ln (\sec 0 + \tan 0) \right)
 \end{aligned}$$

Evaluate

$$\begin{aligned}
 \sec \frac{\pi}{4} &= \sqrt{2} & \tan \frac{\pi}{4} &= 1 \\
 \sec 0 &= 1 & \tan 0 &= 0
 \end{aligned}$$

Therefore

$$\int_0^{\frac{b}{a}} \frac{1}{\sqrt{a^2 x^2 + b^2}} dx = \frac{1}{a} \left(\ln(\sqrt{2} + 1) - \ln(1) \right) = \frac{1}{a} \ln(\sqrt{2} + 1)$$

□

The following two are mainly difficult due to the emergence of integrating $\sec^3 \theta$. Otherwise, the techniques are the same.

Example 53. (*hard*) Using a trigonometric substitution, compute

$$\int_1^2 \sqrt{x^2 - 1} dx$$

Solution. It is possible to evaluate this integral using a different substitution. However, since the question asks for a trigonometric substitution, we proceed in that way.

The expression $x^2 - 1$ matches the identity

$$\sec^2 \theta - 1 = \tan^2 \theta$$

This suggests the substitution

$$x = \sec \theta$$

We need to replace all instances of x with θ , including the bounds and dx , therefore we compute

$$\begin{aligned}
 x &= \sec \theta \\
 dx &= \sec \theta \tan \theta d\theta \\
 \sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} = \sqrt{\tan^2 \theta} = \tan \theta
 \end{aligned}$$

Convert the bounds

$$\begin{aligned}x &= 1 \therefore \sec \theta = 1 \therefore \theta = 0 \\x &= 2 \therefore \sec \theta = 2 \therefore \theta = \frac{\pi}{3}\end{aligned}$$

Substitute from left to right

$$\begin{aligned}\int_1^2 \sqrt{x^2 - 1} \, dx &= \int_{x=1}^{x=2} \sqrt{x^2 - 1} \, dx \\&= \int_{\theta=0}^{\theta=\pi/3} \tan \theta \cdot \sec \theta \tan \theta \, d\theta \\&= \int_0^{\pi/3} \sec \theta \tan^2 \theta \, d\theta\end{aligned}$$

Use $\tan^2 \theta = \sec^2 \theta - 1$

$$\begin{aligned}\int_0^{\pi/3} \sec \theta \tan^2 \theta \, d\theta &= \int_0^{\pi/3} \sec \theta (\sec^2 \theta - 1) \, d\theta \\&= \int_0^{\pi/3} (\sec^3 \theta - \sec \theta) \, d\theta\end{aligned}$$

The integral of $\sec \theta$ is in the formula book. To integrate $\sec^3 \theta$, we use integration by parts.

Write

$$\sec^3 \theta = \sec \theta \cdot \sec^2 \theta$$

Choose

$$\begin{array}{ll}u = \sec \theta & v' = \sec^2 \theta \\u' = \sec \theta \tan \theta & v = \tan \theta\end{array}$$

Apply integration by parts

$$\begin{aligned}\int \sec^3 \theta \, d\theta &= \sec \theta \tan \theta - \int \tan \theta (\sec \theta \tan \theta) \, d\theta \\&= \sec \theta \tan \theta - \int \sec \theta \tan^2 \theta \, d\theta \\&= \sec \theta \tan \theta - \int \sec \theta (\sec^2 \theta - 1) \, d\theta \\&= \sec \theta \tan \theta - \int \sec^3 \theta \, d\theta + \int \sec \theta \, d\theta\end{aligned}$$

Rearranging

$$\begin{aligned}2 \int \sec^3 \theta \, d\theta &= \sec \theta \tan \theta + \int \sec \theta \, d\theta \\\therefore \int \sec^3 \theta \, d\theta &= \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| + c\end{aligned}$$

Return to the definite integral

$$\int_0^{\pi/3} (\sec^3 \theta - \sec \theta) \, d\theta = \left[\frac{1}{2} \sec \theta \tan \theta - \frac{1}{2} \ln |\sec \theta + \tan \theta| \right]_0^{\pi/3}$$

Evaluate

$$\begin{aligned}\sec \frac{\pi}{3} &= 2 & \tan \frac{\pi}{3} &= \sqrt{3} \\ \sec 0 &= 1 & \tan 0 &= 0\end{aligned}$$

Therefore

$$\int_1^2 \sqrt{x^2 - 1} \, dx = \sqrt{3} - \frac{1}{2} \ln(2 + \sqrt{3})$$

□

Example 54. (*hard*) Let $a > 0$ and $b > 0$. Using a trigonometric substitution, compute

$$\int_0^{\frac{b}{a}} \sqrt{a^2 x^2 + b^2} \, dx$$

Solution. We want to get it into a form where we can spot a term like $1 \pm x^2$ or $x^2 \pm 1$.

Factor out b^2

$$\sqrt{a^2 x^2 + b^2} = b \sqrt{1 + \left(\frac{ax}{b}\right)^2}$$

Using $1 + \tan^2 \theta = \sec^2 \theta$, we set

$$\frac{ax}{b} = \tan \theta$$

We need to replace all instances of x with θ , including the bounds and dx , therefore we compute

$$\begin{aligned}x &= \frac{b}{a} \tan \theta \\ dx &= \frac{b}{a} \sec^2 \theta \, d\theta \\ \sqrt{a^2 x^2 + b^2} &= b \sec \theta\end{aligned}$$

Convert the bounds

$$\begin{aligned}x = 0 &\therefore \tan \theta = 0 \therefore \theta = 0 \\ x = \frac{b}{a} &\therefore \tan \theta = 1 \therefore \theta = \frac{\pi}{4}\end{aligned}$$

Substitute from left to right

$$\begin{aligned}\int_0^{\frac{b}{a}} \sqrt{a^2 x^2 + b^2} \, dx &= \int_0^{\pi/4} b \sec \theta \cdot \frac{b}{a} \sec^2 \theta \, d\theta \\ &= \frac{b^2}{a} \int_0^{\pi/4} \sec^3 \theta \, d\theta\end{aligned}$$

Using the result derived above in Example 53

$$\int \sec^3 \theta \, d\theta = \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| + c$$

Therefore

$$\int_0^{\frac{b}{a}} \sqrt{a^2 x^2 + b^2} \, dx = \frac{b^2}{a} \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right]_0^{\pi/4}$$

Evaluate

$$\sec \frac{\pi}{4} = \sqrt{2} \quad \tan \frac{\pi}{4} = 1$$

Hence

$$\int_0^{\frac{b}{a}} \sqrt{a^2x^2 + b^2} dx = \frac{b^2}{2a} \left(\sqrt{2} + \ln(\sqrt{2} + 1) \right)$$

□

4.3.2 Additional Exercises

Example 55. (*hard*)

Calculate the indefinite integral $\int \operatorname{cosec} x dx$.

Rearrange your answer to be in the form found in your formula book.

$$\int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x + \cot x|$$

Hence deduce the indefinite integral $\int \sec x dx$.

Rearrange your answer to be in the form found in your formula book.

$$\int \sec x dx = \ln |\sec x + \tan x|$$

Solution. There is a reason that this is included in your formula book! There is no “nice” way of approaching this (without complex numbers).

If you were attempting this for yourself without knowing the answer, there are really three approaches:

1. Substitution, e.g. with $u = \sin x$
2. By parts with $u = \frac{1}{\sin x}$ and $v' = 1$
3. More rearranging before trying the above two again

The first option, if you work through it gives that

$$\int \operatorname{cosec} x dx = \int \frac{1}{u} \frac{1}{\sqrt{1-u^2}} du$$

But this integral is again very difficult, and probably requires more substitutions! This can be done, but it is good to attempt another approach if your first approach seems to be complicating the integral.

The second option, if you work through it gives that

$$\int \operatorname{cosec} x dx = \frac{x}{\sin x} + \int \frac{\cos x}{\sin^2 x} dx$$

Which gives us a disgusting integral to solve, which is definitely more complicated than the one we started with! That leaves us with the third option, which is to rearrange before integrating.

We will use multiple tricks to massage this into an integral we can approach, starting with the least intuitive trick.

$$\begin{aligned}\int \frac{1}{\sin x} dx &= \int \frac{\sin x}{\sin^2 x} dx && \text{(multiply top and bottom by } \sin x \text{)} \\ &= \int \frac{\sin x}{1 - \cos^2 x} dx && \text{(using } \sin^2 x + \cos^2 x = 1 \text{)}\end{aligned}$$

Now let's use the substitution $u = \cos x$. Therefore $\frac{du}{dx} = -\sin x$. Pretending to rearrange this for $-du$ gives $-du = \sin x dx$.

We can substitute this into our final line above since the numerator is $\sin x dx$.

$$\begin{aligned}\int \frac{1}{\sin x} dx &= - \int \frac{1}{1 - \cos^2 x} du && \text{(using } -du = \sin x dx \text{)} \\ &= - \int \frac{1}{1 - u^2} du && \text{(using } u = \cos x \text{)} \\ &= - \int \frac{1}{(1 - u)(1 + u)} du && \text{(difference of two squares)}\end{aligned}$$

We now use partial fractions to separate the final fraction into two easier to integrate fractions.

$$\begin{aligned}\frac{1}{(1 - u)(1 + u)} &= \frac{a}{1 - u} + \frac{b}{1 + u} && \text{(set up partial fractions)} \\ &= \frac{a(1 + u) + b(1 - u)}{(1 - u)(1 + u)} && \text{(add the fractions)} \\ &= \frac{(a + b) + (a - b)u}{(1 - u)(1 + u)} && \text{(collecting terms)}\end{aligned}$$

This gives us the following equation to solve:

$$a(1 + u) + b(1 - u) = 1$$

Or equivalently

$$(a + b) + (a - b)u = 1$$

There are two approaches to proceed to solve for a and b . One is to substitute $u = 1$ and $u = -1$ to cancel a or b in succession. This gives us:

$$\begin{aligned}b(1 - (-1)) &= 1 && (u = -1) \\ a(1 + 1) &= 1 && (u = 1)\end{aligned}$$

Which can be solved to find $a = b = \frac{1}{2}$.

The second approach is to compare like terms with coefficients of u and constants on the left and right side of the equation.

$$(a + b) + (a - b)u = 1 + 0u$$

We now compare the terms and deduce

$$\begin{aligned}a + b &= 1 \\ a - b &= 0\end{aligned}$$

Adding the two together gives $2a = 1$ and therefore $a = \frac{1}{2}$. The second equation shows that $a = b$ therefore they are both $\frac{1}{2}$.

Returning to our integral, we can now split our denominator into two fractions.

$$\begin{aligned}\int \frac{1}{\sin x} dx &= - \int \frac{1}{(1-u)(1+u)} du && \text{(where we left off)} \\ &= - \int \left(\frac{\frac{1}{2}}{1-u} + \frac{\frac{1}{2}}{1+u} \right) du && \text{(substituting the partial fraction coefficients)} \\ &= -\frac{1}{2} \left(\int \frac{1}{1-u} du + \int \frac{1}{1+u} du \right) && \text{(factorising)}\end{aligned}$$

Recall that the integral of $\frac{1}{u}$ is $\ln|u|$. Therefore by the chain rule (or by substitution of $y = 1 - u$ and $z = 1 + u$)

$$\begin{aligned}\int \frac{1}{\sin x} dx &= -\frac{1}{2} (-\ln|1-u| + \ln|1+u|) + c && \text{(integrating)} \\ &= -\frac{1}{2} \left(\ln \left| \frac{1+u}{1-u} \right| \right) + c && \text{(recall } \ln x - \ln y = \ln \frac{x}{y} \text{)} \\ &= -\frac{1}{2} \ln \left| \frac{1+\cos x}{1-\cos x} \right| + c && \text{(substituting } u = \cos x \text{)} \\ &= -\frac{1}{2} \ln \left| \frac{(1+\cos x)^2}{(1-\cos x)(1+\cos x)} \right| + c && \text{(multiplying top and bottom by } (1+\cos x) \text{)} \\ &= -\frac{1}{2} \ln \left| \frac{(1+\cos x)^2}{1-\cos^2 x} \right| + c && \text{(multiplying denominator)} \\ &= -\frac{1}{2} \ln \left| \frac{(1+\cos x)^2}{\sin^2 x} \right| + c && \text{(using } \sin^2 x + \cos^2 x = 1 \text{)} \\ &= -\frac{1}{2} \ln \left| \left(\frac{1+\cos x}{\sin x} \right)^2 \right| + c && \text{(using } (xy)^2 = x^2 y^2 \text{)} \\ &= -\ln \left| \frac{1+\cos x}{\sin x} \right| + c && \text{(recall } a \ln x = \ln(x^a) \text{)} \\ &= -\ln |\operatorname{cosec} x + \cot x| + c && \text{(separating the fraction)}\end{aligned}$$

Which is finally in the form of your formula book.

What about sec? Recall that $\sin(x - \frac{\pi}{2}) = -\cos(x)$. Therefore

$$\int \sec x dx = - \int \operatorname{cosec}(x - \frac{\pi}{2}) dx$$

Now, using the chain rule, we can observe that adding a constant to the argument of a function also adds the same constant to the argument of the derivative. To see this, differentiate $f(x+a)$ using the chain rule – you get $f'(x+a)$.

Therefore,

$$\begin{aligned}
\int \sec x dx &= - \int \operatorname{cosec}\left(x - \frac{\pi}{2}\right) dx && \text{(using } \sec x = -\operatorname{cosec}(x - \frac{\pi}{2}) \text{)} \\
&= \ln \left| \operatorname{cosec}\left(x - \frac{\pi}{2}\right) + \cot\left(x - \frac{\pi}{2}\right) \right| + c && \text{(using our result for } \int \operatorname{cosec} \text{)} \\
&= \ln \left| -\sec x + \cot\left(x - \frac{\pi}{2}\right) \right| + c && \text{(using our relation for cosec and sec again)} \\
&= \ln \left| -\sec x + \frac{\cos\left(x - \frac{\pi}{2}\right)}{\sin\left(x - \frac{\pi}{2}\right)} \right| + c && \text{(definition of cot)} \\
&= \ln \left| -\sec x + \frac{\sin x}{-\cos x} \right| + c && \text{(shifting sin and cos)} \\
&= \ln |-\sec x - \tan x| + c && \text{(definition of tan)} \\
&= \ln |\sec x + \tan x| + c && \text{(using } |-x| = |x| \text{)}
\end{aligned}$$

Which is finally in the form of the formula book!

□

Example 56. Compute the following indefinite integral, assuming that $4 - x^2 \geq 0$.

$$\int \frac{1}{x^2 \sqrt{4 - x^2}} dx$$

Solution. I will approach this from the point of view of discovering what the correct approach is, if it is not obvious from the start, and ignoring that we are in the substitution section of these notes.

Firstly, it is not obvious to “see” an easy way to integrate it directly, so we move on to trying to rearrange it into something we can integrate.

The only way I can spot to rearrange the integral is by spotting a difference of two squares in the denominator:

$$\int \frac{1}{x^2 \sqrt{(2 - x)(2 + x)}} dx$$

The issue is that this didn’t lead to any additional simplifications, so it is unclear whether this helped or not.

Instead let’s consider an integration by parts, which might seem like a natural way to progress.

Using LATE does not help too much as a choice of u , so let’s just choose the left side as u :

$$u = \frac{1}{x^2} = x^{-2} \qquad v = ? \qquad (4.3.1)$$

$$u' = -2x^{-3} = \frac{-2}{x^3} \qquad v' = \frac{1}{\sqrt{4 - x^2}} \qquad (4.3.2)$$

This has two problems: one is that the u' has introduced an additional power of x , making our life worse, and the other problem is that we still need to integrate something difficult to find v . This should clue us in to the idea that maybe this approach is wrong.

Next let’s try integration by substitution. For example, we could try $u = x^2 \sqrt{4 - x^2}$, but finding what dx is will be an issue since we will have a very complicated derivative via the product rule.

Again this is complicating our work. At some point we must realise that if this is an exam question, we are probably not expected to spend this much time on it.

Instead, let's try a substitution of the "bad" part: $u = \sqrt{4 - x^2}$. If we were to continue, we would realise that the chain rule Section 3.4 is needed for $\frac{du}{dx}$.

The other part of the integral is x^2 , which we can rearrange for by squaring both sides and finding $x^2 = 4 - u^2$.

This would turn the integral into the following.

$$\int \frac{1}{u(4 - u^2)} \cdot (\text{whatever } dx \text{ is})$$

This is also a complicated integral, but potentially more manageable than the original. So far, this is our best option.

However, if you have recently looked at the trigonometric substitution Section 4.3.1, you might recognise the $\sqrt{4 - x^2}$ as containing something that looks like $a \pm bx^2$.

This is the correct approach.

Let's find a trig identity which looks like $4 - x^2$, following the procedure outlined in that section.

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ \therefore 4 \sin^2 \theta + 4 \cos^2 \theta &= 4 \\ \therefore 4 \sin^2 \theta &= 4 - 4 \cos^2 \theta \\ \therefore 4 \cos^2 \theta &= 4 - 4 \sin^2 \theta\end{aligned}$$

This gives us two options which look like $4 - (\text{something})^2$. The point is that when we take the square root, the square on the left-hand-side will cancel and leave us with something much simpler. Let's choose the first one⁸ and deduce what substitution we need.

$$\begin{aligned}4 \sin^2 \theta &= 4 - 4 \cos^2 \theta \\ \therefore 2 \sin \theta &= \sqrt{4 - 4 \cos^2 \theta} && \text{(what we want to use)} \\ &= \sqrt{4 - x^2} && \text{(what we want to replace)} \\ \therefore x^2 &= 4 \cos^2 \theta && \text{(what we need to be true)} \\ \therefore x &= 2 \cos \theta && \text{(what the substitution is)} \\ \therefore \frac{dx}{d\theta} &= -2 \sin \theta && \text{(for } dx) \\ \therefore dx &= -2 \sin \theta d\theta && \text{(pretending to rearrange for } dx)\end{aligned}$$

Now we can substitute the relevant equations from above into the integral from left to right:

⁸Choosing the other one and working through ends up with needing to integrate $\operatorname{cosec}^2 \theta$, which is not in the Edexcel formula book, but is possible to continue through regardless. We choose the other approach for reasons that will become clear at the end of the example.

$$\begin{aligned}
\int \frac{1}{x^2\sqrt{4-x^2}} dx &= \int \frac{1}{4\cos^2\theta\sqrt{4-x^2}} dx && \text{(using } x^2 = 4\cos^2\theta\text{)} \\
&= \int \frac{1}{4\cos^2\theta \cdot 2\sin\theta} dx && \text{(using } 2\sin\theta = \sqrt{4-4\cos^2\theta}\text{)} \\
&= \int \frac{1}{4\cos^2\theta \cdot 2\sin\theta} (-2\sin\theta) d\theta && \text{(using } dx = -2\sin\theta d\theta\text{)} \\
&= -\frac{1}{4} \int \sec^2\theta d\theta && \text{(rearranging, using } \frac{1}{\cos\theta} = \sec\theta\text{)}
\end{aligned}$$

Now we could continue with integrating this, but for Edexcel A-level, it is always worth checking if a trigonometric integral is just in the formula book. The formula book indeed tells us that:

$$\int \sec^2 kx dx = \frac{1}{k} \tan kx + c$$

Setting $k = 1$ and $x = \theta$ in the formula gives:

$$\begin{aligned}
\int \frac{1}{x^2\sqrt{4-x^2}} dx &= -\frac{1}{4} \int \sec^2\theta && \text{(where we left off)} \\
&= -\frac{1}{4} \tan\theta + c
\end{aligned}$$

We are not done however, since we introduced θ as part of our substitution – the question was in terms of x so we should give this in terms of x too.

Recall that $x = 2 \cos \theta$, and therefore $\cos \theta = \frac{1}{2}x$

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{4-x^2}} dx &= -\frac{1}{4} \tan \theta + c \\
 &= -\frac{1}{4} \frac{\sin \theta}{\cos \theta} + c && \text{(using } \tan \theta = \frac{\sin \theta}{\cos \theta} \text{)} \\
 &= -\frac{1}{4} \frac{\sqrt{1-\cos^2 \theta}}{\cos \theta} + c && \text{(using } \sin \theta = \sqrt{1-\cos^2 \theta} \text{)} \\
 &= -\frac{1}{4} \frac{\sqrt{1-(\frac{1}{2}x)^2}}{\frac{1}{2}x} + c && \text{(using } \cos \theta = \frac{1}{2}x \text{)} \\
 &= -\frac{2}{4} \frac{\sqrt{1-(\frac{1}{2}x)^2}}{x} + c && \text{(multiplying top and bottom by 2)} \\
 &= -\frac{1}{2} \frac{\sqrt{1-(\frac{1}{2}x)^2}}{x} + c && \text{(simplifying fraction)} \\
 &= -\frac{1}{2} \frac{\sqrt{1-\frac{1}{4}x^2}}{x} + c && \text{(expanding the square)} \\
 &= -\frac{1}{2} \frac{\sqrt{\frac{4}{4}-\frac{1}{4}x^2}}{x} + c && \text{(writing } 1 = \frac{4}{4} \text{)} \\
 &= -\frac{1}{2} \frac{\sqrt{\frac{1}{4}(4-x^2)}}{x} + c && \text{(factorising)} \\
 &= -\frac{1}{2} \frac{\sqrt{\frac{1}{4}}\sqrt{4-x^2}}{x} + c && \text{(using } \sqrt{ab} = \sqrt{a}\sqrt{b} \text{)} \\
 &= -\frac{1}{2} \frac{\frac{1}{2}\sqrt{4-x^2}}{x} + c && \text{(using } \sqrt{a/b} = \sqrt{a}/\sqrt{b} \text{)} \\
 &= -\frac{1}{4} \frac{\sqrt{4-x^2}}{x} + c && \text{(multiplying top and bottom by 2)}
 \end{aligned}$$

And finally, we find that the integral is as follows.

$$\int \frac{1}{x^2 \sqrt{4-x^2}} dx = -\frac{1}{4} \frac{\sqrt{4-x^2}}{x} + c$$

□

4.4 Mixed Exercises

These questions use a range of techniques. Remember that the main way to approach an integral question is as follows:

1. Do we know what function has a derivative which is what we are trying to find the integral of? I.e. can we “just see” the answer? This can include the formula book.
2. Can we rearrange the integral into something we know how to integrate?
3. Can we use integration by parts?
4. Can we use a substitution?

Where the numbering corresponds to difficulty. If you can do an earlier step you probably should.

The following exercises will demonstrate this.

Example 57. Find the following indefinite integral, where $a \neq 0$ is a constant.

$$\int \frac{1}{x} \cdot x^a dx$$

What happens if $a = 0$?

Solution. If we skipped the stepped approach, we might get tunnel-visioned into jumping into a integration by parts. This would have us choose $u = x^a$ and $v' = \frac{1}{x}$ via LATE, and therefore $v = \ln x$, and we could continue. This unfortunately is jumping the gun and complicating our lives, when if we had followed the steps the answer would be much clearer.

Firstly, it is not directly obvious to see a function which differentiates into $\frac{1}{x} \cdot x^a$, but we *can* rearrange it into something we can handle.

$$\begin{aligned} \int \frac{1}{x} \cdot x^a dx &= \int x^{-1} \cdot x^a dx \\ &= \int x^{a-1} dx \end{aligned} \quad (\text{using } x^a x^b = x^{a+b})$$

Now this one you can recognise as something we can integrate. Since we know x^a differentiates to become ax^{a-1} , we can divide by a to find:

$$\begin{aligned} \int \frac{1}{x} \cdot x^a dx &= \int x^{a-1} dx \\ &= \frac{1}{a} x^a + c \end{aligned}$$

Therefore, simplifying the integral first allowed us to find that

$$\int \frac{1}{x} \cdot x^a dx = \frac{1}{a} x^a + c$$

Now this holds whenever $a \neq 0$, which can be seen by the right-hand-side being divided by a . If $a = 0$, the integral becomes:

$$\int \frac{1}{x} dx = \ln |x| + c$$

Which again is possible by first simplifying the integral before continuing. Notice that $\ln x$ only occurs when we have exactly $\frac{1}{x}$ in the integral⁹, otherwise we also get a power of x as the integral as seen above.

This completes the question by considering the two cases separately. □

Example 58. Find

$$\int_0^9 \frac{x}{1 + \sqrt{x}} dx$$

Solution. This might seem to be obviously a substitution question, but I recommend to at least attempt to go through the steps above in order first.

Can we just integrate it? Well since we have a $\frac{1}{1+\sqrt{x}}$ term in the integrand, it is useful to think of any integral that might give us this term. One you might try is $\ln(1 + \sqrt{x})$. This differentiates to give $\frac{\frac{1}{2\sqrt{x}}}{1+\sqrt{x}}$, which isn't exactly the form we need.

⁹In general the integral needs to be of the form $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$.

Secondly, can we rearrange or simplify it? I cannot see a way to rearrange this in an obvious way.

Next you might have tried by parts with:

$$\begin{array}{ll} u = \frac{1}{1 + \sqrt{x}} & v = \frac{1}{2}x^2 \\ u' = -\frac{1}{2\sqrt{x}} \frac{1}{(1 + \sqrt{x})^2} & v' = x \end{array}$$

Where we would then need to integrate $u'v$ as per the by parts formula. This new integral is most likely more complicated than the one we started with! So we should consider another method. We need to have clarity of mind to see that we are going down the wrong path with this method for this question.

So we proceed with a substitution. The first thing that should jump out at you is using the substitution $u = 1 + \sqrt{x}$

We now calculate everything in terms of u that we need.

$$\begin{array}{ll} \frac{du}{dx} = \frac{1}{2\sqrt{x}} & \text{(differentiating)} \\ dx = 2\sqrt{x}du & \text{(rearranging for } dx) \\ u = 1 + \sqrt{x} & \text{(our substitution)} \\ x = (u - 1)^2 & \text{(rearranging } u = 1 + \sqrt{x}) \end{array}$$

Now let's substitute each instance of x from left to right with u in our integral, making sure to remember the integral bounds.

$$\begin{aligned} \int_0^9 \frac{x}{1 + \sqrt{x}} dx &= \int_{x=0}^{x=9} \frac{x}{1 + \sqrt{x}} dx && \text{(reminding ourselves of } x) \\ &= \int_{u=1}^{u=4} \frac{x}{1 + \sqrt{x}} dx && \text{(using } u = 1 + \sqrt{x}) \\ &= \int_{u=1}^{u=4} \frac{(u - 1)^2}{1 + \sqrt{x}} dx && \text{(using } x = (u - 1)^2) \\ &= \int_{u=1}^{u=4} \frac{(u - 1)^2}{u} dx && \text{(using } u = 1 + \sqrt{x}) \\ &= \int_{u=1}^{u=4} \frac{(u - 1)^2}{u} 2\sqrt{x} du && \text{(using } dx = 2\sqrt{x} du) \\ &= \int_{u=1}^{u=4} \frac{(u - 1)^2}{u} 2\sqrt{(u - 1)^2} du && \text{(using } x = (u - 1)^2) \\ &= \int_{u=1}^{u=4} \frac{(u - 1)^2}{u} 2(u - 1) du && \text{(using } (a^2)^{\frac{1}{2}} = a) \\ &= 2 \int_{u=1}^{u=4} \frac{(u - 1)^3}{u} du && \text{(simplifying)} \end{aligned}$$

Now we again ask ourselves the questions we ask ourselves whenever there is an integral.

Can we see what the integral is? Due to the $\frac{1}{u}$, we might expect there to be some $\ln$ in the answer, but it is not obvious.

Then we ask: can we rearrange this into something we *do* know how to integrate?¹⁰

¹⁰It is possible to approach this with by parts with $f = (u - 1)^3$ and $g' = \frac{1}{u}$, but you would need to do by parts multiple times with the resulting integral. There are no obvious substitutions either. It is essential to be able to identify when you can "just do" the integral and not waste time with a more complicated method.

Well, the only thing we can try is expanding:

$$\begin{aligned}
 2 \int_{u=1}^{u=4} \frac{(u-1)^3}{u} du &= 2 \int_{u=1}^{u=4} \frac{u^3 + 3u^2 \times (-1) + 3u \times (-1)^2 + (-1)^3}{u} du && \text{(using binomial expansion formula)} \\
 &= 2 \int_{u=1}^{u=4} \frac{u^3 - 3u^2 + 3u - 1}{u} du && \text{(simplifying)} \\
 &= 2 \int_{u=1}^{u=4} \frac{u^3}{u} - 3 \frac{u^2}{u} + 3 \frac{u}{u} - \frac{1}{u} du && \text{(separating the fraction)} \\
 &= 2 \int_{u=1}^{u=4} u^2 - 3u + 3 - \frac{1}{u} du && \text{(simplifying)}
 \end{aligned}$$

Which we can actually solve by computing each of the integrals one-by-one!

$$\begin{aligned}
 2 \int_{u=1}^{u=4} u^2 - 3u + 3 - \frac{1}{u} du &= 2 \left[\frac{1}{3} u^3 - \frac{3}{2} u^2 + 3u - \ln |u| \right]_{u=1}^{u=4} \\
 &= 2 \left(\frac{1}{3} 4^3 - \frac{3}{2} 4^2 + 3 \times 4 - \ln 4 \right) && \text{("just doing" the integrals)} \\
 &\quad - 2 \left(\frac{1}{3} 1^3 - \frac{3}{2} 1^2 + 3 \times 1 - \ln 1 \right) && \text{(using } \ln 1 = 0 \text{)}
 \end{aligned}$$

Where you could expand all of this or put it in a calculator for the answer. □

Example 59. Find

$$\int_1^{e^2} x^2 \ln x dx$$

Solution. It's always good to think whether you can spot the integral before starting. Since it is not clear or in the formula book, we then ask ourselves whether we can rearrange it in any sensible way. Since there is no real way to rearrange this obviously, we continue with by parts or substitution.

There is an obvious calling for by parts since it is two functions multiplied together.

Using LATE, we declare that $u = \ln x$ and $v' = x^2$. Writing this in a square and filling in the missing u' and v via differentiation and integration gives us:

$$\begin{array}{ll}
 u = \ln x & v = \frac{1}{3} x^3 \\
 u' = \frac{1}{x} & v' = x^2
 \end{array}$$

Recall (or derive from the product rule as in Section 4.2) the by parts formula:

$$\int_a^b uv' dx = [uv]_a^b - \int_a^b u'v dx$$

Substituting our square into this formula gives the following:

$$\begin{aligned}
\int_1^{e^2} x^3 \ln x dx &= \left[\ln x \frac{1}{3} x^3 \right]_1^{e^2} - \int_1^{e^2} \frac{1}{x} \times \frac{1}{3} x^3 dx \\
&= \left(\ln(e^2) \frac{1}{3} (e^2)^3 - \ln(1) \frac{1}{3} (1)^3 \right) - \int_1^{e^2} \frac{1}{x} \times \frac{1}{3} x^3 dx && \text{(substituting bounds)} \\
&= \left(2 \ln(e) \frac{1}{3} e^6 - \ln(1) \frac{1}{3} \right) - \int_1^{e^2} \frac{1}{x} \times \frac{1}{3} x^3 dx && \text{(using } \ln a^b = b \ln a \text{ and } \ln 1 = 0) \\
&= \frac{2}{3} e^6 - \int_1^{e^2} \frac{1}{x} \times \frac{1}{3} x^3 dx && \text{(using } \ln e = 1) \\
&= \frac{2}{3} e^6 - \frac{1}{3} \int_1^{e^2} x^2 dx && \text{(simplifying the integral)} \\
&= \frac{2}{3} e^6 - \frac{1}{3} \left[\frac{1}{3} x^3 \right]_1^{e^2} && \text{(evaluating the integral)} \\
&= \frac{2}{3} e^6 - \frac{1}{3} \left(\frac{1}{3} (e^2)^3 - \frac{1}{3} 1^3 \right) && \text{(evaluating the bounds)}
\end{aligned}$$

Which you could go on to simplify and or plug into a calculator.

□

Example 60. Find

$$\int_0^1 8x^2 e^{-3x} dx$$

Then state the indefinite integral

$$\int 8x^2 e^{-3x} dx$$

Solution. Again we start by trying to see if we can see an answer immediately. Since the derivative of $e^{f(x)}$ has a term like $e^{f(x)}$ in it, we might try something like $x^3 e^{-3x}$, which differentiates to become $3x^2 e^{-3x} - 3x^3 e^{-3x}$, where this is almost like what we want except the second term has an x^3 in it, ruining our fun.

Since there is no obvious way to rearrange this, and we could not integrate it directly, we try by parts. Using LATE, choose $u = 8x^2$ and $v' = e^{-3x}$.

Let us differentiate u and integrate v'^{11} to complete our square:

$$\begin{array}{ll}
u = 8x^2 & v = -\frac{1}{3} e^{-3x} \\
u' = 16x & v' = e^{-3x}
\end{array}$$

Recall (or derive from the product rule as in Section 4.2) the by parts formula:

$$\int_a^b uv' dx = [uv]_a^b - \int_a^b u'v dx$$

Substituting our square into this formula gives the following:

¹¹Observe that e^{-3x} differentiates to give $-3e^{-3x}$, so we must divide by -3 to get the answer we want.

$$\begin{aligned}
\int_0^1 8x^2 e^{-3x} dx &= \left[8x^2 \times \left(-\frac{1}{3}e^{-3x}\right) \right]_0^1 - \int_0^1 16x \times \left(-\frac{1}{3}e^{-3x}\right) dx \\
&= \left(8 \times 1^2 \times \left(-\frac{1}{3}e^{-3 \times 1}\right) - \cancel{8 \times 0^2 \times \left(-\frac{1}{3}e^{-3 \times 0}\right)} \right) - \int_0^1 16x \times \left(-\frac{1}{3}e^{-3x}\right) dx \\
&\hspace{15em} \text{(substituting the integral bounds)} \\
&= -\frac{8}{3}e^{-3} - \int_0^1 16x \times \left(-\frac{1}{3}e^{-3x}\right) dx \hspace{5em} \text{(simplifying)} \\
&= -\frac{8}{3}e^{-3} + \frac{16}{3} \int_0^1 x e^{-3x} dx \hspace{5em} \text{(rearranging the integral)}
\end{aligned}$$

Now how can we approach this final integral? If we ask ourselves the same questions again, we realise that we must do integration by parts again! We should not worry however, since although this integral looks similar to the one we started with, the power of x has decreased which means we are making progress – if the power decreases one more time then we will be left with just e^{-3x} , which we know how to integrate happily.

We continue by repeating our process of integration by parts on

$$\int_0^1 x e^{-3x} dx.$$

Using LATE again, choose $u = x$ and $v' = e^{-3x}$. As before, integrating e^{-3x} gives $v = -\frac{1}{3}e^{-3x}$ and differentiating x gives $u' = 1$. Our square is therefore:

$$\begin{array}{ll}
u = x & v = -\frac{1}{3}e^{-3x} \\
u' = 1 & v' = e^{-3x}
\end{array}$$

Applying the by parts formula on $[0, 1]$:

$$\begin{aligned}
\int_0^1 x e^{-3x} dx &= \left[x \times \left(-\frac{1}{3}e^{-3x}\right) \right]_0^1 - \int_0^1 1 \times \left(-\frac{1}{3}e^{-3x}\right) dx \\
&= \left[-\frac{1}{3}x e^{-3x} \right]_0^1 + \frac{1}{3} \int_0^1 e^{-3x} dx \\
&= \left(-\frac{1}{3}e^{-3} - 0 \right) + \frac{1}{3} \int_0^1 e^{-3x} dx \hspace{5em} \text{(evaluating the boundary term)}
\end{aligned}$$

Now we integrate the remaining exponential:

$$\begin{aligned}
\int_0^1 e^{-3x} dx &= \left[-\frac{1}{3}e^{-3x} \right]_0^1 \\
&= -\frac{1}{3}e^{-3} + \frac{1}{3}
\end{aligned}$$

Substituting this back in, we get:

$$\begin{aligned}
\int_0^1 x e^{-3x} dx &= -\frac{1}{3}e^{-3} + \frac{1}{3} \left(-\frac{1}{3}e^{-3} + \frac{1}{3} \right) \\
&= -\frac{1}{3}e^{-3} - \frac{1}{9}e^{-3} + \frac{1}{9} \\
&= -\frac{4}{9}e^{-3} + \frac{1}{9}
\end{aligned}$$

Now recall that our original integral was:

$$\int_0^1 8x^2 e^{-3x} dx = -\frac{8}{3}e^{-3} + \frac{16}{3} \int_0^1 x e^{-3x} dx$$

Substituting our result for $\int_0^1 x e^{-3x} dx$:

$$\begin{aligned}
\int_0^1 8x^2 e^{-3x} dx &= -\frac{8}{3}e^{-3} + \frac{16}{3} \left(-\frac{4}{9}e^{-3} + \frac{1}{9} \right) \\
&= -\frac{8}{3}e^{-3} - \frac{64}{27}e^{-3} + \frac{16}{27} \\
&= -\frac{72}{27}e^{-3} - \frac{64}{27}e^{-3} + \frac{16}{27} && \text{(writing } \frac{8}{3} = \frac{72}{27} \text{)} \\
&= \frac{16}{27} - \frac{136}{27}e^{-3}
\end{aligned}$$

You could leave your answer as

$$\int_0^1 8x^2 e^{-3x} dx = \frac{16}{27} - \frac{136}{27}e^{-3}$$

or factor it slightly as

$$\int_0^1 8x^2 e^{-3x} dx = \frac{8}{27} (2 - 17e^{-3})$$

or simply plug this into a calculator for a decimal approximation.

What if we wanted to know the indefinite integral of $8x^2 e^{-3x}$? The only real difference between definite and indefinite integral is whether you write the square brackets or not. To convert from a definite integral to an indefinite integral, you can work up until the square brackets part, and then remove the square brackets and add c .

For example, we can do some chasing around to find:

$$\begin{aligned}
\int 8x^2 e^{-3x} dx &= 8x^2 \times \left(-\frac{1}{3}e^{-3x} \right) + \frac{16}{3} \int x e^{-3x} dx && \text{(reusing work up until square brackets)} \\
&= 8x^2 \times \left(-\frac{1}{3}e^{-3x} \right) + \frac{16}{3} \left(x \times \left(-\frac{1}{3}e^{-3x} \right) + \frac{1}{3} \int e^{-3x} dx \right) \\
&= 8x^2 \times \left(-\frac{1}{3}e^{-3x} \right) + \frac{16}{3} \left(x \times \left(-\frac{1}{3}e^{-3x} \right) + \frac{1}{3} \left(-\frac{1}{3}e^{-3x} \right) \right) + c
\end{aligned}$$

Where c is a constant.

If you really want to, you could differentiate this to verify the result. □

Example 61. Let a be a positive number. Using $x = a \sin^2 \theta$, find

$$\int_0^a x^{\frac{1}{2}} \sqrt{a-x} dx$$

Give your answer in terms of a .

Solution. As usual, we first ask whether we can “see” an obvious answer which differentiates into the term inside the integral. Because of the square roots and the $a-x$ term, this does not look straightforward, so we follow the hint and use a substitution

Let

$$x = a \sin^2 \theta$$

Since $a > 0$, we can work out everything we need in terms of θ

$$\begin{aligned} \sqrt{x} &= \sqrt{a \sin^2 \theta} = \sqrt{a} \sin \theta && \text{(since } \sin \theta \geq 0 \text{ on } [0, \frac{\pi}{2}]) \\ a-x &= a - a \sin^2 \theta = a(1 - \sin^2 \theta) = a \cos^2 \theta && \text{(using } \sin^2 \theta + \cos^2 \theta = 1) \\ \sqrt{a-x} &= \sqrt{a \cos^2 \theta} = \sqrt{a} \cos \theta \\ \frac{dx}{d\theta} &= a \cdot 2 \sin \theta \cos \theta = a \sin 2\theta && \text{(using } \sin 2\theta = 2 \sin \theta \cos \theta) \\ \therefore dx &= a \sin 2\theta d\theta \end{aligned}$$

We also need to change the integral bounds

$$\begin{aligned} x = 0 &\therefore a \sin^2 \theta = 0 \therefore \sin^2 \theta = 0 \therefore \theta = 0 \\ x = a &\therefore a \sin^2 \theta = a \therefore \sin^2 \theta = 1 \therefore \theta = \frac{\pi}{2} \end{aligned}$$

Now we substitute everything into the integral, step by step, from left to right

$$\begin{aligned} \int_0^a x^{\frac{1}{2}} \sqrt{a-x} dx &= \int_{x=0}^{x=a} \sqrt{x} \sqrt{a-x} dx \\ &= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sqrt{x} \sqrt{a-x} dx && \text{(changing to } \theta \text{ bounds)} \\ &= \int_{\theta=0}^{\theta=\frac{\pi}{2}} (\sqrt{a} \sin \theta) (\sqrt{a} \cos \theta) (a \sin 2\theta) d\theta && \text{(substituting } \sqrt{x}, \sqrt{a-x}, \text{ and } dx) \\ &= \int_{\theta=0}^{\theta=\frac{\pi}{2}} a \sin \theta \cos \theta \cdot a \sin 2\theta d\theta && \text{(since } \sqrt{a} \sqrt{a} = a) \\ &= a^2 \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sin \theta \cos \theta \sin 2\theta d\theta \end{aligned}$$

Now we simplify the integrand using a double angle formula again. Recall

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Therefore

$$\begin{aligned}
\sin \theta \cos \theta \sin 2\theta &= \sin \theta \cos \theta \cdot 2 \sin \theta \cos \theta \\
&= 2 \sin^2 \theta \cos^2 \theta \\
&= \frac{1}{2} (2 \sin \theta \cos \theta)^2 \\
&= \frac{1}{2} \sin^2 2\theta
\end{aligned}$$

Substituting this back into our integral gives

$$\begin{aligned}
\int_0^a x^{\frac{1}{2}} \sqrt{a-x} \, dx &= a^2 \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sin \theta \cos \theta \sin 2\theta \, d\theta \\
&= a^2 \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{2} \sin^2 2\theta \, d\theta \\
&= \frac{a^2}{2} \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sin^2 2\theta \, d\theta
\end{aligned}$$

To integrate $\sin^2 2\theta$, we use the standard identities

$$\begin{aligned}
\cos 2u &= \cos^2 u - \sin^2 u \\
\sin^2 u + \cos^2 u &= 1
\end{aligned}$$

Which can be combined to deduce

$$\sin^2 u = \frac{1}{2}(1 - \cos 2u)$$

Here $u = 2\theta$, so

$$\sin^2 2\theta = \frac{1}{2}(1 - \cos 4\theta)$$

Substituting this into our integral

$$\begin{aligned}
\frac{a^2}{2} \int_{\theta=0}^{\theta=\frac{\pi}{2}} \sin^2 2\theta \, d\theta &= \frac{a^2}{2} \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{2} (1 - \cos 4\theta) \, d\theta \\
&= \frac{a^2}{4} \int_{\theta=0}^{\theta=\frac{\pi}{2}} (1 - \cos 4\theta) \, d\theta
\end{aligned}$$

Now we can integrate term by term

$$\begin{aligned}
\frac{a^2}{4} \int_{\theta=0}^{\theta=\frac{\pi}{2}} (1 - \cos 4\theta) d\theta &= \frac{a^2}{4} \left[\theta - \frac{1}{4} \sin 4\theta \right]_{\theta=0}^{\theta=\frac{\pi}{2}} && \text{(integrating 1 and } \cos 4\theta) \\
&= \frac{a^2}{4} \left(\frac{\pi}{2} - \frac{1}{4} \sin 4 \cdot \frac{\pi}{2} - \left(0 - \frac{1}{4} \sin 0 \right) \right) && \text{(substituting the bounds)} \\
&= \frac{a^2}{4} \left(\frac{\pi}{2} - \frac{1}{4} \sin 2\pi + \frac{1}{4} \sin 0 \right) \\
&= \frac{a^2}{4} \cdot \frac{\pi}{2} && \text{(since } \sin 2\pi = 0 \text{ and } \sin 0 = 0) \\
&= \frac{\pi a^2}{8}
\end{aligned}$$

Therefore

$$\int_0^a x^{\frac{1}{2}} \sqrt{a-x} dx = \frac{\pi a^2}{8}$$

□

Example 62. Find

$$\int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} dx$$

Solution. As usual, we first ask: can we “just see” an obvious answer which differentiates into the term inside the integral? Because of the $\sin \sqrt{x}$ term, this is not obvious, and it does not match anything in the formula book directly either.

Next, we ask whether we can rearrange this into something simpler. Again, nothing jumps out.

However, the $\sqrt{x}$ inside the sine is a strong hint to try a substitution to simplify the integrand. In particular, if we set the square root equal to a new variable, the inside of the sine becomes much simpler.

Let us define

$$z = \sqrt{x}$$

We now want to rewrite everything in terms of z . Rearranging this gives

$$x = z^2$$

Differentiating both sides with respect to x gives

$$\frac{dz}{dx} = \frac{1}{2\sqrt{x}}$$

Now (abusing notation slightly)¹² we “multiply” both sides by dx to get

$$dz = \frac{1}{2\sqrt{x}} dx$$

Substituting $\sqrt{x} = z$ into this, we obtain

$$dz = \frac{1}{2z} dx \quad \therefore \quad dx = 2z dz$$

Let us collect the key relations:

$$\begin{aligned}
z &= \sqrt{x} \\
x &= z^2 \\
dx &= 2z dz
\end{aligned}$$

¹²This is the usual informal way to think about differentials; it can be made rigorous, but that is beyond our scope here.

We also need to update the integral bounds. When $x = 0$,

$$z = \sqrt{0} = 0$$

and when $x = \frac{\pi^2}{4}$

$$z = \sqrt{\frac{\pi^2}{4}} = \frac{\pi}{2}$$

Now we carefully substitute everything from left to right so that the integral becomes entirely in terms of z :

$$\begin{aligned} \int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx &= \int_{x=0}^{x=\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx \\ &= \int_{z=0}^{z=\frac{\pi}{2}} x \sin \sqrt{x} \, dx && \text{(using } z = \sqrt{x} \text{ for the bounds)} \\ &= \int_{z=0}^{z=\frac{\pi}{2}} z^2 \sin z \, dz && \text{(using } x = z^2 \text{ and } \sqrt{x} = z) \\ &= \int_{z=0}^{z=\frac{\pi}{2}} z^2 \sin z \cdot 2z \, dz && \text{(using } dx = 2z \, dz) \\ &= 2 \int_{z=0}^{z=\frac{\pi}{2}} z^3 \sin z \, dz \end{aligned}$$

So the original integral has now become

$$\int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx = 2 \int_0^{\frac{\pi}{2}} z^3 \sin z \, dz$$

The substitution has done its job: the awkward $\sqrt{x}$ has disappeared, and we are left with a polynomial times a trig function, which is a classic situation for integration by parts (see Section 4.2).

Let us define

$$J = \int_0^{\frac{\pi}{2}} z^3 \sin z \, dz$$

so that our original integral is

$$\int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx = 2J$$

We now tackle J using integration by parts, and we will need to do this repeatedly, noticing that each time the power of z goes down by 1, so we are making progress.

First integration by parts

Following LATE, we treat z^3 as algebra and $\sin z$ as trig, and choose

$$u = z^3, \quad v' = \sin z$$

Differentiating and integrating gives:

$$\begin{aligned} u &= z^3 & v &= -\cos z \\ u' &= 3z^2 & v' &= \sin z \end{aligned}$$

(since $(-\cos z)' = \sin z$).

Recall the by-parts formula for definite integrals:

$$\int_a^b uv' dz = [uv]_a^b - \int_a^b u'v dz$$

Substituting our choices in, we obtain

$$\begin{aligned} J &= \int_0^{\frac{\pi}{2}} z^3 \sin z dz \\ &= [z^3 \cdot (-\cos z)]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 3z^2 \cdot (-\cos z) dz \\ &= [-z^3 \cos z]_0^{\frac{\pi}{2}} + 3 \int_0^{\frac{\pi}{2}} z^2 \cos z dz \end{aligned} \quad (\text{rearranging the integral})$$

The integral bound term on the left simplifies nicely: since $\cos(\frac{\pi}{2}) = 0$ and $\cos(0) = 1$,

$$[-z^3 \cos z]_0^{\frac{\pi}{2}} = \left(-\left(\frac{\pi}{2}\right)^3 \cdot 0 \right) - (-0^3 \cdot 1) = 0.$$

So

$$J = 3 \int_0^{\frac{\pi}{2}} z^2 \cos z dz$$

Let us call the new integral

$$K = \int_0^{\frac{\pi}{2}} z^2 \cos z dz$$

so that $J = 3K$.

Second integration by parts

We now compute K by integration by parts again. This time, we choose

$$u = z^2, \quad v' = \cos z$$

Differentiating and integrating:

$$\begin{aligned} u &= z^2 & v &= \sin z \\ u' &= 2z & v' &= \cos z \end{aligned}$$

(since $(\sin z)' = \cos z$).

By the same formula,

$$\begin{aligned} K &= \int_0^{\frac{\pi}{2}} z^2 \cos z dz \\ &= [z^2 \sin z]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2z \sin z dz \end{aligned}$$

Let us evaluate the boundary term:

$$[z^2 \sin z]_0^{\frac{\pi}{2}} = \left(\left(\frac{\pi}{2}\right)^2 \sin\left(\frac{\pi}{2}\right) \right) - (0^2 \sin 0) = \frac{\pi^2}{4} \cdot 1 - 0 = \frac{\pi^2}{4}$$

So we obtain

$$K = \frac{\pi^2}{4} - 2 \int_0^{\frac{\pi}{2}} z \sin z dz$$

Let us define

$$L = \int_0^{\frac{\pi}{2}} z \sin z \, dz$$

so that $K = \frac{\pi^2}{4} - 2L$.

Third integration by parts

The integral L is simpler but still a product, so we perform integration by parts one more time. Take

$$u = z, \quad v' = \sin z$$

Then

$$\begin{aligned} u &= z & v &= -\cos z \\ u' &= 1 & v' &= \sin z \end{aligned}$$

(again, since $(-\cos z)' = \sin z$).

Applying the formula:

$$\begin{aligned} L &= \int_0^{\frac{\pi}{2}} z \sin z \, dz \\ &= [z \cdot (-\cos z)]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 1 \cdot (-\cos z) \, dz \\ &= [-z \cos z]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos z \, dz \end{aligned}$$

Evaluate each part:

$$[-z \cos z]_0^{\frac{\pi}{2}} = \left(-\frac{\pi}{2} \cos\left(\frac{\pi}{2}\right)\right) - (-0 \cdot \cos 0) = -\frac{\pi}{2} \cdot 0 - 0 = 0$$

and

$$\int_0^{\frac{\pi}{2}} \cos z \, dz = [\sin z]_0^{\frac{\pi}{2}} = \sin\left(\frac{\pi}{2}\right) - \sin(0) = 1 - 0 = 1$$

Hence

$$L = 0 + 1 = 1$$

Putting it all back together

We found:

$$K = \frac{\pi^2}{4} - 2L = \frac{\pi^2}{4} - 2 \cdot 1 = \frac{\pi^2}{4} - 2$$

Then

$$J = 3K = 3\left(\frac{\pi^2}{4} - 2\right) = \frac{3\pi^2}{4} - 6$$

Finally, recall that the original integral is

$$\int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx = 2J$$

so

$$\begin{aligned} \int_0^{\frac{\pi^2}{4}} x \sin \sqrt{x} \, dx &= 2\left(\frac{3\pi^2}{4} - 6\right) \\ &= \frac{3\pi^2}{2} - 12 \end{aligned}$$

Which finishes the question. □

Example 63. Compute the following integral

$$\int_0^1 \frac{2x}{\sqrt{1-x^4}} dx$$

Then state the indefinite integral of $\frac{2x}{\sqrt{1-x^4}}$.

Solution. As usual, we first ask whether we can “see” an obvious answer which differentiates into the term inside the integral

$$\frac{2x}{\sqrt{1-x^4}}$$

There is a $\sqrt{1-x^4}$ in the denominator, and the numerator is $2x$. This looks like something that might appear from the chain rule, but it is not in a form we recognise directly

Next we ask if integration by parts would help. If we tried by parts, there is no obvious way to choose u and v' so that the resulting integral is simpler. For example, if we chose

$$u = 2x, \quad v' = \frac{1}{\sqrt{1-x^4}}$$

then we would need to find an expression for v by integrating $\frac{1}{\sqrt{1-x^4}}$, which is *harder* than the question we started with. So integration by parts is not a good choice here

This suggests we should try substitution. The x^4 inside the square root comes from $(x^2)^2$, and the numerator has a factor of $2x$. This suggests using

$$u = x^2$$

We now work out everything we need in terms of u

$$\begin{aligned} u &= x^2 \\ \frac{du}{dx} &= 2x && \text{(differentiating)} \\ du &= 2x dx \\ \therefore 2x dx &= du \end{aligned}$$

We also change the bounds from x to u

$$\begin{aligned} x = 0 &\therefore u = 0^2 = 0 \\ x = 1 &\therefore u = 1^2 = 1 \end{aligned}$$

Now we substitute, step by step, from left to right

$$\begin{aligned}
\int_0^1 \frac{2x}{\sqrt{1-x^4}} dx &= \int_{x=0}^{x=1} \frac{2x}{\sqrt{1-x^4}} dx \\
&= \int_{u=0}^{u=1} \frac{2x}{\sqrt{1-(x^2)^2}} dx && \text{(using } u = x^2\text{)} \\
&= \int_{u=0}^{u=1} \frac{2x}{\sqrt{1-u^2}} dx && \text{(substituting } x^2 = u \text{ inside the square root)} \\
&= \int_{u=0}^{u=1} \frac{1}{\sqrt{1-u^2}} du && \text{(using } 2x dx = du\text{)}
\end{aligned}$$

Now our integral is

$$\int_0^1 \frac{1}{\sqrt{1-u^2}} du$$

This is a standard form which you might recognise from the formula book if you have looked at the further maths section as the derivative of $\sin^{-1} u$. Without access to this knowledge, we can still proceed using a second substitution. The integrand has the form $\frac{1}{\sqrt{1-u^2}}$, which suggests a trigonometric substitution. One way to reason through this is to recall that $1 - \sin^2 x = \cos^2 x$, therefore using $u = \sin \theta$ will result in some simplification here.

We choose

$$u = \sin \theta$$

Then we work out everything we need in terms of θ

$$\begin{aligned}
u &= \sin \theta \\
\frac{du}{d\theta} &= \cos \theta && \text{(differentiating)} \\
du &= \cos \theta d\theta \\
1 - u^2 &= 1 - \sin^2 \theta = \cos^2 \theta && \text{(using } \sin^2 \theta + \cos^2 \theta = 1\text{)} \\
\sqrt{1 - u^2} &= \sqrt{\cos^2 \theta} = \cos \theta && \text{(on } 0 \leq \theta \leq \frac{\pi}{2}, \cos \theta \geq 0\text{)}
\end{aligned}$$

We also convert the bounds from u to θ

$$\begin{aligned}
u = 0 &\therefore \sin \theta = 0 \therefore \theta = 0 \\
u = 1 &\therefore \sin \theta = 1 \therefore \theta = \frac{\pi}{2}
\end{aligned}$$

Now we substitute everything into the integral, again from left to right

$$\begin{aligned}
\int_{u=0}^{u=1} \frac{1}{\sqrt{1-u^2}} du &= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{\sqrt{1-u^2}} du && \text{(using } u = \sin \theta \text{)} \\
&= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{\sqrt{1-\sin^2 \theta}} du && \\
&= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{\cos \theta} du && \text{(using } \sqrt{1-\sin^2 \theta} = \cos \theta \text{)} \\
&= \int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{1}{\cos \theta} \cdot \cos \theta d\theta && \text{(using } du = \cos \theta d\theta \text{)} \\
&= \int_{\theta=0}^{\theta=\frac{\pi}{2}} 1 d\theta && \text{(simplifying)}
\end{aligned}$$

This is now very simple to integrate

$$\begin{aligned}
\int_{\theta=0}^{\theta=\frac{\pi}{2}} 1 d\theta &= [\theta]_{\theta=0}^{\theta=\frac{\pi}{2}} \\
&= \frac{\pi}{2} - 0 \\
&= \frac{\pi}{2}
\end{aligned}$$

Therefore

$$\int_0^1 \frac{2x}{\sqrt{1-x^4}} dx = \frac{\pi}{2}$$

Now we state the indefinite integral. Looking back at our first substitution, we had

$$\int \frac{2x}{\sqrt{1-x^4}} dx = \int \frac{1}{\sqrt{1-u^2}} du = \int 1 d\theta = \theta + c$$

Unravelling, we use that $u = \sin \theta$ to write that $\theta = \sin^{-1} u$, and substitute into the above to find that

$$\int \frac{2x}{\sqrt{1-x^4}} dx = \sin^{-1} u + c$$

Now we substitute back $u = x^2$

$$\int \frac{2x}{\sqrt{1-x^4}} dx = \sin^{-1}(x^2) + c$$

You can check this by differentiating $\sin^{-1}(x^2)$ (e.g. using the formula book in the further maths section) and confirming that you get $\frac{2x}{\sqrt{1-x^4}}$

□

Example 64. Compute the following indefinite integral.

$$\int x e^{x^2} dx$$

Solution. If you are rushing, you might immediately try to jump in with integration by parts or substitution. Integration by parts would lead you to needing to integrate e^{x^2} , which you might try a substitution of $u = x^2$ to try to simplify, but you find that $dx = \frac{1}{2x}du$, and your integral ends up becoming even more complicated.

Instead – slowing down – we can try to “just integrate” it.

Since we know that (by the chain rule Section 3.4) differentiating an exponential gives the same exponential times something, we know our integral needs to have e^{x^2} as one of the terms to get something that differentiates to give xe^{x^2} .

Let’s try differentiating e^{x^2} and see if it is close to xe^{x^2} . We can do this by using the chain rule, see Section 3.4.

$$\begin{aligned}(e^{f(x)})' &= f'(x)e^{f(x)} && \text{(using the chain rule)} \\ &= 2xe^{x^2} && \text{(using } f(x) = x^2\text{)}\end{aligned}$$

This is close to our desired xe^{x^2} , so we can divide both sides by 2 to get

$$\left(\frac{1}{2}e^{x^2}\right)' = xe^{x^2}$$

This gives us a function that differentiates to give the function inside the integral, which is exactly what we are looking for.

Therefore, the indefinite integral of xe^{x^2} is, using $f(x) = x^2$,

$$\int xe^{x^2} dx = \frac{1}{2}e^{x^2} + c$$

Where c is a constant. □

Example 65. Compute the following indefinite integral.

$$\int \frac{5x}{1+2x^2} dx$$

Solution. If you are rushing, you might immediately try to jump in with integration by parts or substitution. Alternatively, one might try to factorise the denominator and use partial fractions. The most likely successful approach of those is substitution, but it would lead to a lengthy computation.

Instead – slowing down – we can try to “just integrate” it.

Since we know that (by the chain rule Section 3.4) differentiating $\ln(f(x))$ gives something times $\frac{1}{f(x)}$, we might guess our integral needs to have $\ln(1+2x^2)$ as one of the terms to get something that differentiates to give $\frac{5x}{1+2x^2}$.

Let’s try differentiating $\ln(1+2x^2)$ and see if it is close to $\frac{5x}{1+2x^2}$. We can do this by using the chain rule, see Section 3.4.

$$\begin{aligned}(\ln(f(x)))' &= \frac{f'(x)}{f(x)} && \text{(using the chain rule)} \\ &= \frac{4x}{1+2x^2} && \text{(using } f(x) = 1+2x^2\text{)}\end{aligned}$$

This is close to our desired $\frac{5x}{1+2x^2}$, except we have a factor of 4 instead of the 5 that we want. We can multiply both sides by $\frac{5}{4}$ to convert our 4 to a 5 and find that

$$\left(\frac{5}{4}\ln(1+2x^2)\right)' = \frac{5x}{1+2x^2}$$

This gives us a function that differentiates to give the function inside the integral, which is exactly what we are looking for.

Therefore, the indefinite integral of $\frac{5x}{1+2x^2}$ is, using $f(x) = 1 + 2x^2$,

$$\int \frac{5x}{1+2x^2} dx = \frac{5}{4}\ln(1+2x^2) + c$$

Where c is a constant. □

5 Kinematics

5.1 Displacement, Velocity, and Acceleration

I will avoid using SUVAT equations immediately, since they hide a lot of the fundamentals.

We can relate displacement, velocity, and acceleration as functions of time t via integration and differentiation. This is by recalling that velocity is the rate of change of displacement, and acceleration is the rate of change of velocity, so we can write:

$$v(t) = \frac{ds}{dt}$$
$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

And you can go the ‘other way’ via integration:

$$v(t) = \int a(t)dt + c_1$$
$$s(t) = \int v(t)dt + c_2$$

Where we can find the constants of integration c_1 and c_2 if we know some initial conditions, e.g. the starting velocity and the starting position.

Let’s say we have a particle moving with **constant acceleration** a , with an initial velocity u and initial displacement s_0 .

We can start with the acceleration and integrate¹³ to find:

$$v(t) = \int a dt + c = at + c_1$$

And use that $v(0) = u$ to find:

$$v(0) = u$$
$$= a \cdot 0 + c_1$$
$$\therefore c_1 = u$$
$$\therefore v(t) = at + u$$

Now we can integrate again to find the displacement:

$$s(t) = \int v(t)dt + c_2 = \int (at + u)dt + c_2 = \frac{1}{2}at^2 + ut + c_2$$

And use that $s(0) = s_0$ to find:

$$s(0) = s_0$$
$$= \frac{1}{2}a \cdot 0^2 + u \cdot 0 + c_2$$
$$\therefore c_2 = s_0$$
$$\therefore s(t) = \frac{1}{2}at^2 + ut + s_0$$
$$\therefore s(t) - s_0 = ut + \frac{1}{2}at^2$$

¹³It is essential to understand that this integral relies on a being constant in t .

We can write this in terms of v instead of u by substituting the first equation $v = at + u$ to get:

$$\begin{aligned} s(t) - s_0 &= ut + \frac{1}{2}at^2 \\ &= (v - at)t + \frac{1}{2}at^2 \\ &= vt - at^2 + \frac{1}{2}at^2 \\ &= vt - \frac{1}{2}at^2 \end{aligned}$$

Now we can notice that the last two equations have a similar term $\pm \frac{1}{2}at^2$, so we can add them together to get:

$$\begin{aligned} 2(s(t) - s_0) &= ut + vt + \cancel{\frac{1}{2}at^2} - \cancel{\frac{1}{2}at^2} \\ 2(s(t) - s_0) &= t(u + v) \\ \therefore s(t) - s_0 &= \frac{t(u + v)}{2} \end{aligned}$$

All of the last equations contain t so it would be useful to eliminate t ¹⁴. If we assume $a \neq 0$, we can rearrange the first equation to get $t = \frac{v-u}{a}$, and substitute this into the last equation to get:

$$\begin{aligned} s(t) - s_0 &= \frac{t(u + v)}{2} \\ &= \frac{\frac{v-u}{a}(u + v)}{2} \\ &= \frac{(v - u)(u + v)}{2a} \\ &= \frac{v^2 - u^2}{2a} \\ \therefore v^2 &= u^2 + 2a(s - s_0) \end{aligned}$$

These combine to give us our SUVAT equations from the formula book, explaining motion in a straight line with constant acceleration a , with initial velocity u , initial displacement s_0 , and displacement $s(t)$ and velocity $v(t)$ at time t :

$$\begin{aligned} v(t) &= u + at \\ s(t) - s_0 &= ut + \frac{1}{2}at^2 \\ s(t) - s_0 &= v(t)t - \frac{1}{2}at^2 \\ s(t) - s_0 &= \frac{t(u + v(t))}{2} \\ v(t)^2 &= u^2 + 2a(s(t) - s_0) \end{aligned}$$

In particular, if $s_0 = 0$ ¹⁵, i.e. we are measuring relative to the starting point, and we set v as our final velocity and s as our displacement (ignoring the time variable), then we get the SUVAT

¹⁴The neatest way to do this is via chain rule and writing $a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = v \frac{dv}{ds}$, but we can also do it via substitution, which I suspect students will find simpler to understand.

¹⁵This highlights that the SUVAT equations describe displacement from the starting position, rather than absolute position relative to a fixed origin.

equations from the formula booklet:

$$\begin{aligned}v &= u + at \\s &= ut + \frac{1}{2}at^2 \\s &= vt - \frac{1}{2}at^2 \\s &= \frac{t(u + v)}{2} \\v^2 &= u^2 + 2as\end{aligned}$$

I highly recommend understanding where the equations come from, especially relating to differentiating and integrating acceleration, velocity, and displacement, rather than just blindly using the equations.

Despite this, the equations can still speed up calculations, since they always allow you to find the missing variable without needing to do any differentiation or integration. This is because each equation has 4 of the variables s , u , v , a , and t , so if you know any 3 of these variables, you can find the missing variable by using the appropriate SUVAT equation.

The only place you might get tripped up is forgetting that s is relative to the starting point, in the case where you are not starting from $s_0 = 0$.

Example 66. A particle starts from rest and moves with constant acceleration a for t seconds, covering a displacement of s metres.

Find an equation for the final velocity v of the particle in terms of s and t , assuming $t \neq 0$.

You are told that the particle covers 5 metres before reaching a final velocity of 2 m/s. Find what time t is at the time of reaching the final velocity, and then deduce what the acceleration a is.

Solution. Since the particle starts from rest, we have $u = 0$. We are asked to find an equation for v in terms of s and t and we know what u is, so let's look for a SUVAT equation which has v , s , t , and u . The only equation which has all of these variables is the fourth equation, so we can rearrange this to get:

$$\begin{aligned}s &= \frac{t(u + v)}{2} \\2s &= t(u + v) \\2s &= t(0 + v) \\2s &= tv \\\therefore v &= \frac{2s}{t} \text{ m/s}\end{aligned}$$

We are told that $v = 2$ m/s after the object has covered 5 metres, so we can substitute this into the equation we just found to get:

$$\begin{aligned}2 &= \frac{2 \cdot 5}{t} \\t &= 5 \text{ seconds}\end{aligned}$$

Now we can use the first SUVAT equation to find a :

$$\begin{aligned}v &= u + at \\2 &= 0 + a \cdot 5 \\a &= \frac{2}{5} \text{ m/s}^2\end{aligned}$$

□

Example 67. A particle starts with velocity 6 m/s and moves with constant acceleration a , covering a distance of 32 metres before reaching a velocity v .

Find an equation for the acceleration a in terms of v .

You are told the particle reaches a final velocity of 10 m/s. Find the acceleration a , and then deduce the time t taken.

Solution. We know that $u = 6$ and $s = 32$. We are asked to find an equation for a in terms of v , so we look for a SUVAT equation involving a , v , u , and s . The only equation with these variables is:

$$v^2 = u^2 + 2as$$

Rearranging this to make a the subject gives:

$$\begin{aligned} v^2 &= u^2 + 2as \\ v^2 - u^2 &= 2as \\ a &= \frac{v^2 - u^2}{2s} \end{aligned}$$

Substituting $u = 6$ and $s = 32$, we get:

$$\begin{aligned} a &= \frac{v^2 - 6^2}{2 \cdot 32} \\ &= \frac{v^2 - 36}{64} \end{aligned}$$

So the required equation is:

$$a = \frac{v^2 - 36}{64} \text{ m/s}^2$$

We are told that $v = 10$ m/s, so:

$$\begin{aligned} a &= \frac{10^2 - 36}{64} \\ &= \frac{100 - 36}{64} \\ &= \frac{64}{64} \\ &= 1 \text{ m/s}^2 \end{aligned}$$

Now we use the first SUVAT equation to find the time:

$$\begin{aligned} v &= u + at \\ 10 &= 6 + 1 \cdot t \\ t &= 4 \text{ seconds} \end{aligned}$$

Hence the five variables are:

$$u = 6, \quad v = 10, \quad s = 32, \quad a = 1, \quad t = 4$$

□

Example 68. A particle moves with constant acceleration a , changing its velocity from u m/s to 14 m/s in 6 seconds, while travelling a distance s .

Find an equation for the displacement s in terms of u .

You are told that the initial velocity is 2 m/s. Find the displacement s , and then deduce the acceleration a .

Solution. We know that $v = 14$ and $t = 6$. We are asked to find an equation for s in terms of u , so we look for a SUVAT equation involving s , u , v , and t . The only equation with these variables is:

$$s = \frac{t(u + v)}{2}$$

Substituting $v = 14$ and $t = 6$, we get:

$$\begin{aligned} s &= \frac{6(u + 14)}{2} \\ &= 3(u + 14) \end{aligned}$$

So the required equation is:

$$s = 3(u + 14) \text{ metres}$$

We are told that $u = 2$ m/s, so:

$$\begin{aligned} s &= 3(2 + 14) \\ &= 3 \cdot 16 \\ &= 48 \text{ metres} \end{aligned}$$

Now we use the first SUVAT equation to find the acceleration:

$$\begin{aligned} v &= u + at \\ 14 &= 2 + a \cdot 6 \\ 12 &= 6a \\ a &= 2 \text{ m/s}^2 \end{aligned}$$

Hence the five variables are:

$$u = 2, \quad v = 14, \quad t = 6, \quad s = 48, \quad a = 2$$

□

Example 69. A particle starts at position $s_0 = 4$ metres with initial velocity $u = 3$ m/s and moves with constant acceleration $a = 2$ m/s² for t seconds, reaching position s .

Find the position s after 5 seconds.

Solution. Since the particle does not start at the origin, we must be careful to use the SUVAT equation for displacement relative to the starting position:

$$s - s_0 = ut + \frac{1}{2}at^2$$

We are given:

$$s_0 = 4, \quad u = 3, \quad a = 2, \quad t = 5$$

So substituting into the equation gives:

$$\begin{aligned} s - 4 &= 3 \cdot 5 + \frac{1}{2} \cdot 2 \cdot 5^2 \\ s - 4 &= 15 + 25 \\ s - 4 &= 40 \\ s &= 44 \end{aligned}$$

Therefore the position of the particle after 5 seconds is:

$$s = 44 \text{ metres}$$

□

Example 70. A particle at rest which has a mass of 2 kg has a force applied to it of 4 N.

What is the acceleration of the particle?

Deduce an equation for the velocity v of the particle in terms of time t .

Solution. We know that the force F and mass m are related to acceleration a by Newton's second law:

$$F = ma$$

Substituting $F = 4$ N and $m = 2$ kg, we get:

$$\begin{aligned}4 &= 2a \\ a &= 2 \text{ m/s}^2\end{aligned}$$

Since the particle starts from rest, the initial velocity $u = 0$. Using the first SUVAT equation:

$$\begin{aligned}v &= u + at \\ v &= 0 + 2t \\ v &= 2t\end{aligned}$$

So the velocity of the particle in terms of time t is:

$$v = 2t \text{ m/s}$$

□

Example 71. A particle has a mass of 2 kg and has been moving at a constant speed of 3 m/s. At time $t = 0$ seconds, a force of $4t$ N is applied opposing the motion.

What is the acceleration of the particle in terms of time t ?

Deduce an equation for the velocity v of the particle in terms of time t .

At what time t does the particle come to rest?

Solution. Let's set our positive direction to be in the direction of the initial motion. Using Newton's second law, we have:

$$\begin{aligned}F &= ma \\ -4t &= 2a \\ a &= -2t \text{ m/s}^2\end{aligned}$$

Since the particle is moving at a constant speed of 3 m/s before the force is applied, we have $u = 3$.

Now the question is trying to trick us; we do not have constant acceleration here. Instead we must remember where the equations came from:

$$v(t) = \int a(t) dt + c$$

Substituting $a(t) = -2t$ gives:

$$\begin{aligned}v(t) &= \int -2t dt + c \\ &= -t^2 + c\end{aligned}$$

Since the initial velocity is $u = 3$ m/s, we have $v(0) = 3$, so $c = 3$. Therefore:

$$v(t) = -t^2 + 3$$

The particle comes to rest when $v(t) = 0$, so we can solve the equation:

$$\begin{aligned} 0 &= -t^2 + 3 \\ t^2 &= 3 \\ t &= \sqrt{3} \text{ seconds} \end{aligned}$$

We used the positive root since the force was applied at $t = 0$ seconds and the particle comes to rest at a later time, so t must be positive.

□

Example 72. A star is being consumed by a black hole at a large distance. The star is losing mass, and its mass is described by $\frac{M}{t+1}$ kg at time t seconds, where M is the initial mass of the star. The black hole is exerting a force of $100t^2$ N on the star, and the star is accelerating towards the black hole.

Assume for this question that, despite the changing mass, the motion may be modelled using $F = m(t)a(t)$ ¹⁶.

Modelling the star as a particle, find an equation for the acceleration a of the star in terms of time t and initial mass M .

Assuming the star starts from rest, find an equation for the velocity v of the star in terms of time t and initial mass M .

Solution. Let's choose the direction towards the black hole as positive. Using Newton's second law, we have:

$$\begin{aligned} F &= ma \\ 100t^2 &= \frac{M}{t+1}a \\ a &= \frac{100t^2(t+1)}{M} \\ a &= \frac{100t^3 + 100t^2}{M} \text{ m/s}^2 \end{aligned}$$

Since the star starts from rest, the initial velocity $u = 0$. Using the definition of acceleration:

$$\begin{aligned} v(t) &= \int a(t) dt + c \\ &= \int \frac{100t^3 + 100t^2}{M} dt + c \\ &= \frac{100t^4}{4M} + \frac{100t^3}{3M} + c \\ &= \frac{25t^4}{M} + \frac{100t^3}{3M} + c \end{aligned}$$

Since $v(0) = 0$, we have

$$\begin{aligned} 0 &= \frac{25 \cdot 0^4}{M} + \frac{100 \cdot 0^3}{3M} + c \\ \therefore c &= 0 \end{aligned}$$

Therefore:

$$v(t) = \frac{25t^4}{M} + \frac{100t^3}{3M}$$

□

¹⁶More generally, Newton's second law is $\mathbf{F}_{\text{ext}} = \frac{d\mathbf{p}}{dt}$, where $\mathbf{p} = m\mathbf{v}$ is the momentum. When the mass is constant, this reduces to $\mathbf{F}_{\text{ext}} = m\mathbf{a}$. For systems with changing mass, extra care is needed, so in this question we explicitly assume that the simpler form $F = m(t)a(t)$ may be used.

5.2 Additional Exercises

Sometimes the motion is not obviously in a straight line, so we need to be careful to apply the SUVAT equations in the correct direction. This often involves finding components of forces or velocities, as in Section 2, and applying SUVAT in each component separately.

Example 73. A cannon fires a 10 kg cannonball with initial speed 20 m/s at an angle of 20° above the horizontal from a point 50 metres above the ground.

Model the cannonball as a particle. Ignore air resistance, and assume that after firing, the only force acting on the cannonball is gravity.

Find the time taken for the cannonball to hit the ground.

Hence find how far from the cannon the ball lands horizontally.

Finally, state what the actual distance from the cannon to the landing point is. Hint: use Pythagoras' theorem.

Solution. We resolve the initial velocity into horizontal and vertical components.

The horizontal component is:

$$u_x = 20 \cos(20^\circ)$$

The vertical component is:

$$u_y = 20 \sin(20^\circ)$$

Since gravity acts vertically downward, the vertical acceleration is:

$$a_y = -9.8 \text{ m/s}^2$$

Let the ground be at height $y = 0$, and let the cannon fire from height $y_0 = 50$ metres.

Using the SUVAT equation for vertical displacement relative to the starting point,

$$y - y_0 = u_y t + \frac{1}{2} a_y t^2$$

we get

$$\begin{aligned} 0 - 50 &= (20 \sin 20^\circ)t + \frac{1}{2}(-9.8)t^2 \\ -50 &= 20 \sin(20^\circ)t - 4.9t^2 \end{aligned}$$

Rearranging gives:

$$4.9t^2 - 20 \sin(20^\circ)t - 50 = 0$$

Using the quadratic formula,

$$t = \frac{20 \sin(20^\circ) \pm \sqrt{(20 \sin(20^\circ))^2 - 4(4.9)(-50)}}{2(4.9)}$$

Now,

$$20 \sin(20^\circ) \approx 6.84$$

so

$$\begin{aligned} t &= \frac{6.84 \pm \sqrt{6.84^2 + 980}}{9.8} \\ &= \frac{6.84 \pm \sqrt{1026.79}}{9.8} \\ &= \frac{6.84 \pm 32.04}{9.8} \end{aligned}$$

This gives two solutions:

$$t \approx \frac{6.84 + 32.04}{9.8} = 3.97$$
$$t \approx \frac{6.84 - 32.04}{9.8} = -2.57$$

Since time cannot be negative in this context, we reject the negative solution. Therefore the cannonball hits the ground after

$$t \approx 3.97 \text{ seconds}$$

Now we find how far the ball travels horizontally in this time.

Since there is no horizontal acceleration, the horizontal motion has constant velocity:

$$x = u_x t$$

So

$$x = 20 \cos(20^\circ) \times 3.97$$

Now,

$$20 \cos(20^\circ) \approx 18.79$$

so

$$x \approx 18.79 \times 3.97$$
$$\approx 74.6$$

Therefore the cannonball lands

$$74.6 \text{ metres}$$

from the cannon horizontally.

To find the actual distance, we use that the vertical distance travelled is 50 metres along with the horizontal distance 74.6 metres. This forms a right angled triangle so we can use Pythagoras' theorem to find the actual distance d from the cannon:

$$d = \sqrt{74.6^2 + 50^2} \approx 89.8 \text{ metres}$$

□

Example 74. A cannon fires a 10 kg cannonball with initial speed 20 m/s at an angle of θ to the horizontal from ground level.

Model the cannonball as a particle. Ignore air resistance, and assume that after firing, the only force acting on the cannonball is gravity.

Find, in terms of θ , the time taken for the cannonball to hit the ground.

Hence find, in terms of θ , how far from the cannon the ball lands horizontally.

Finally, find the value of θ which maximises this horizontal distance.

Hint: Differentiate.

Solution. We resolve the initial velocity into horizontal and vertical components.

The horizontal component is:

$$u_x = 20 \cos \theta$$

The vertical component is:

$$u_y = 20 \sin \theta$$

Since the cannonball is fired from the ground and lands back on the ground, its initial and final vertical positions are both 0.

The vertical acceleration is

$$a_y = -9.8 \text{ m/s}^2$$

Using the SUVAT equation for vertical displacement,

$$s = ut + \frac{1}{2}at^2$$

we get

$$\begin{aligned} 0 &= (20 \sin \theta)t + \frac{1}{2}(-9.8)t^2 \\ 0 &= 20 \sin \theta t - 4.9t^2 \end{aligned}$$

Factorising gives:

$$0 = t(20 \sin \theta - 4.9t)$$

So either

$$t = 0$$

or

$$20 \sin \theta - 4.9t = 0$$

The solution $t = 0$ corresponds to the instant the cannonball is fired, so the time taken to hit the ground is:

$$\begin{aligned} t &= \frac{20 \sin \theta}{4.9} \\ &= \frac{200 \sin \theta}{49} \end{aligned}$$

Therefore the time of flight is

$$t = \frac{200 \sin \theta}{49} \text{ seconds}$$

Now we find the horizontal distance travelled.

Since there is no horizontal acceleration, horizontal motion has constant velocity:

$$x = u_x t$$

So

$$\begin{aligned} x &= (20 \cos \theta) \left(\frac{200 \sin \theta}{49} \right) \\ &= \frac{4000 \sin \theta \cos \theta}{49} \end{aligned}$$

Using the identity

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

we can write this as

$$x = \frac{2000}{49} \sin(2\theta)$$

Therefore the horizontal distance travelled is

$$x = \frac{4000 \sin \theta \cos \theta}{49} = \frac{2000}{49} \sin(2\theta) \text{ metres}$$

To maximise this distance, we differentiate with respect to θ ¹⁷. Since

$$x = \frac{2000}{49} \sin(2\theta),$$

we have

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{2000}{49} \cdot 2 \cos(2\theta) \\ &= \frac{4000}{49} \cos(2\theta) \end{aligned}$$

For a maximum, we set

$$\begin{aligned} \frac{dx}{d\theta} &= 0 \\ \cos(2\theta) &= 0 \end{aligned}$$

So

$$2\theta = \frac{\pi}{2}$$

and hence

$$\theta = \frac{\pi}{4} = 45^\circ$$

Therefore the value of θ which maximises the horizontal distance is

$$45^\circ$$

□

Example 75. A tortoise and a hare move along the same straight track.

At time $t = 0$, the tortoise is 10 metres ahead of the hare and is moving at a constant speed of 3 m/s.

At the same instant, the hare is moving in the same direction with speed 10 m/s and constant deceleration 2 m/s².

Show that the tortoise and the hare meet twice after $t = 0$.

Find the distance travelled by the tortoise between the two meeting points.

Solution. Let the hare's starting point be taken as the origin.

We first find an equation for the position of the tortoise.

At time $t = 0$, the tortoise is already 10 metres ahead of the hare, so its initial position is

$$s_0 = 10$$

¹⁷Strictly speaking, differentiation assumes θ is measured in radians. The final maximising angle is still 45° .

It is moving at constant speed 3 m/s, so:

$$u = 3, \quad a = 0$$

Since we know the initial position, initial velocity, acceleration, and time, the most useful SUVAT equation is:

$$s - s_0 = ut + \frac{1}{2}at^2$$

Substituting in the values for the tortoise gives:

$$\begin{aligned} s_T - 10 &= 3t + \frac{1}{2}(0)t^2 \\ s_T &= 10 + 3t \end{aligned}$$

Now we find an equation for the position of the hare.

At time $t = 0$, the hare is at the origin, so:

$$s_0 = 0$$

Its initial speed is 10 m/s and its acceleration is

$$a = -2 \text{ m/s}^2$$

since it is decelerating.

Again using

$$s - s_0 = ut + \frac{1}{2}at^2$$

we get:

$$\begin{aligned} s_H - 0 &= 10t + \frac{1}{2}(-2)t^2 \\ s_H &= 10t - t^2 \end{aligned}$$

The tortoise and hare meet when they have the same position, so we set

$$s_T = s_H$$

Hence:

$$10 + 3t = 10t - t^2$$

Rearranging gives:

$$t^2 - 7t + 10 = 0$$

Factorising:

$$(t - 2)(t - 5) = 0$$

So:

$$t = 2 \quad \text{or} \quad t = 5$$

Both values are positive, so the tortoise and the hare meet twice after $t = 0$.

Now we find the distance travelled by the tortoise between the two meeting points.

We know that the tortoise's position is

$$s_T = 10 + 3t$$

At the first meeting point, when $t = 2$:

$$\begin{aligned} s_T &= 10 + 3(2) \\ &= 16 \end{aligned}$$

At the second meeting point, when $t = 5$:

$$\begin{aligned} s_T &= 10 + 3(5) \\ &= 25 \end{aligned}$$

So the distance travelled by the tortoise between the two meeting points is:

$$25 - 16 = 9$$

Therefore the tortoise travels

9 metres

between the two meeting points.

□

6 Probability and Statistics

6.1 Averages

Let $x_1, \dots, x_n$ be numeric recorded data values.

Definition 6.1. The mean (commonly referred to as average) of $x_1, \dots, x_n$ is:

$$\bar{x} = \frac{x_1 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Note that

$$\bar{x} = \frac{1}{n}x_1 + \dots + \frac{1}{n}x_n$$

A weighted average is an average in which some of the data is given higher or lower priority. More rigorously, if we have weights $w_1, \dots, w_n$ where each w_i is between 0 and 1, and $w_1 + \dots + w_n = 1$, then

Definition 6.2. A weighted average of $x_1, \dots, x_n$ using weights $w_1, \dots, w_n$ is defined as:

$$\bar{x} = w_1x_1 + \dots + w_nx_n = \sum_{i=1}^n w_ix_i$$

Note that the mean can be viewed as a weighted average with each weight being $\frac{1}{n}$.

What weights should we choose?

An important use-case of weighted averages is the “correct” way to take averages of averages (or percentages). If we **have two averages already** $\bar{x}$ and $\bar{y}$, and the **number of records (frequency) used to compute those averages**, n and m , as in

$$\bar{x} = \frac{x_1 + \dots + x_n}{n}$$
$$\bar{y} = \frac{y_1 + \dots + y_m}{m}$$

Then we can compute the average of both via the formula for a weighted average of averages:

$$\bar{z} = \frac{n\bar{x} + m\bar{y}}{n + m}$$

Where $\bar{z}$ is the average for the row-level numeric data values for x and y combined (union).

Note that this is the same as

$$\bar{z} = \frac{n}{n+m}\bar{x} + \frac{m}{n+m}\bar{y}$$

Where you can notice that $\frac{n}{n+m} \times 100$ and $\frac{m}{n+m} \times 100$ are the percentages of the total number of records!

Why this formula? We know that

$$\frac{n\bar{x} + m\bar{y}}{n+m} = \frac{\cancel{n} \frac{x_1 + \dots + x_n}{\cancel{n}} + \cancel{m} \frac{y_1 + \dots + y_m}{\cancel{m}}}{n+m} = \frac{x_1 + \dots + x_n + y_1 + \dots + y_m}{n+m}$$

Which is exactly the formula for the average of the combined data field of x and y ! Therefore, this is the correct way to take averages of averages.

This is **not** equal to

$$\text{naive average} = \frac{\bar{x} + \bar{y}}{2}$$

Note that similar logic can be applied for when you have more than two averages.

6.2 Distributions

A probability distribution is a function which satisfies two things:

1. The area underneath the function is equal to 1.
2. The function is always non-negative.

Then we can interpret the function in a probabilistic sense in the following way. The area between two points is how ‘likely’ that sampling a random variable with that probability distribution results in a value from that range.

For example, the normal distribution has this property, where the integral of the entire distribution is 1, and it is always positive.

6.2.1 Normal Distribution

The normal distribution is a probability distribution characterised by two numbers: the mean μ and the variance σ^2 . Note that the variance is denoted with a square to remind us that taking the (positive) square root gives us the standard deviation σ . It is denoted $N(\mu, \sigma^2)$.

The normal distribution has the shape of a bell curve around the mean, which can be interpreted as in Section 6.2 that the majority of samples will occur around the mean, and that the further away you get from the mean, the less likely a sample is to come from that area. This can be seen visually by which parts of the bell curve have more area underneath, and therefore have a higher likelihood of occurring.

For example, let’s say that human adult weights can be modelled as a normal distribution. We sketch it as a bell curve, symmetric around 75 as in Figure 34.

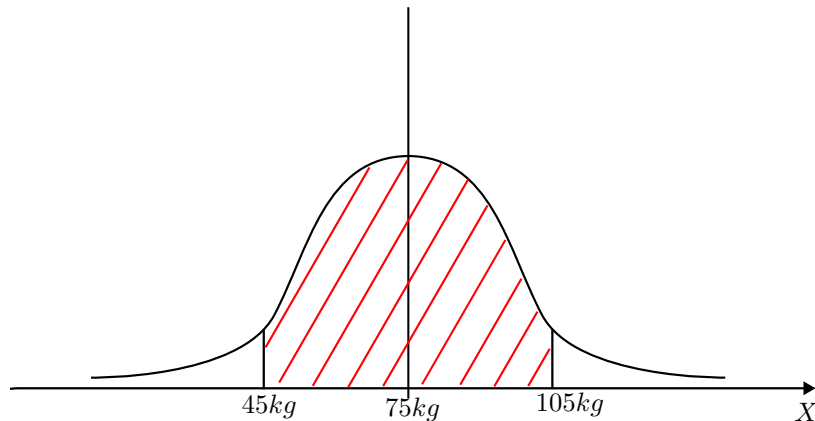


Figure 34: A sketch of the normal distribution where $\mu = 75$, $\sigma = 30$

Since it is symmetric, we know that the probability that a random sample X is 75 kg or below is 0.5 – this can be denoted as

$$P(X \leq 75) = 0.5$$

In fact, it is always true for normal distributions that

$$P(X \leq \mu) = 0.5$$

This is because $P(X \leq \mu) = P(X \geq \mu)$, and $P(X \leq \mu) + P(X \geq \mu) = 1$, therefore

$$P(X \geq \mu) = 1 - P(X \leq \mu)$$

Substituting back into the first equation gives that $P(X \leq \mu) = 1 - P(X \leq \mu)$, which can be solved to find $P(X \leq \mu) = 0.5$.

Moreover, due to symmetries and summing to 1, we can figure out some other fun things:

$$\begin{aligned} P(45 \leq X \leq 105) &= 1 - (P(X < 45) + P(X > 105)) && \text{(the full area is 1)} \\ &= 1 - 2P(X < 45) && \text{(symmetry around the mean)} \end{aligned}$$

All of the above are using symmetry around the mean! Again, this is true for any normal distribution – you can use symmetries to convert to other regions of the bell curve:

Exercise 7. Consider the normal distribution $N(\mu, \sigma^2)$. Using symmetry and the area being 1, show the following equalities:

$$P(X \leq a) = P(X \geq 2\mu - a) = 1 - P(X > a) = 1 - P(X < 2\mu - a)$$

Have a think if you can reason your way through any of the above! Hint: what is the midpoint of a and $2\mu - a$? What do these equalities become when $\mu = 0$ and $\sigma = 1$?

In your exam cheat sheet, you get told a lot of values for probabilities of the form $P(X > a)$, where X comes from a specific normal distribution called the standard normal distribution.

Definition 6.3. The *standard normal distribution* is a normal distribution with $\mu = 0$ and $\sigma^2 = 1$, denoted $N(0, 1)$.

Using symmetries and the area being 1, you can always rearrange any probability question to be of the form $P(X < a)$. Then you can look up the nearest value from the cheat sheet to use for your probability.

The problem is that often you are not given a normal distribution which has this mean and variance, so how can you use this cheat sheet?

It turns out that transforming X by shifting and rescaling gives us a new normal distribution with the mean shifted and the standard deviation rescaled. This can be understood intuitively if you understand that, for a normal distribution, the mean represents the centre of the distribution, and the standard deviation represents the spread of the distribution. It turns out that 68% of random samples will occur within one standard deviation of the mean, 95% of random samples will occur within two standard deviations of the mean, and 99.7% of random samples will occur within three standard deviations of the mean. This gives a visual interpretation of the mean being the middle and the standard deviation being the width.

So our goal is to think of a way to transform X so that the mean is shifted to become 0 and the standard deviation is scaled to be 1. A little bit of thought would lead us to the formula

$$Z = \frac{X - \mu}{\sigma}$$

Which will shift the mean μ to 0 (you can substitute $X = \mu$ to see this), and will make the entire thing “thinner” by a factor of σ .

It can be proven that this transformation indeed converts $N(\mu, \sigma^2)$ to the standard normal distribution $N(0, 1)$. So the process in a normal distribution question is:

1. Identify if it is a normal distribution question.
2. Convert the normal distribution to a standard normal distribution by $Z = \frac{X - \mu}{\sigma}$.
3. Use symmetries and that the area is 1 to rearrange the probability to one that can be found in the lookup sheet.

For point 1, we can use a little heuristic to figure out if we are dealing with a normal distribution.

1. Does the question deal with *continuous* data? As in, the gaps between points can be infinitely small. For example, the weight of a person or a moment in time is continuous, but the value a die can take cannot be 4.5 – this is a discrete value set. The normal distribution has to be continuous.
2. Does the question give you or mention the mean and variance or standard deviation?

If the answer to both of those is yes, you are probably going to use a normal distribution. If no, you’ll need to use something else! Usually a question spells it out for you.

Often, the question asks you about the mean of the normal distribution, and in particular, how you might model the mean itself.

Given a set of samples $x_1, \dots, x_n$, we can find the sample mean via:

$$\frac{x_1 + \dots + x_n}{n}$$

If we consider a set of n random variables, $X_1, \dots, X_n$, where each is independently distributed with the same normal distribution $N(\mu, \sigma^2)$, we can ask ourselves what would the distribution of the mean of those random variables look like:

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n}$$

It turns out that this can be considered itself a random variable having a normal distribution with the same mean, but a rescaled variance $\bar{X} \sim N(\mu, (\sigma/\sqrt{n})^2)$.

Why this is true requires a few facts out of scope, one being that the sum of scaled normal distributions is itself a normal distribution, combined with some facts about how the mean and variance behave under summing and scaling.¹⁸

One more intuitive way to think about this is that the mean of the mean should itself be the mean with enough points, but the standard deviation of a mean should be smaller since the more points we have the closer to the mean we expect to be, and the mean is bringing those points with it. The additional points help to cancel out any extremes.

Example 76. A baby's weight at birth can be modelled as a normal distribution with mean 3.5 kg and standard deviation 0.5 kg.

Your sibling tells you that a healthy range of baby weights is between 3.37 kg and 3.76 kg.

You are given the following values for $P(Z > z)$ where Z is a standard normal distribution:

p	z
0.5000	0.0000
0.4000	0.2533
0.3000	0.5244

Using the closest values from the above table, find the probability that the weight of the baby is between 3.37 kg and 3.76 kg.

Solution. We are only told values for $P(Z > z)$ where Z is a standard normal distribution. So we will need to do two things.

1. Convert to a standard normal distribution
2. Use symmetries to write our region in terms of $P(Z > z)$ for some z

Firstly, let $X \sim N(3.5, 0.5^2)$ be the random variable representing the weight of the baby. We can convert to a standard normal distribution by using the formula $Z = \frac{X-\mu}{\sigma}$, which gives us $Z = \frac{X-3.5}{0.5}$.

We are interested in the region $3.37 \leq X \leq 3.76$. We can rearrange this to be in terms of Z by substituting $X = 0.5Z + 3.5$ (the rearrangement of the previous formula) to get

$$3.37 \leq 0.5Z + 3.5 \leq 3.76$$

Which can be rearranged to find

$$\frac{3.37 - 3.5}{0.5} \leq Z \leq \frac{3.76 - 3.5}{0.5}$$

This can be simplified to

$$-0.26 \leq Z \leq 0.52$$

¹⁸Precisely, we can use Expectation Algebra (see formula book) to realise that $E(\sum X/n) = 1/n \sum E(X) = E(X)$ and $\text{Var}(\sum X/n) = 1/n^2 \sum \text{Var}(X) = 1/n \text{Var}(X)$, which we can combine with the assumption that a linear combination of normal distributions is itself normal to get the result.

This is not in the form of $P(Z > z)$, but we can use symmetries to find that $P(Z > 0.26) = P(Z < -0.26)$ and $P(Z > 0.52) = 1 - P(Z < 0.52)$.

The region we want is found by $P(Z > -0.26) - P(Z > 0.52)$, which is converted to $1 - P(Z > 0.26) - P(Z > 0.52)$.

The closest values we have from the table are $P(Z > 0.2533)$ and $P(Z > 0.5244)$, which we can use to find that the probability of the baby weight being between 3.37 kg and 3.76 kg is approximately

$$(1 - P(Z > 0.2533)) - P(Z > 0.5244) = (1 - 0.4000) - 0.3000 = 0.3000$$

This finishes the question. □

For the standard normal distribution, the following geometrical techniques are essential:

1. Reflecting about the mean at 0 corresponds to putting a minus sign in front of the value z .
2. Choosing which side of a value to find the area under corresponds to finding the probability p or $1 - p$ for the other side.

6.2.2 Why do we care about the standard normal distribution?

This requires a little bit of background. Using the symmetries of the normal distribution covered in Section 6.2.1, having the mean be 0 is very useful since reflecting amounts to putting a minus sign in front of the value. Therefore, it is simpler to work with a normal distribution whose mean is 0.

Secretly, the function which describes a normal distribution is

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

And since we know that the area under the curve between two points represents the probability of our normally distributed random variable being observed between those two points, this means that:

$$P(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

The correct approach to an integral like this is to cry. The second best approach is to use a numerical method. It turns out that this is impossible to solve using techniques in A-Level¹⁹, and you would need to approach this using numerical methods such as the trapezium rule (see Section 12.1).

In a real-world setting, this might take a long time. Thus collectively, we agreed to use the standard normal distribution so that we could pre-compute a bunch of probability values for various regions, and store them in a large table.

Then to find the probability of a normally distributed random variable, you can find the answer by converting it to a standard normal distribution and then look up the value from the table.

Due to symmetries, the table only needs to contain values of $P(Z > z)$ for $z \geq 0$, since we can use symmetries to find any other region.

¹⁹More precisely, there is no analytical solution to this integral.

Nowadays, even handheld calculators have the ability to compute the probability of a normal distribution (secretly using a numerical method like the trapezium rule), but since the exam cannot assume you have such a device, we still need to use the standard normal distribution.

This means for the exam, it is best practice to convert to the standard normal distribution, and only then use your calculator if you have one. Otherwise, you must use the lookup table provided in the formula booklet.

6.2.3 Binomial Distribution

The binomial distribution is a useful probability distribution, which gives a probability of a given number of successes in a fixed number of trials. This means we know the number of times something has been repeated, and know that in each case only two outcomes are possible.

It is to be used when there is a fixed number of trials, n , each with one of two outcomes (success or failure), and we want to count how many successes occur. We also need the probability of success to be constant and for the trials to be independent (not influence the next trial).

Examples of typical question wordings may include “out of 50 customers...” “in a sample of 20 items...” “each component is defective with a probability of 0.05...”

In the formula book, you can find statistical tables of values for the probability that the number of successes, X is less than or equal to x with the probability of a success being p and the number of trials being n .

We denote a random variable which is distributed with a binomial distribution as $X \sim B(n, p)$.

Example 77. A chair factory produces a defective chair with a probability of 5%. Using the formula book, what is the probability there is no more than 1 defective chair in a batch of 50 chairs, to four significant figures?

Solution. Let’s model this with a “success” being a defective chair, so that we want the number of successes to be 1 or less.

Our random variable is thus modelled as $X \sim B(50, 0.05)$. To use the formula book, we must notice that it gives us a list of values for $P(X \leq x)$, thus since we want 1 or less defective chairs, we are interested in

$$P(X \leq 1) = 0.2794$$

Where we read from the 0.05 column.

Therefore the probability that there is no more than 1 defective chair in a batch of 50 is 28.94% $\square$

In fact, the probability of exactly x successes is given in the formula book:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

So the previous question could be worked out directly as

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Exercise 8. Analysis shows that 30% of the time, a randomly chosen cereal ball from a cereal manufacturer is chocolate. A box of cereal contains 100 cereal balls. Using an appropriate choice of distribution, what is the probability of getting exactly 30 chocolate balls in the cereal box?

Instead, what is the probability of getting between 29 and 31 chocolate balls in this sample, inclusive? Then state the probability of getting a number of chocolate balls not between 29 and 31 inclusive.

Finally, if you are told that the probability of getting less than 29 chocolate balls is 0.37678, what is the probability of getting more than 31 chocolate balls in your cereal? You may assume that they use the same model as you.

Solution. Since we have a fixed probability of success (a ball being chocolate), a fixed number of trials, can assume independence of trials, and we are interested in the number of successes, a reasonable choice of distribution is a binomial distribution. We can model the number of chocolate balls as X , where $X \sim B(100, 0.3)$.

Then we can use the formula to find $P(X = 30)$, which is

$$P(X = 30) = \binom{100}{30} (0.3)^{30} (0.7)^{70} \approx 0.08678$$

This number might be smaller than you expected! Considering that 30% of the time a ball is chocolate, you should expect to find on average around 30 chocolate balls in a sample of 100.

Continuing, we have that

$$\begin{aligned} P(29 \leq X \leq 31) &= P(X = 29) + P(X = 30) + P(X = 31) \\ &= \binom{100}{29} (0.3)^{29} (0.7)^{71} + \binom{100}{30} (0.3)^{30} (0.7)^{70} + \binom{100}{31} (0.3)^{31} (0.7)^{69} \\ &\approx 0.08556 + 0.08678 + 0.08398 \\ &\approx 0.25632 \end{aligned}$$

This accounts for a quarter of all possibilities, which might be more than expected considering the previous number.

We were also asked the probability of being outside this range:

$$\begin{aligned} P(X < 29 \text{ or } X > 31) &= 1 - P(\text{not } (X < 29 \text{ or } X > 31)) && \text{(the full area is 1)} \\ &= 1 - P(29 \leq X \leq 31) \\ &\approx 1 - 0.25632 \\ &\approx 0.74368 \end{aligned}$$

For the final part of the question, we are given that $P(X < 29) = 0.37678$. We can use this to find $P(X > 31)$ by using the fact that $P(X < 29 \text{ or } X > 31) = P(X < 29) + P(X > 31)$, since these two events are mutually exclusive (cannot happen at the same time).

$$\begin{aligned} P(X > 31) &= P(X < 29 \text{ or } X > 31) - P(X < 29) \\ &\approx 0.74368 - 0.37678 \\ &\approx 0.3669 \end{aligned}$$

This finishes the question. □

The previous question highlights the importance of viewing things geometrically, and deducing how to split the region we want into different chunks we care about.

6.3 Approximating the Binomial Distribution with a Normal Distribution

It turns out that the binomial distribution looks more and more like a bell curve when you increase n ²⁰.

When n is large enough, we can approximate the binomial distribution from Section 6.2.3 with a normal distribution from Section 6.2.1.

Why would we want to do this? The statistical tables only go up to $n = 50$, and therefore if n is much larger, it would be very difficult to compute for example

$$P(X \leq 500) = P(X = 0) + P(X = 1) + \cdots + P(X = 500)$$

Instead, we can use that we have a statistical table for the standard normal distribution.

The normal distribution requires both the mean μ and the variance σ^2 , not to be confused with the standard deviation σ . The only possible reasonable choices of mean and variance to approximate a binomial distribution are the mean and variance of the binomial distribution itself. These can be found via the formula for mean and variance, but are handily given in the formula book too:

1. For a binomial distribution $B(n, p)$, the mean is $\mu = np$
2. For a binomial distribution $B(n, p)$, the variance is $\sigma^2 = np(1 - p)$

Therefore the standard deviation is $\sigma = \sqrt{np(1 - p)}$

This gives us our approach for tackling a question where we have a binomial distribution with very large n and want to answer it by approximating it with a normal distribution:

1. Write the binomial distribution
2. Approximate it with a normal distribution $\mu = np, \sigma^2 = np(1 - p)$: $N(np, np(1 - p))$
3. Convert it to the standard normal distribution with $Z = \frac{X - \mu}{\sigma}$, remembering that $\sigma = \sqrt{np(1 - p)}$
4. Look up the result from the table
5. Convert back to X if needed

Example 78. Our breakfast cereal factory has a 50% chance of producing a chocolate ball or a non-chocolate ball for the cereal, independently.

We want to know if we look at 10000 balls, what is the greatest number of chocolate balls that could occur where it is still in the lowest 10% of possible outcomes?

As in find x so that $P(X \leq x) = 0.1$ where X is the number of chocolate balls²¹, choosing your distribution appropriately. You may approximate your answer using a normal distribution if justified appropriately.

Solution. Firstly, since we have two independent outcomes and a fixed number of trials, a binomial distribution $X \sim B(10000, 0.5)$ is appropriate, where X is the number of chocolate balls.

Then we want to find x so that $P(X \leq x) = P(X < x - 1) = 0.1$ Since n is large, we may use a normal distribution as approximation.

²⁰See the Central Limit Theorem for further reading

²¹We are trying to find the width of the one-sided tail which contains 10% of the lower outcomes of chocolate balls, related to hypothesis testing.

We will use $\mu = np$, $\sigma^2 = np(1 - p)$ so that $X \sim N(np, np(1 - p))$.

$$\begin{aligned}\mu &= np = 5000 \\ \sigma^2 &= np(1 - p) = 5000(0.5) = 2500\end{aligned}$$

Therefore we can approximate $X \sim N(5000, 2500)$

We want to convert this to $N(0, 1)$, which can be done with the formula $Z = \frac{X - \mu}{\sigma}$. Therefore

$$Z = \frac{X - 5000}{\sqrt{2500}} = \frac{X - 5000}{50}$$

Now, using symmetries of the standard normal distribution see Section 6.2.1, we know that $P(Z < z) = P(Z > -z)$.

Therefore, we are interested in when $P(Z > -z) = 0.1$, which we can find from the formula book is 1.2816.

This gives us that $z = -1.2816$. We wish to match this to our original scalings, which we can achieve by undo-ing our conversion to the standard normal distribution:

$$\begin{aligned}P(X < x - 1) &= 0.1 \\ \therefore P(Z < \frac{x - 1 - \mu}{\sigma}) &= 0.1 \\ \text{and since } P(Z < z) = 0.1 \text{ has answer } z &= -1.2816 \\ \frac{x - 1 - \mu}{\sigma} &= -1.2816 \\ \therefore \frac{x - 1 - 5000}{50} &= -1.2816 \\ \therefore -1.2816 \cdot 50 &= x - 5001 \\ \therefore x &= 5001 - 1.2816 \cdot 50 \\ \therefore x &\approx 4935.42.\end{aligned}$$

Since x must be an integer and we want the greatest integer with $P(X \leq x) \approx 0.1$, we take

$$x = 4935.$$

□

6.4 Hypothesis Testing

The goal of hypothesis testing is to decide whether the data provides enough evidence to doubt a particular claim about a population.

More precisely:

We start with a claim, collect sample data, and then would like to check how wrong we were with our claim – we assess whether the sample is inconsistent with that claim, to some level of significance.

Since we are dealing with probabilities, we cannot fully disprove (or prove) anything, rather only find that it is exceedingly unlikely to be true given what we've seen, like if we claimed 80% of people

had sentient hair, but only experienced people without sentient hair. After a while of seeing no sentience in hair, we would like to find a way to reassess our original beliefs. Conversely, if we somehow started spotting people with sentient hair every second person, we could never be certain that our claim is true, but only get more certain that other claims are false.

To do this, we will have two hypotheses, the original baseline hypothesis that we assume to be true, called the *null hypothesis*, and the *alternative hypothesis*, which is usually “not the null hypothesis” or some directional version of not (like more than or less than).

The null hypothesis is denoted H_0 and the alternative hypothesis is denoted H_1 . In our hair sentience example, the null hypothesis would be that 50% of people have sentient hair, and we are doubting this so the alternative hypothesis might be that less than 50% of people have sentient hair.

*To test the validity of the null hypothesis, we calculate the probability of the observed data occurring (or more extreme than the observed data), assuming that H_0 is true.*²²

The way to think about how this helps us is to realise this is the same as asking “if we rank all possible outcomes from most to least likely given our null hypothesis, what percentile of least likely things happening does the observed data lie in?” Then if our observation is in the e.g. bottom 1% of least likely outcomes if the null hypothesis was true, we should start to doubt it.

The key thing that makes this all feasible to compute in an exam question is that the null hypothesis will allow you to either calculate the probabilities directly or look them up using the statistical tables in the formula book.

Remember that you have tables for the following:

1. Binomial Cumulative Distribution Function
2. Percentage Points of The Normal Distribution
3. Poisson Cumulative Distribution Function
4. Percentage Points of the χ^2 Distribution
5. Critical Values for Correlation Coefficients

Not all of these are required for A-level hypothesis testing, but they appear in the formula book.

It is useful to know when to use each of the distributions above.

1. Binomial Distribution

Gives a probability of a given number of successes in a fixed number of trials.

To be used when there is a fixed number of trials, n , each with one of two outcomes (success or failure), and we want to count how many successes occur. We also need the probability of success to be constant and for the trials to be independent (not influence the next trial).

Examples of typical question wordings may include “out of 50 customers...” “in a sample of 20 items...” “each component is defective with a probability of 0.05...” “test whether the proportion has changed...”

2. Poisson Distribution²³

Gives the probability of a given number of events occurring in a fixed interval of time or space.

To be used when we are counting how many events occur in a fixed time period or region, and events occur randomly and independently at a constant average rate. There is no fixed number of trials; instead, we are modelling a rate of occurrence.

²²This is a conditional probability, and is normally denoted $P(\text{observed data (or more extreme)} \mid H_0 \text{ is true})$

²³Currently not mentioned in the syllabus for hypothesis testing, but left here for completeness.

Examples of typical question wordings may include “an average of 3 calls per minute...” “defects occur at a rate of 0.4 per metre...” “customers arrive at a rate of...” “test whether the rate has changed...”

3. Normal Distribution

Gives probabilities for values of a continuous variable, or for the sampling distribution of a mean.

To be used when data are stated to be normally distributed, or when testing a population mean using a sample mean (particularly when a standard deviation is given). It is used when we are standardising values to find a Z -score.

Examples of typical question wordings may include “heights are normally distributed...” “the mean lifetime is 500 hours...” “a sample of size 36 has mean...” “test whether the mean has increased...”

4. Correlation Coefficients

Gives critical values for testing whether there is evidence of correlation between two variables.

To be used when we have paired quantitative data and have calculated a product moment correlation coefficient r , and we wish to test whether there is evidence of positive or negative linear correlation in the population.

Examples of typical question wordings may include “test whether there is evidence of correlation...” “test whether there is negative correlation...” “a scatter diagram suggests...” “calculate the product moment correlation coefficient...”

The template we will follow is roughly:

1. Figure out the distribution needed.
2. State the hypotheses clearly.
3. State the significance level.
4. Determine the test statistic (Normal / Correlation) or find the critical region (Binomial).
5. Compare the observed value with the critical value (or compute the p -value).
6. State a conclusion in context.

For the alternative hypothesis, for the purposes of the exam, we can choose either not equal, less than, or greater than, depending of the wording of the question. E.g. if we wanted to check that the number of chocolate balls is not 50% in our chocolate ball cereal warehouse, we could have our alternative hypothesis of the probability being not equal to 50%, whereas if we wanted to check that the number of chocolate balls is more than expected, we would use the alternative hypothesis of the probability being more than 50%.

An alternative hypothesis which looks like $H_1 : p \neq a$ is called a two-tailed test, whereas an alternative hypothesis which looks like $H_1 : p > a$ or $H_1 : p < a$ is called a one-tailed test. The two-tailed test has this name since it can be split into two regions on either side of the null value, e.g. $p < a$ and $p > a$, which when combined account for all of $p \neq a$. In practice, a two-tailed test involves calculating probabilities for both tails separately. Note that we have just used p as a notational example.

We choose (usually are told) a *level of significance*, which amounts to “assuming H_0 is true, how extreme of a result do we want to choose as too unexpected before we doubt it?” More rigorously, it is the maximum probability (in the case of one tail) we are allowing for an event to occur in the extreme of our null hypothesis. For a two-tailed test, we consider it as two one-tailed tests with half the allowed maximum probability.

What about critical regions? The critical region is the set of possible observations in which we would have evidence to reject the null hypothesis, i.e. a value which lies within our one or two-tailed region which would give us reason to doubt our null hypothesis for some level of significance.

In the case of the binomial distribution (and Poisson if it were examined, or any discrete distribution for that matter), since we cannot have a decimal number of successes, the critical region does not perfectly line up with the percentages we have set as the maximum allowed percentage for a tail via the level of significance. For example, if we have a one-tailed test and a level of significance of 2.5%, we might find that the probability of being in our tail is 2.4%, if the next value jumps up to e.g. 3.7%. This means that the actual maximum probability we are allowing to be in our critical region is 2.4% in this case, rather than 2.5%. We call the actual percentage likelihood of being in the critical region the *actual level of significance*. In the case of two tails, we would sum the one-tailed probabilities of being in the critical region.

For the level of significance, we normally require to be strictly within the tail, as in for a one-tailed test with a level of significance of 2.5% for a binomial distribution, we only accept a critical region which has a probability less than 2.5%, not equal to. This should be clarified by any exam question, however.

Another way to assess how extreme a result is under the null hypothesis is via the *p-value*.

The *p-value* is the probability, assuming H_0 is true, of obtaining a result at least as extreme as the one observed. In other words, it tells us how surprising our observation would be if the null hypothesis were correct. We can then compare the *p-value* with the level of significance, and if the *p-value* is less than the level of significance, then we reject the null hypothesis. It is equivalent to checking the critical region, except we are comparing the actual probabilities instead of the values.

Example 79. A supervisor at the chocolate ball cereal warehouse wants to ensure that there are an equal percentage of chocolate balls to non-chocolate balls in our chocolate ball cereal.

During production, the balls are mixed together in bulk randomly and then portioned into a cereal box. We can consider the probability each ball is chocolate as independent.

The supervisor randomly samples 50 cereal balls from a box of cereal, and wants to know what the critical region is for a two-tailed test with a significance level of 1%. The probability of being within the critical region should be less than 1%, with each tail being less than 0.5%

Stating your hypotheses clearly, conduct an appropriate hypothesis test, stating clearly the critical region and the actual level of significance.

If the supervisor finds that there are 16 chocolate balls in his sample, what should his conclusion be?

Solution. Since we have independent events, a probability of “success” of 0.5 since there should be an equal percentage of chocolate to non-chocolate balls, we are using a fixed number of events, and we want to count the number of successes, we should use a binomial distribution $X \sim B(50, 0.5)$, where X is the number of chocolate balls. We want to test if the probability p really is 0.5, therefore since we are using a two-tailed test, our hypotheses are as follows:

$$\begin{aligned} H_0 : p &= 0.5 && \text{(null hypothesis)} \\ H_1 : p &\neq 0.5 && \text{(alternative hypothesis)} \end{aligned}$$

To find the critical region for a significance level of 1%, we consider the two tails separately, which each have a maximum percentage of 0.5%.

Therefore we are interested in the values x and y such that

$$P(X \leq x) < 0.005$$

$$P(X \geq y) < 0.005$$

Then together, our critical region would be $X \leq x$ or $X \geq y$.

Start with x :

Using the formula book for $P(X \leq x)$ for $B(50, p)$, in the column corresponding to $p = 0.5$, we find that

$$P(X \leq 15) = 0.0033$$

$$P(X \leq 16) = 0.0077$$

Therefore, since each tail must have a probability less than 0.005, we choose $x = 15$ as our first (left) part of the critical region, with a probability of 0.0033.

Now for y : We must first convert this to the form of less than:

$$P(X \geq y) < 0.005$$

$$\therefore 1 - P(X < y) < 0.005$$

$$\therefore 1 - P(X \leq y - 1) < 0.005$$

$$\therefore P(X \leq y - 1) > 0.995$$

So let's find the value where the probability exceeds 0.995:

$$P(X \leq 33) = 0.9923$$

$$P(X \leq 34) = 0.9967$$

Thus $y - 1 = 34$, making $y = 35$. One way to think about this, even though it seems like we have overshoot the actual answer from the table, is that the table tells us specifically the ' $\leq$ ' region, and we want to find the '>' region, which is from one more than the ' $\leq$ ' region.

Therefore, since each tail must have a probability less than 0.005, we found the point at which the probability becomes more than 0.995, and therefore we have found that the second (right) part of the critical region is $y = 35$, with a probability of $1 - 0.9967 = 0.0033$.

That makes our critical region $X \leq 15$ or $X \geq 35$, with an actual level of significance of the sum of the individual probabilities:

$$0.0033 + 0.0033 = 0.0066 = 0.66\%$$

This completes the first part of the question. Now if the supervisor found that in his sample there were 16 chocolate balls, this is (just about) not within the critical region of $X \leq 15$ or $X \geq 35$, and therefore this sample does not provide enough evidence to reject the null hypothesis, at this level of significance.

I.e. there is not enough evidence to suggest that the percentage of chocolate balls produced by the factory differs from 50%, to this level of significance. $\square$

Example 80. A consumer of the previous chocolate ball cereal warehouse suspects that the chocolate ball cereal warehouse is stingy with their chocolate balls, which are claimed to be an equal chance of having chocolate or non-chocolate balls.

This consumer takes a sample of 50 balls from their cereal box, and counts the chocolate balls, finding there to be 16.

Using an appropriate one-tailed test with a significance level of 1%, what should the consumer's conclusion be? The probability of the tail should be less than 1%.

Stating your hypotheses clearly, conduct an appropriate hypothesis test, stating clearly the critical region and the actual level of significance.

Then comment on the comparison to the conclusion from the previous question.

Solution. We can use the same setup for the most part from the previous question: X the number of chocolate balls, $X \sim B(50, 0.5)$ assumed, $H_0 p = 0.5$, but since the consumer is doing a single tailed test, wanting to know whether they are stingy, the alternative hypothesis is instead

$$H_1 p < 0.5$$

Therefore, since the probability of the tail must be less than 0.01, we use the formula book for $P(X \leq x)$ for $B(50, p)$, in the column corresponding to $p = 0.5$, to find that

$$P(X \leq 16) = 0.0077$$

$$P(X \leq 17) = 0.0164$$

We therefore find that the critical region is $X \leq 16$, with an actual significance level of 0.0077

This completes the first part of the question.

For the consumer's observation, they found 16 chocolate balls in their sample of 50 from the cereal, which is within the critical region.

Therefore there is evidence to reject the null hypothesis, i.e. there is evidence to suggest that the percentage of chocolate balls produced by the factory is less than 50%, to this level of significance.

Commenting on the previous example, we found that the amount of tails alone can change the conclusion; the supervisor found nothing to be wrong whereas the consumer found they were stingier than claimed. $\square$

Example 81. The chocolate ball cereal warehouse claims that the mean mass of each chocolate ball is 5.40 g. Past records suggest that the masses of chocolate balls are normally distributed with standard deviation 0.68 g.

A consumer suspects that the mean mass of the chocolate balls has changed.

A random sample of 52 chocolate balls is taken and found to have mean mass 5.72 g.

Using a 5% level of significance, conduct an appropriate hypothesis test to assess the consumer's claim.

Stating your hypotheses clearly, conduct an appropriate hypothesis test, stating clearly the critical region

Find the p -value for this test and interpret it in context.

Solution. Firstly, let's model the mass of the chocolate ball as X , so that $X \sim N(5.4, 0.68^2)$

Since we are taking a sample of 52 chocolate balls and using the sample mean, we instead model the sample mean $\bar{X}$:

$$\bar{X} \sim N\left(5.4, \frac{0.68^2}{52}\right)$$

See Section 6.2.1.

Then our hypothesis should be around μ itself:

$$H_0 : \mu = 5.4$$

$$H_1 : \mu \neq 5.4$$

We are then interested in the two-tailed test, finding the values x, y such that $P(\bar{X} < x) = 0.025$ and $P(\bar{X} > y) = 0.025$, using that we have a 5% significance.

To compute these values, we want to use the formula book, which tells us values related to the standard normal distribution, so we transform:

$$Z = \frac{\bar{X} - 5.4}{0.68/\sqrt{52}}$$

Then $Z \sim N(0, 1)$. Since the formula book tells us values for $P(Z > z)$, we can use symmetries to know that $P(Z < z) = P(Z > -z)$. We continue, reading from the formula book:

$$P(Z > z) = 0.025$$

$$\therefore z = 1.9600$$

Now let's work out what this means for our two cases:

Starting with y :

$$\begin{aligned} P(\bar{X} > y) &= 0.025 \\ \therefore P\left(Z > \frac{y - 5.4}{0.68/\sqrt{52}}\right) &= 0.025 \\ \therefore \frac{y - 5.4}{0.68/\sqrt{52}} &= 1.9600 \\ \therefore y - 5.4 &= 1.9600 \left(\frac{0.68}{\sqrt{52}}\right) \\ \therefore y - 5.4 &\approx 0.1849 \\ \therefore y &\approx 5.5849 \end{aligned}$$

Continuing with x :

$$\begin{aligned}
P(\bar{X} < x) &= 0.025 \\
\therefore P\left(Z < \frac{x - 5.4}{0.68/\sqrt{52}}\right) &= 0.025 \\
\therefore P\left(Z > -\frac{x - 5.4}{0.68/\sqrt{52}}\right) &= 0.025 \\
\therefore -\frac{x - 5.4}{0.68/\sqrt{52}} &= 1.9600 \\
\therefore x - 5.4 &= -1.9600 \left(\frac{0.68}{\sqrt{52}}\right) \\
\therefore x - 5.4 &\approx -0.1849 \\
\therefore x &\approx 5.2151
\end{aligned}$$

Therefore the critical region for $\bar{X}$ is

$$\bar{X} < 5.215 \quad \text{or} \quad \bar{X} > 5.585.$$

Now compare the observed value with this critical region. The sample mean mass is 5.72 g, which lies in the critical region since

$$5.72 > 5.585.$$

Therefore we reject H_0 . There is sufficient evidence at the 5% level to suggest that the mean mass of chocolate balls differs from 5.4 g.

We now find the p -value using the standard normal tables.

$$\begin{aligned}
Z &= \frac{5.72 - 5.4}{0.68/\sqrt{52}} \\
&\approx 3.39
\end{aligned}$$

From the formula book,

$$P(Z > 3.2905) = 0.0005.$$

Since $3.39 > 3.2905$, the probability in one tail must be smaller:

$$P(Z > 3.39) < 0.0005.$$

As this is a two-tailed test,

$$p\text{-value} = 2P(Z > 3.39) < 2(0.0005) = 0.001.$$

Hence the p -value is less than 0.001, which is far smaller than 0.05, giving very strong evidence against H_0 . $\square$

Example 82. A cereal factory is investigating whether increasing the number of chocolate balls in each serving improves customer satisfaction.

Marc selects a random sample of 12 customers. Each customer is given a bowl of cereal and then asked to rate their happiness on a scale from 1 to 10.

For each customer he records:

- x : the number of chocolate balls in their serving
- y : their happiness rating

The results are shown in the table below.

x	8	10	12	6	14	9	11	7	13	15	5	16
y	4	6	7	3	8	5	6	4	8	9	2	9

1. Use your calculator to find the product moment correlation coefficient between x and y for this data.
2. Test whether or not there is evidence of a positive correlation between the number of chocolate balls in a serving and the happiness rating.

You should:

- state your hypotheses clearly
- use a 5% level of significance
- state your conclusion in context.

Solution. We are given paired data (x, y) , so we use the product moment correlation coefficient.

From the formula book:

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n}, \quad S_{yy} = \sum y_i^2 - \frac{(\sum y_i)^2}{n}, \quad S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

and

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

Here $n = 12$.

Using a calculator to obtain the required totals:

$$\sum x = 126, \quad \sum y = 71, \quad \sum x^2 = 1466, \quad \sum y^2 = 481, \quad \sum xy = 838.$$

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 1466 - \frac{126^2}{12} = 1466 - 1323 = 143,$$

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 481 - \frac{71^2}{12} = 481 - 420.0833 \approx 60.9167,$$

$$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 838 - \frac{(126)(71)}{12} = 838 - 745.5 = 92.5.$$

Therefore,

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{92.5}{\sqrt{(143)(60.9167)}} \approx 0.991.$$

We now test for positive correlation.

$$H_0 : \rho = 0$$

$$H_1 : \rho > 0$$

From the formula book table of critical values for correlation coefficients, for $n = 12$ at the 5% level (one-tailed), the critical value is

$$r = 0.4973.$$

Since

$$0.991 > 0.4973,$$

we reject H_0 .

There is evidence at the 5% level of a positive correlation between the number of chocolate balls in a serving and the happiness rating.

□

6.4.1 Additional Exercises

Example 83. After spotting many people with orange hair, James believes that the percentage of orange-haired people has increased to 75% in recent times.

To support his claim, James randomly selects 50 people and finds that 29 of them have orange hair. You may assume that the selections are independent.

This is less than his expected amount, making him doubt his claim.

Using an appropriate one-tailed test with a significance level of 1%, conduct an appropriate hypothesis test. The probability of the tail should be less than 1%.

Stating your hypotheses clearly, conduct an appropriate hypothesis test, stating clearly the critical region and the actual level of significance.

Solution. Let's model the number of non-orange-haired people, so that the null hypothesis is that 25% of people have non-orange hair. This is to allow us to use the formula book, which has values between $p = 0.05$ and $p = 0.5$.

If X is the number of non-orange-haired people, we will model $X \sim B(50, 0.25)$, and our hypotheses are that either James is right or the number of non-orange-haired people has increased, letting p be the probability a person has non-orange hair:

$$H_0 : p = 0.25$$

$$H_1 : p > 0.25$$

The critical region is the set of values that are in the 1% most extreme tail of non-orange-haired people under H_0 , therefore we wish to find x so that $P(X \geq x) < 0.01$.

We convert to $\leq$ for the formula book:

$$\begin{aligned}
P(X \geq x) &< 0.01 \\
\therefore 1 - P(X < x) &< 0.01 \\
\therefore 1 - P(X \leq x - 1) &< 0.01 \\
\therefore 1 - 0.01 &< P(X \leq x - 1) \\
\therefore 0.99 &< P(X \leq x - 1)
\end{aligned}$$

From the formula book, the first value which exceeds 0.99 is:

$$P(X \leq 20) = 0.9937$$

Therefore we find that

$$x - 1 = 20$$

Which gives us our critical region of $X \geq 21$. This makes the actual level of significance $P(X \geq 21) = 1 - 0.9937 = 0.0063$.

Recall that X was the number of non-orange-haired people.

James found that 29 out of his 50 people had orange hair, meaning that 21 people had non-orange hair. Since this is within the critical region, there is evidence to reject the null hypothesis that 75% of people have orange hair, in favour of there being less than 75% of people having orange hair, to this level of significance. $\square$

7 Trigonometry

Some key concepts were covered in Section 2, where we used standard trig identities to find components of vectors.

7.1 Small angle approximations

Using something called the Taylor expansion of trigonometric functions, we can see how each of the functions behave near 0.

$$\begin{aligned}\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \\ \tan x &= x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots\end{aligned}$$

Where these come from can be researched by an interested reader via Taylor series.

Using the above, we can observe that if x is really small (close to 0), x^2 is a small number times a small number, which is even smaller itself. Similarly, all higher powers of x will also be (exponentially) smaller.

Therefore, one way of approximating these functions when x is close to 0 is by removing any power of x above 2, since they are all tiny.

From this, you get the small angle approximations of the trigonometric functions.²⁴

$$\begin{aligned}\sin x &\approx x \\ \cos x &\approx 1 - \frac{x^2}{2} \\ \tan x &\approx x\end{aligned}$$

Note that if you remember only two of them, a handy way of remembering the third is by rearranging

$$\tan x = \frac{\sin x}{\cos x}$$

Which also holds for small angle approximations, and using instead that $\cos x \approx 1$ (check this)!

This can be used to approximate complicated functions within each trig function.

Example 84. Let $f(x)$ be a function of x , and let $f(x)$ be close to 0 when x is near some number a .

Using small angle approximations, write $\sin(f(x))$, $\cos(f(x))$, and $\tan(f(x))$ in terms of $f(x)$, and make sure to state when these small angle approximations hold.

Solution. Using the formulae for small angle approximations, using $f(x)$ in place of x in the formulae, we find that – **as long as $f(x)$ is close to zero** – the following small angle approximations hold:

²⁴This technically requires that x is in radians too, but we will assume that in this section.

$$\begin{aligned}\sin(f(x)) &\approx f(x) \\ \cos(f(x)) &\approx 1 - \frac{f(x)^2}{2} \\ \tan(f(x)) &\approx f(x)\end{aligned}$$

Since **this only holds when $f(x)$ is near 0**, this holds as long as x is near a . This is because the question told us that $f(x)$ is close to 0 when x is near a .

This completes the question.

□

This can be used to get approximate solutions to problems, but only holds when the argument of the trigonometric functions is close to zero!

Now for some questions.

Example 85. Let x be a small non-zero number in radians.

Using small angle approximations and justifying why you can, simplify the following.

$$x \frac{\cos(x)}{\sqrt{2}x - \sin(x^2)}$$

Show that, as long as $x \neq \sqrt{2}$, this can be simplified to become:

$$\frac{\sqrt{2} + x}{2}$$

Using this, estimate the integral:

$$\int_{0.05}^{0.1} x \frac{\cos(x)}{\sqrt{2}x - \sin(x^2)} dx$$

Solution. First note that we are told x is small, and therefore x^2 is smaller. We then proceed with calculating our small angle approximations:

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

And using that x^2 is small,

$$\sin(x^2) \approx x^2$$

$$\begin{aligned}
\frac{x \cos(x)}{\sqrt{2x} - \sin(x^2)} &\approx x \frac{1 - \frac{x^2}{2}}{\sqrt{2x} - x^2} && \text{(using our small angle approximations)} \\
&= x \frac{1 - \frac{x^2}{\sqrt{2}^2}}{\sqrt{2x} - x^2} && \text{(writing } 2 = \sqrt{2}^2\text{)} \\
&= x \frac{1 - (\frac{x}{\sqrt{2}})^2}{\sqrt{2x} - x^2} && \text{(using } a^2/b^2 = (a/b)^2\text{)} \\
&= x \frac{(1 - \frac{x}{\sqrt{2}})(1 + \frac{x}{\sqrt{2}})}{\sqrt{2x} - x^2} && \text{(using difference of two squares)} \\
&= \cancel{x} \frac{(1 - \frac{x}{\sqrt{2}})(1 + \frac{x}{\sqrt{2}})}{\cancel{x}(\sqrt{2} - x)} && \text{(factorising denominator, and using } x \neq 0\text{)} \\
&= \frac{(1 - \frac{x}{\sqrt{2}})(1 + \frac{x}{\sqrt{2}})}{\sqrt{2}(1 - \frac{x}{\sqrt{2}})} && \text{(factorising again)} \\
&= \frac{1}{\sqrt{2}} + \frac{x}{2} \\
&= \frac{\sqrt{2}}{2} + \frac{x}{2} \\
&= \frac{\sqrt{2} + x}{2}
\end{aligned}$$

This finishes the first part of the question. Now for the integral, we use the small result from the first question since the integral bounds are small:

$$\begin{aligned}
\int_{0.05}^{0.1} x \frac{\cos(x)}{\sqrt{2x} - \sin(x^2)} dx &\approx \int_{0.05}^{0.1} \frac{\sqrt{2} + x}{2} dx \\
&= \int_{0.05}^{0.1} \frac{\sqrt{2}}{2} dx + \int_{0.05}^{0.1} \frac{x}{2} dx && \text{(splitting the integral)} \\
&= \left[\frac{\sqrt{2}}{2} x \right]_{x=0.05}^{x=0.1} + \left[\frac{1}{4} x^2 \right]_{x=0.05}^{x=0.1} && \text{(integrating the parts)} \\
&= \frac{\sqrt{2}}{2} \cdot 0.1 - \frac{\sqrt{2}}{2} \cdot 0.05 + \frac{1}{4} (0.1)^2 - \frac{1}{4} (0.05)^2
\end{aligned}$$

Which can be solved using a calculator.

□

8 Parametric Equations

Parametric equations are when you are given multiple equations which use shared variables, and are calculated from the shared variables.

Commonly in exams this will be given in a form where you have x and y in terms of t separately.

$$\begin{aligned}x &= f(t) \\ y &= g(t)\end{aligned}$$

You will normally be asked to find y in terms of x . From here, there are two main approaches:

1. Rearrange one of the equations for t and substitute this into the other equation to eliminate t . This is sometimes not possible!
2. Rearrange one of the equations so that it is in terms of the other equation, and then substitute the variable x or y in directly.

We will see both approaches below.

Example 86. A curve has equations:

$$\begin{aligned}x &= 2t + 1 \\ y &= 4t - 5 + \frac{3}{t}, \quad t \neq 0\end{aligned}$$

Let $x \neq 1$. Then show that the curve described by the above equations can be written as:

$$y = \frac{ax^2 + bx + c}{x - 1}$$

Where a , b , and c are constants to be found.

Solution. We can approach this by writing the simplest equation in terms of t . Note that we can't easily write y in terms of t , so if we were not able to do this we would need to find another approach.

$$\begin{aligned}x &= 2t + 1 \\ x - 1 &= 2t \\ \frac{x - 1}{2} &= t\end{aligned}$$

And then substituting what t is into y .

$$\begin{aligned}
y &= 4t - 5 + \frac{3}{t} \\
&= 4\frac{x-1}{2} - 5 + \frac{3}{(\frac{x-1}{2})} && \text{(replacing each } t \text{ with } \frac{x-1}{2}) \\
&= 2(x-1) - 5 + \frac{3}{\frac{1}{2}(x-1)} && \text{(taking out a factor of } \frac{1}{2}) \\
&= 2(x-1) - 5 + \frac{2}{2} \cdot \frac{3}{\frac{1}{2}(x-1)} && \text{(multiplying by 1)} \\
&= 2(x-1) - 5 + \frac{6}{\frac{1}{2}(x-1)} && \text{(multiplying fractions)} \\
&= 2(x-1) - 5 + \frac{6}{x-1} \\
&= \frac{2(x-1)(x-1)}{(x-1)} - \frac{5(x-1)}{(x-1)} + \frac{6}{x-1} && \text{(common denominator)} \\
&= \frac{2(x-1)(x-1) - 5(x-1) + 6}{(x-1)} && \text{(putting over one denominator)} \\
&= \frac{2(x^2 - 2x + 1) - 5x + 5 + 6}{(x-1)} && \text{(expanding brackets)} \\
&= \frac{2x^2 - 4x + 2 - 5x + 11}{(x-1)} && \text{(expanding brackets)} \\
&= \frac{2x^2 - 9x + 13}{(x-1)} && \text{(simplifying)}
\end{aligned}$$

Which finishes the question, finding $a = 2$, $b = -9$, and $c = 13$ so that

$$y = \frac{ax^2 + bx + c}{x-1} = \frac{2x^2 + (-9)x + 13}{(x-1)}$$

□

Example 87. A curve has equations:

$$\begin{aligned}
x &= \frac{1}{t} + 5t \\
y &= \frac{1}{t^2} + 10 + 25t^2, \quad t \neq 0
\end{aligned}$$

Show that the curve described by the above equations can be written as:

$$y = f(x)$$

Where $f(x)$ is a function of x to be found.

Solution. We might want to approach this by writing the simplest equation in terms of t , but in this case it is not easily possible. For example, trying to rearrange the x equation for t you might get stuck at the point

$$x = t\left(\frac{1}{t^2} + 5\right)$$

And not able to continue so simply.

Instead, we attempt to rearrange the more complicated equation to be written in terms of the simpler one.

What can we do with the second equation? Firstly, we can turn it into something that is easier to work with:

$$y = \frac{1}{t^2} + 10 + 25t^2$$

$$\therefore t^2 y = 1 + 10t^2 + 25t^4$$

Here, this almost looks familiar as a quadratic, except we have this pesky t^2 hanging around. Let's briefly use a variable $u = t^2$ for convenience. Then

$$t^2 y = 1 + 10t^2 + 25t^4$$

$$\therefore uy = 1 + 10u + 25u^2 \quad (\text{using } u = t^2)$$

$$= (1 + 5u)(1 + 5u) \quad (\text{factorising})$$

$$= (1 + 5u)^2$$

$$\therefore uy = (1 + 5u)^2 \quad (\text{reminding ourselves})$$

$$\therefore y = \frac{1}{u}(1 + 5u)^2 \quad (\text{rearranging})$$

$$\therefore y = \frac{1}{t^2}(1 + 5t^2)^2 \quad (\text{using } u = t^2)$$

$$= \left(\frac{1}{t}\right)^2(1 + 5t^2)^2 \quad (\text{using } a^2/b^2 = (a/b)^2)$$

$$= \left(\frac{1}{t}(1 + 5t^2)\right)^2 \quad (\text{using } a^2b^2 = (ab)^2)$$

$$= \left(\frac{1}{t} + 5t\right)^2 \quad (\text{expanding the bracket inside})$$

This is as much simplification as we can get, and we have found that

$$y = \frac{1}{t^2} + 10 + 25t^2 = \left(\frac{1}{t} + 5t\right)^2$$

Note that we could have gone straight to the answer if we noticed that we could factorise directly.

But the last equality contains the equation for x directly:

$$x = \frac{1}{t} + 5t$$

Therefore we can replace any instance of $\frac{1}{t} + 5t$ with x to find:

$$y = x^2$$

This completes the question, finding $f(x) = x^2$.

□

Example 88. A curve has equations:

$$x = \cos(t)$$

$$y = \tan(t) + \sec(t)$$

Where $0 \leq t < \frac{\pi}{2}$.

Show that the curve described by the above equations can be written as:

$$y = f(x)$$

Where $f(x)$ is a function of x to be found.

Comment on whether the final function is the same as the curve for all x .

Solution. We might want to approach this by writing the simplest equation in terms of t , but in this case we would need to invert $\cos$ and be careful with when this is allowed.

Instead, we attempt to rearrange the more complicated equation to be written in terms of the simpler one.

What can we do with the second equation? Firstly, we can write the definitions and then try to extract $\cos(t)$ wherever possible:

$$\begin{aligned} y &= \tan(t) + \sec(t) \\ &= \frac{\sin t}{\cos t} + \frac{1}{\cos t} \end{aligned}$$

This is almost in terms of $\cos t$, except we have this pesky $\sin t$ lingering. We can make use of familiar equations to fix this (using that $0 \leq t < \frac{\pi}{2}$):

$$\begin{aligned} \sin^2 t + \cos^2 t &= 1 \\ \therefore \sin^2 t &= 1 - \cos^2 t \\ \therefore \sin t &= \sqrt{1 - \cos^2 t} \end{aligned}$$

We can now substitute everything in to find:

$$\begin{aligned} y &= \frac{\sin t}{\cos t} + \frac{1}{\cos t} \\ &= \frac{\sqrt{1 - \cos^2 t}}{\cos t} + \frac{1}{\cos t} && \text{(substituting } \sin t \text{ above)} \\ &= \frac{\sqrt{1 - x^2}}{x} + \frac{1}{x} && \text{(using } x = \cos t \text{)} \end{aligned}$$

This finishes the question, finding

$$y = f(x) = \frac{\sqrt{1 - x^2}}{x} + \frac{1}{x}$$

Note that the question was careful to restrict t between $0 \leq t < \frac{\pi}{2}$. This means that only a portion of the *full* curve given by

$$y = \frac{\sqrt{1 - x^2}}{x} + \frac{1}{x}$$

is covered by the original equations for x and y , specifically between the ranges of x allowed by $\cos t$ inside this range, which is $0 < x \leq 1$.

Therefore, the curve is actually the portion of the graph

$$y = \frac{\sqrt{1 - x^2}}{x} + \frac{1}{x}$$

Where $0 < x \leq 1$. The expression defines a larger graph, but the parametric curve is only the part in this range of x . $\square$

The next question makes use of small angle approximations, covered in Section 7.1.

Example 89. Let $t > 0$ be small, and measured in radians. A curve is given parametrically via

$$\begin{aligned}x &= \cos(t^2) + 1 \\y &= \frac{1}{t^4} + t^8\end{aligned}$$

Using small angle approximations, show that the curve described by the above equations can be written as:

$$y \approx f(x)$$

Where $f(x)$ is a function of x to be found. Comment on what portion of the curve is valid for this approximation.

Solution. There is no nice way to rearrange y for t , and the question instructs us to use small angle approximations. Let's begin with recalling that if θ is small,

$$\cos(\theta) \approx 1 - \frac{\theta^2}{2}$$

And since t is given to be small, we know that t^2 must be also small. Therefore,

$$\cos(t^2) \approx 1 - \frac{(t^2)^2}{2}$$

And therefore, we have that

$$x = \cos(t^2) + 1 \approx 1 - \frac{t^4}{2} + 1 = 2 - \frac{t^4}{2}$$

This is simple enough to rearrange for t :

$$\begin{aligned}x + \frac{t^4}{2} &\approx 2 \\ \therefore \frac{t^4}{2} &\approx 2 - x \\ \therefore t^4 &\approx 4 - 2x \\ \therefore t &\approx (4 - 2x)^{\frac{1}{4}}\end{aligned}$$

We can plug this into the equation for y :

$$\begin{aligned}
y &= \frac{1}{t^4} + t^8 \\
&\approx \frac{1}{((4-2x)^{\frac{1}{4}})^4} + ((4-2x)^{\frac{1}{4}})^8 \\
&= \frac{1}{4-2x} + (4-2x)^2 && \text{(using } (a^b)^c = a^{bc} \text{)} \\
&= \frac{1}{4-2x} + \frac{(4-2x)^2(4-2x)}{4-2x} && \text{(common denominator)} \\
&= \frac{1 + (4-2x)^3}{4-2x}
\end{aligned}$$

This completes the question, as long as $x \neq 2$, which is always true. This is because the approximation is valid whenever t is positive and small, which means that x is close to 2 but not equal to 2 since $x = \cos(t^2) + 1$ is only equal to 2 at $t = 0$. Moreover, we deduce that x must be slightly less than 2 since $\cos(t^2) + 1$ is slightly less than 2 when t is slightly bigger than 0.

This means that the approximation is valid slightly below $x = 2$.

□

Example 90. A curve is defined by

$$\begin{aligned}
x &= \frac{1}{t} + 1, \quad t \neq 0 \\
y &= \cos t
\end{aligned}$$

Show that we can write

$$y = \cos \left(\frac{1}{ax+b} \right)$$

Where a and b are constants to be found. Then find the derivative:

$$\frac{dy}{dx}$$

Solution. Firstly we can rearrange x in terms of t :

$$\begin{aligned}
x &= \frac{1}{t} + 1 \\
x - 1 &= \frac{1}{t} \\
t(x - 1) &= 1 \\
t &= \frac{1}{x - 1} && \text{(as long as } x \neq 1 \text{)}
\end{aligned}$$

Then we can substitute t into y to find:

$$y = \cos \left(\frac{1}{x - 1} \right)$$

Which finds that $a = 1$ and $b = -1$ so that

$$y = \cos \left(\frac{1}{ax+b} \right) = \cos \left(\frac{1}{x-1} \right)$$

Now to differentiate y in terms of x , we must use the chain rule. We can ask our question of: *if we could replace any part with u , how can we turn it into a function we know how to differentiate?*

Note that we know how to differentiate $\cos u$, so we write:

$$\begin{aligned} y &= \cos\left(\frac{1}{x-1}\right) \\ y &= \cos u && \text{(what we want)} \\ u &= \frac{1}{x-1} && \text{(to get what we want)} \end{aligned}$$

Now remind ourselves of the chain rule Section 3.4, e.g. by pretending that the du cancel:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

And therefore we can differentiate the equations to find the right-hand side of the above:

$$\frac{dy}{du} = -\sin u$$

But what about differentiating $u = \frac{1}{x-1}$? For this, we can do another chain rule! After asking ourselves the same question, we proceed with:

$$\begin{aligned} u &= \frac{1}{x-1} \\ u &= \frac{1}{v} && \text{(what we want)} \\ v &= x-1 && \text{(to get what we want)} \end{aligned}$$

And the chain rule in this case would be:

$$\frac{du}{dx} = \frac{du}{dv} \frac{dv}{dx}$$

We can differentiate each equation to get the right-hand side of the above:

$$\begin{aligned} \frac{du}{dv} &= -\frac{1}{v^2} && \text{(writing } \frac{1}{v} = v^{-1}) \\ \frac{dv}{dx} &= 1 \\ \therefore \frac{du}{dx} &= -\frac{1}{v^2} && \text{(using the chain rule)} \\ &= -\frac{1}{(x-1)^2} && \text{(using } v = x-1) \end{aligned}$$

Therefore, reminding ourselves, we have found:

$$\begin{aligned} \frac{du}{dx} &= -\frac{1}{(x-1)^2} \\ \frac{dy}{du} &= -\sin u && \text{(from above)} \\ \therefore \frac{dy}{dx} &= -\sin u \left(-\frac{1}{(x-1)^2}\right) && \text{(using the chain rule)} \\ &= \frac{\sin u}{(x-1)^2} \\ &= \frac{\sin\left(\frac{1}{x-1}\right)}{(x-1)^2} && \text{(using } u = \frac{1}{x-1}) \end{aligned}$$

Therefore, if $y = \cos\left(\frac{1}{x-1}\right)$, then

$$\frac{dy}{dx} = \frac{\sin\left(\frac{1}{x-1}\right)}{(x-1)^2}$$

This completes the question.

□

9 Vectors

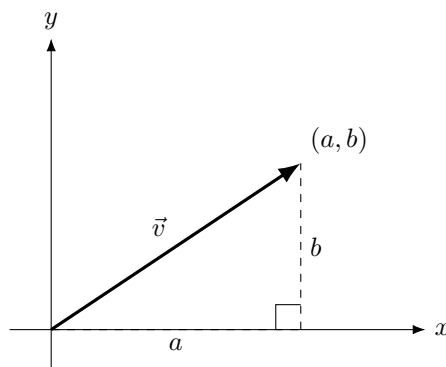
Vectors encode a length and a direction. It is good to visualise them as an arrow from a base point, and adding a vector takes you from the base point to the end of that arrow.

In this way, you can visualise an ant travelling around by adding and taking away vectors. The output of the adding and taking away will give you a single vector, which encodes the length and direction from the starting point to the final point the ant ended at.

Vectors are written as a column of numbers, where each number corresponds to moving an amount in that axis' direction.

This allows us to visualise moving in a vector as moving each axis individually.

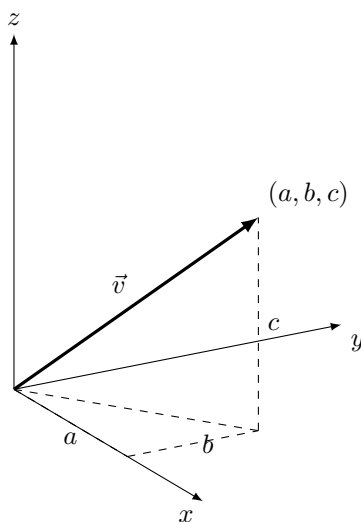
To get the overall length of the vector, we can now ask ourselves how we can find the length of the line formed by drawing the shape formed by going in each axis' direction as instructed by the vector.



Drawn like this, it should be obvious that the length of $\vec{v}$ can be found using Pythagoras:

$$|\vec{v}| = \sqrt{a^2 + b^2}$$

We can extend this to 3 dimensions:



And some geometry (think about why this is true by splitting into two different triangles!) can let you find out that:

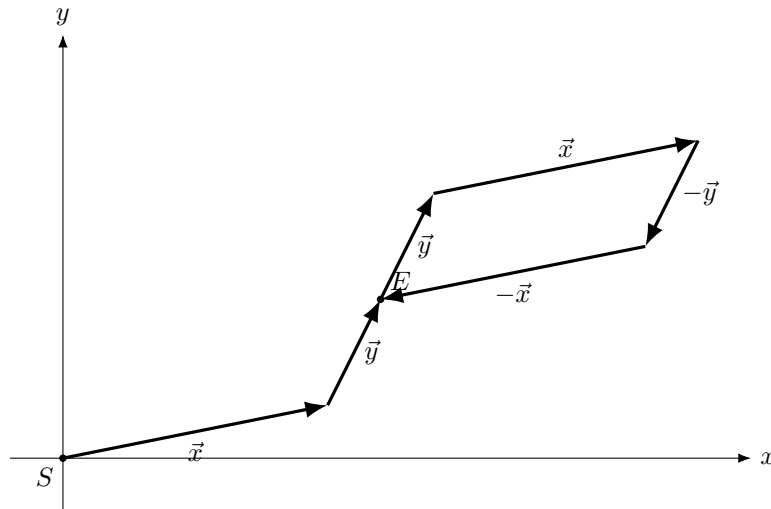
$$|\vec{v}| = \sqrt{\sqrt{a^2 + b^2}^2 + c^2} = \sqrt{a^2 + b^2 + c^2}$$

This gives us the equations for lengths of two and three-dimensional vectors as the square root of the sum of squares of the components! This should now be clear that it is the same as Pythagoras.

To continue the discussion about imagining vectors as a path you can follow from a starting point to an end point, let's imagine we have two vectors $\vec{x}$ and $\vec{y}$. We are an ant that starts at the origin and can only move using these vectors. We choose to do the following:

1. Go along $\vec{x}$
2. Go along $\vec{y}$
3. Go along $\vec{y}$
4. Go along $\vec{x}$
5. Go along the reverse of $\vec{y}$
6. Go along the reverse of $\vec{x}$

This can be visualised as the following, labelling the start as S and the end as E:



If we wanted to write the outcome of this path as a single vector, we can sum all of the steps we took:

$$\vec{SE} = \vec{x} + \vec{y} + \vec{y} + \vec{x} - \vec{y} - \vec{x}$$

Which we can simplify as usual:

$$\vec{SE} = \vec{x} + \vec{y}$$

This gives us exactly what the arrow between the origin and the end point would be as a vector, which in this case is much simpler than the path we actually took to get there.

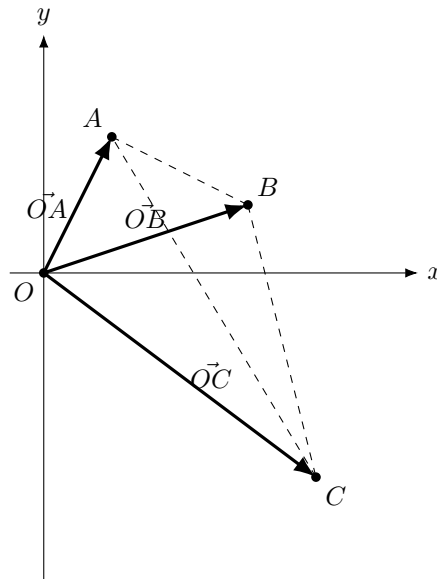
Example 91. Jim wants to test two routes to his work. Jim lives at point A , and works at point C , with a point B as a potential place to stop on his way. Jim has a map that shows distances to the town centre, which he marks as O , the origin. Jim knows the vectors:

$$\vec{OA} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \vec{OB} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \vec{OC} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Jim compares the two routes by going to work via B but going directly from work to his home without stopping at B .

By calculating his journey as vectors, find the distances he travelled to and from work.

Solution.



On the way to work, Jim takes the following path given by $\vec{AB}$ and then $\vec{BC}$.

On the way back, Jim follows the path given by $\vec{CA}$.

To find the path $\vec{AB}$, we imagine ourselves as an ant at point A , where we are only allowed to move along or back along the arrows we have been given in the question. From this we realise we want to travel backwards through $\vec{OA}$ and then forwards through $\vec{OB}$.

This results in:

$$\vec{AB} = -\vec{OA} + \vec{OB}$$

We now substitute the values and add the components:

$$\vec{AB} = -\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

Then repeat for $\vec{BC}$:

$$\begin{aligned} \vec{BC} &= -\vec{OB} + \vec{OC} \\ &= -\begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ -4 \end{pmatrix} \end{aligned}$$

Then to get the lengths, we use the equation for length of a vector (secretly Pythagoras) on each leg of the journey separately to find the distance Jim travelled to work by travelling through B is

$$|\vec{AB}| + |\vec{BC}| = \sqrt{2^2 + 1^2} + \sqrt{1^2 + (-4)^2} = \sqrt{5} + \sqrt{17}$$

What about on the way back? Jim travels directly $\vec{CA}$ on the way back. There are many ways we could calculate this as a vector, using the fact we know if we add the individual vector steps we get the overall vector taking us from the starting point to the end point.

1. Through town: $\vec{CA} = -\vec{OC} + \vec{OA}$
2. Reversing the original route: $\vec{CA} = -\vec{BC} - \vec{AB}$
3. Whatever this route is: $\vec{CA} = -\vec{OC} + \vec{OB} + \vec{BA}$
4. ...

In fact, the right-hand side will be identical for all of these options! Let's calculate the first two:

$$\begin{aligned}\vec{CA} &= -\vec{OC} + \vec{OA} \\ &= -\begin{pmatrix} 4 \\ -3 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ 5 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\vec{CA} &= -\vec{BC} - \vec{AB} \\ &= -\begin{pmatrix} 1 \\ -4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ 5 \end{pmatrix}\end{aligned}$$

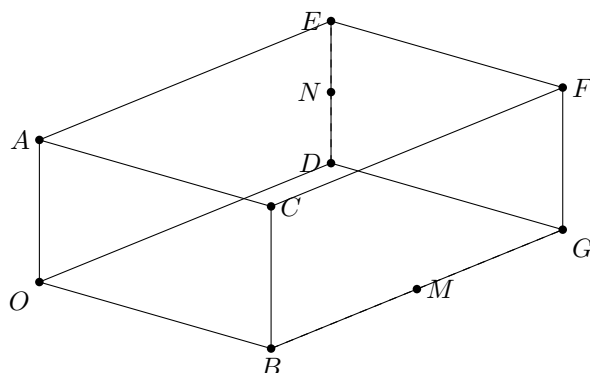
Which suggests that maths is correct.

We may now calculate the distance going directly home would be:

$$\left| \begin{pmatrix} -3 \\ 5 \end{pmatrix} \right| = \sqrt{(-3)^2 + 5^2} = \sqrt{9 + 25} = \sqrt{34}$$

You can check in a calculator that it is a shorter distance to travel directly in this case. In fact, it is always true that adding a point along the journey can only increase the journey distance. $\square$

Example 92. You are given a cuboid with one corner at the origin with the remaining vertices marked A through G , N is at the midpoint of D and E , and M is at the midpoint of B and G as in the diagram below.



You are told that

1. $|\vec{OA}| = 1$
2. $|\vec{OB}| = 2$
3. $|\vec{OD}| = 3$

and that A is on the z axis, B is on the x axis, and D is on the y axis.

Find the vector $\vec{MN}$, and then find its length.

Solution. With the information given, we can write

$$\vec{OA} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (9.0.1)$$

$$\vec{OB} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \quad (9.0.2)$$

$$\vec{OD} = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \quad (9.0.3)$$

$$(9.0.4)$$

Since it is a cuboid, we know that all lines with the same direction (parallel) must have the same vector (they are the same length in the same direction).

For example, we know that

$$\begin{aligned} \vec{BG} &= \vec{OD} \\ \vec{DE} &= \vec{OA} \end{aligned}$$

Equipped with this, let's find a path from M to N via the vectors we know:

$$\vec{MN} = \vec{MB} + \vec{BO} + \vec{OD} + \vec{DN}$$

We know the vector $\vec{OD}$, and we also know $\vec{BO} = -\vec{OB}$ since it is in the opposite direction with the same length.

Now consider $\vec{BM}$. Since it is halfway along $\vec{BG}$, we know that

$$B\vec{M} = \frac{1}{2}B\vec{G}$$

And since we also know $B\vec{G} = O\vec{D}$, we can write

$$B\vec{M} = \frac{1}{2}O\vec{D} = \frac{1}{2} \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}$$

Therefore

$$M\vec{B} = \begin{pmatrix} 0 \\ -3/2 \\ 0 \end{pmatrix}$$

Similarly, since $D\vec{N}$ is halfway along $D\vec{E}$, we can calculate

$$\begin{aligned} D\vec{N} &= \frac{1}{2}D\vec{E} \\ &= \frac{1}{2}O\vec{A} && \text{(using } D\vec{E} = O\vec{A}) \\ &= \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix} \end{aligned}$$

Therefore, we have found that

$$\begin{aligned} M\vec{N} &= M\vec{B} + B\vec{O} + O\vec{D} + D\vec{N} \\ &= \begin{pmatrix} 0 \\ -3/2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ 3/2 \\ 1/2 \end{pmatrix} \end{aligned}$$

Therefore $M\vec{N}$ has length $|M\vec{N}| = \sqrt{(-2)^2 + (3/2)^2 + (1/2)^2} = \sqrt{13/2}$. □

10 Sequences and Series

There are many kinds of series and sequences, but we care about two of them: *arithmetic* and *geometric*. In general, a sequence is an ordered list of numbers, and a series is a sum of a sequence.

10.1 Arithmetic Series

An arithmetic sequence is one in which the distance between consecutive points is the same:

i	1	2	3	4	...
x_i	-6	1	8	15	...

In the above, each term has a gap of 7, which can be worked out by calculating $x_n - x_{n-1}$, and verifying that it is 7 in each case.

What if we wanted to add the terms together? What if we wanted to find a formula for the arithmetic series?

A general arithmetic series with n terms looks like the following:

$$S = a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d)$$

Here a is the first term, and d is the distance between consecutive terms.

Since we are mathematicians, let's give the sum a label S .

$$a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d)$$

Here comes a fun trick, if we write the sum backwards:

$$S = (a + (n - 1)d) + (a + (n - 2)d) + (a + (n - 3)d) + \cdots + a$$

Why would we do this? It turns out that adding together the two simplifies our lives greatly:

$$\begin{aligned} S + S &= && a + && (a + d) + && (a + 2d) + && \cdots + && (a + (n - 1)d) \\ &+ && (a + (n - 1)d) + && (a + (n - 2)d) + && (a + (n - 3)d) + && \cdots + && a \\ &= && (2a + (n - 1)d) + && (2a + (n - 1)d) + && (2a + (n - 1)d) + && \cdots + && (2a + (n - 1)d) \\ &= && n(2a + (n - 1)d) \end{aligned}$$

The last line is since we had n of the same term added together. Therefore we have found:

$$2S = n(2a + (n - 1)d)$$

And dividing gives us the formula for an arithmetic series of n terms, starting at a and with a distance of d :

$$S = \frac{n}{2}(2a + (n - 1)d)$$

Example 93. An arithmetic series is given by

$$-6 + 1 + 8 + 15 + 22 + \dots$$

What is the sum of the first 7 terms?

Solution. If this is an arithmetic series, then each consecutive term must have the same distance apart.

We can calculate that this distance should be $1 - (-6) = 7$, and can verify this is true for the remaining entries.

Then all we need to do is plug into our formula, using that the first term is $a = -6$ and we found that $d = 7$, and we want the first $n = 7$ terms:

$$\begin{aligned} S &= \frac{n}{2}(2a + (n-1)d) \\ &= \frac{7}{2}(2 \cdot (-6) + (7-1)7) \\ &= \frac{7}{2}(-12 + 6 \cdot 7) \\ &= 7(-6 + 3 \cdot 7) \\ &= 7(15) \\ &= 105 \end{aligned}$$

This finishes the question, finding the sum of the first 7 terms to be 105. □

10.2 Geometric Series

A geometric sequence is one in which the ratio between consecutive points is the same:

i	1	2	3	4	...
x_i	32	16	8	4	...

In the above, each term is $1/2$ of the previous, which can be worked out by calculating x_n/x_{n-1} , and verifying that it is $1/2$ in each case.

What if we wanted to add the terms together? What if we wanted to find a formula for the geometric series?

A general geometric series with n terms looks like the following:

$$a + ra + r^2a + \dots + r^{n-2}a + r^{n-1}a$$

Here a is the first term, and r is the ratio between consecutive terms.

Since we are mathematicians, let's give the sum a label S .

$$a + ra + r^2a + \dots + r^{n-2}a + r^{n-1}a$$

Here comes another fun trick, if we write the sum multiplied by r :

$$rS = ra + r^2a + r^3a + \dots + r^{n-1}a + r^na$$

Why would we do this? It turns out that subtracting the two simplifies our lives greatly:

$$\begin{array}{ccccccccc} S - rS & = & a & & & & & & \\ & & & +\cancel{ra} & +\cancel{r^2a} & +\dots & +\cancel{r^{n-1}a} & & \\ & & & -\cancel{ra} & -\cancel{r^2a} & -\dots & -\cancel{r^{n-1}a} & -r^na & \\ & = & a - r^na & & & & & & \end{array}$$

After factorising the left and right side, we find:

$$S(1 - r) = a(1 - r^n)$$

And dividing gives us the formula for geometric series of n terms, starting at a and with a ratio of r :

$$S = \frac{a(1 - r^n)}{1 - r}$$

Example 94. A geometric series is given by

$$32 + 16 + 8 + 4 + \dots$$

What is the sum of the first 5 terms?

Solution. If this is a geometric series, then each consecutive term must have the same ratio.

We can calculate that this ratio should be $16/32 = 1/2$, and can verify this is true for the remaining entries.

Then all we need to do is plug into our formula, using that the first term is $a = 32$ and we found that $r = 1/2$, and we want the first $n = 5$ terms:

$$\begin{aligned} S &= \frac{a(1 - r^n)}{1 - r} \\ &= \frac{32 \left(1 - \left(\frac{1}{2}\right)^5\right)}{1 - \frac{1}{2}} \\ &= 2 \cdot 32 \left(1 - \left(\frac{1}{2}\right)^5\right) \\ &= 2 \cdot 32 \left(1 - \frac{1}{32}\right) \\ &= 2 \cdot 32 \cdot \frac{31}{32} \\ &= 2 \cdot 31 \\ &= 62 \end{aligned}$$

This completes the question, finding the sum of the first 5 terms to be 62. □

11 Proofs

This is the section that is most like what a degree in maths would look like.

There are a few ways to attempt a proof.

11.1 Proof by Contradiction

A proof by contradiction works by assuming the opposite is true, and then applying logically correct steps until we reach a logical absurdity such as $1 = 0$. If we reached a logical absurdity via logically correct steps, then the original assumption must be wrong. Since the opposite of what we wanted to prove is wrong, the original statement was correct.

Example 95. Prove by contradiction that

$$(a - 2\sqrt{ab})(a + 2\sqrt{ab}) \geq -4b^2$$

for any non-negative real numbers a, b .

Solution. First assume the opposite is true:

$$(a - 2\sqrt{ab})(a + 2\sqrt{ab}) < -4b^2$$

Then let us rearrange this until we reach a contradiction.

$$\begin{aligned}(a - 2\sqrt{ab})(a + 2\sqrt{ab}) &< -4b^2 \\ a^2 - 4ab &< -4b^2 && \text{(expanding)} \\ a^2 - 4ab + 4b^2 &< 0 && \text{(rearranging)} \\ (a - 2b)^2 &< 0 && \text{(factorising)}\end{aligned}$$

This is a contradiction because the left-hand side is a square number, and a square number cannot be less than zero. Therefore our original assumption

$$(a - 2\sqrt{ab})(a + 2\sqrt{ab}) < -4b^2$$

was false, which means that the following must be true:

$$(a - 2\sqrt{ab})(a + 2\sqrt{ab}) \geq -4b^2$$

This completes the proof. □

11.2 Proof by Deduction

Proof by deduction is the act of starting with a true statement, applying logically correct steps, and ending on a statement that must therefore be true.

It is like “proof because it is true” but made rigorous.

Example 96. Starting with the statement

$$(a - 2b)^2 \geq 0$$

Prove by deduction that

$$(a - 2\sqrt{ab})(a + 2\sqrt{ab}) \geq -4b^2$$

for any non-negative real numbers a, b .

Solution. We start with the true statement, and rearrange to deduce the desired final statement.

$$\begin{aligned}
(a - 2b)^2 &\geq 0 && \text{(starting with the hint)} \\
a^2 - 4ab + 4b^2 &\geq 0 && \text{(expanding)} \\
a^2 - 4ab &\geq -4b^2 && \text{(rearranging)} \\
a^2 - \sqrt{4ab}^2 &\geq -4b^2 && \text{(using that } a, b \text{ are non-negative real numbers)} \\
(a - \sqrt{4ab})(a + \sqrt{4ab}) &\geq -4b^2 && \text{(difference of two squares)} \\
(a - 2\sqrt{ab})(a + 2\sqrt{ab}) &\geq -4b^2 && \text{(using } \sqrt{xy} = \sqrt{x}\sqrt{y})
\end{aligned}$$

Which is the statement we were trying to prove. This completes the proof. $\square$

11.3 Proof by Exhaustion

Proof by exhaustion is a method of proof that is essentially brute force; you try every case, and verify each case.

This can be used when there are a small number of cases to test.

Often, it is possible to use algebra to reduce what seems like a lot (even infinite) cases to only a few.

For example, if you need to prove that n^2 preserves even and odd for all integers, you can test all integers in two cases by splitting them into even $2m$ numbers and odd $2m - 1$ numbers. In this way we reduce “all integers” to two cases and can prove it by exhaustion.

Example 97. Prove by exhaustion that, for every integer n whenever the left-hand side is defined,

$$\sqrt{5 - n^2} < 3$$

Solution. Since $\sqrt{5 - n^2}$ is only defined when $5 - n^2 \geq 0$, we only need to check the integers $-2, -1, 0, 1, 2$.

Since $(-n)^2 = n^2$, we only need to check 0, 1, and 2,

Starting with $n = 2$,

$$\begin{aligned}
\sqrt{5 - 2^2} &= \sqrt{5 - 4} \\
&= \sqrt{1} \\
&= 1
\end{aligned}$$

And since $1 < 3$, the statement is true for $n = 2$.

For $n = 1$,

$$\begin{aligned}
\sqrt{5 - 1^2} &= \sqrt{5 - 1} \\
&= \sqrt{4} \\
&= 2
\end{aligned}$$

And since $2 < 3$, the statement is true for $n = 1$.

For $n = 0$,

$$\begin{aligned}\sqrt{5-0^2} &= \sqrt{5-0} \\ &= \sqrt{5}\end{aligned}$$

Now to check if $\sqrt{5}$ is less than 3, we can observe that

$$5 < 9$$

And therefore, taking square roots of both sides, we can deduce

$$\sqrt{5} < 3$$

Which completes the proof by exhaustion, because we have checked every case in which

$$\sqrt{5-n^2} < 3$$

is defined. □

Example 98. Using algebra, prove by exhaustion that $n^2 - 3$ is never a multiple of 9 for any integer n .

Solution. We can split integers into three cases: $3m$, $3m + 1$, and $3m + 2$ where m is an integer.

This accounts for every integer.

Case 1: $n = 3m + 2$

Let's substitute and rearrange to see what we can deduce.

$$\begin{aligned}n^2 - 3 &= (3m + 2)^2 - 3 && \text{(using } n = 3m + 2\text{)} \\ &= 9m^2 + 12m + 4 - 3 && \text{(expanding)} \\ &= 9m^2 + 12m + 1 \\ &= 3(3m^2 + 4m) + 1\end{aligned}$$

Looking at the final line, we see that when $n = 3m + 2$, $n^2 - 3$ is one more than a multiple of 3, and therefore never a multiple of 3

Since 9 is a multiple of 3, it is also never a multiple of 9.

Case 2: $n = 3m + 1$

Let's substitute and rearrange to see what we can deduce.

$$\begin{aligned}n^2 - 3 &= (3m + 1)^2 - 3 && \text{(using } n = 3m + 1\text{)} \\ &= 9m^2 + 6m + 1 - 3 && \text{(expanding)} \\ &= 9m^2 + 6m - 2 \\ &= 3(3m^2 + 2m) - 2\end{aligned}$$

Looking at the final line, we see that when $n = 3m + 1$, $n^2 - 3$ is two less than a multiple of 3, and therefore never a multiple of 3.

Since 9 is a multiple of 3, it is also never a multiple of 9.

Case 3: $n = 3m$

Let's substitute and rearrange to see what we can deduce.

$$\begin{aligned}n^2 - 3 &= (3m)^2 - 3 && \text{(using } n = 3m\text{)} \\&= 9m^2 - 3 && \text{(expanding)}\end{aligned}$$

Looking at the final line, we see that when $n = 3m$, $n^2 - 3$ is three less than a multiple of 9, and therefore never a multiple of 9

Since we have exhausted all cases, we can conclude that $n^2 - 3$ is never a multiple of 9 for any integer n . This completes the proof. $\square$

12 Numerical Methods

Numerical methods are ways to approximate the solution to equations using a repeatable procedure, usually designed for computers to use efficiently.

In an exam, you should assume that you are to solve a question exactly unless specifically told otherwise, e.g. “by using the trapezium rule, estimate the integral.”

In fact, it is always preferred to get an exact solution by solving mathematically rather than estimating using one of these techniques, and these should be reserved for when that is not possible or feasible.

12.1 The Trapezium Rule

The goal of the trapezium rule is to estimate the area under the curve²⁵ as in the shaded area of Figure 35, i.e. a way to estimate an integral, using only knowledge of the function we are trying to integrate at a few points.

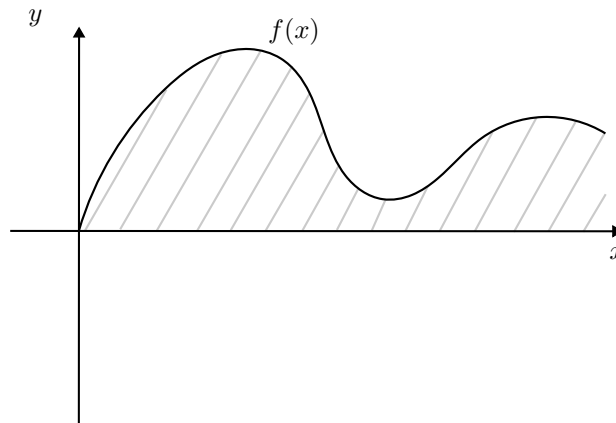


Figure 35: A function shaded below the curve

One approach is to:

1. Choose a bunch of x points we want to know the value of the function at.
2. Find what $f(x)$ is at each point chosen.
3. Connect the $f(x)$ points with straight lines, and draw lines down to the x axis from each $f(x)$ point found. This will make a sequence of trapeziums, where the “top” of the trapezium is approximating the function curve with straight lines. This can be seen in Figure 36.
4. Calculate the area of each trapezium and add them together.

²⁵Actually we are estimating the integral, which is a signed area since the integral of a negative function is also negative. This is left here to assist intuition.

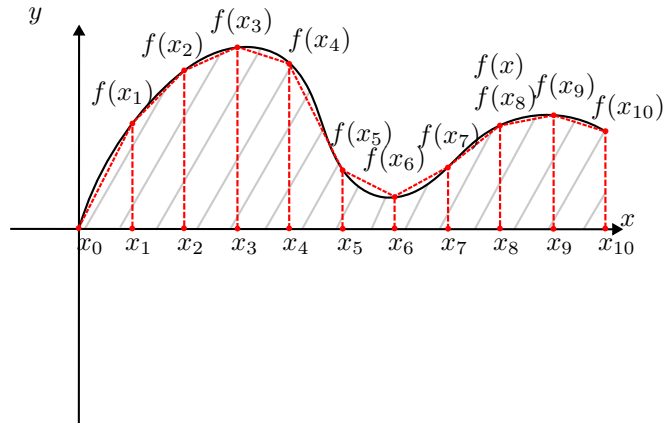


Figure 36: A function shaded below the curve, with trapeziums drawn, equal width.

Recall that for a trapezium with side lengths a and b , and height h , the area of a trapezium is the following.

$$\text{Area of trapezium} = \frac{h(a + b)}{2}$$

This can be proved by formalising the geometric construction seen in Figure 37, creating a copy of the trapezium, rotating it 180° , and then gluing it on the side of the original trapezium. It can be proved²⁶ that this creates a parallelogram with width $a + b$ and height h , which has a total area of $h(a + b)$. Then since we duplicated our trapezium, this must be twice the area of the original trapezium.

²⁶You would approach this by labelling the angles in the outer triangles, and using the fact that they must add up to 180° to find that the construction does indeed create a parallelogram.

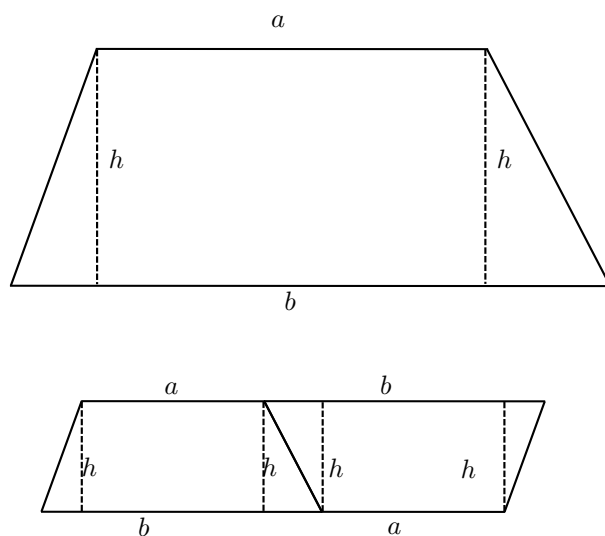


Figure 37: A trapezium glued to itself to form a parallelogram.

So what are the areas of the trapeziums in our problem? For convenience, we will space the x points out evenly, with a distance of h between each one. As in, $x_1 = x_0 + h$, $x_2 = x_1 + h$, and so on for how many x points we chose.

Firstly, let's label the diagram with the widths and heights of the trapeziums, as in Figure 38.

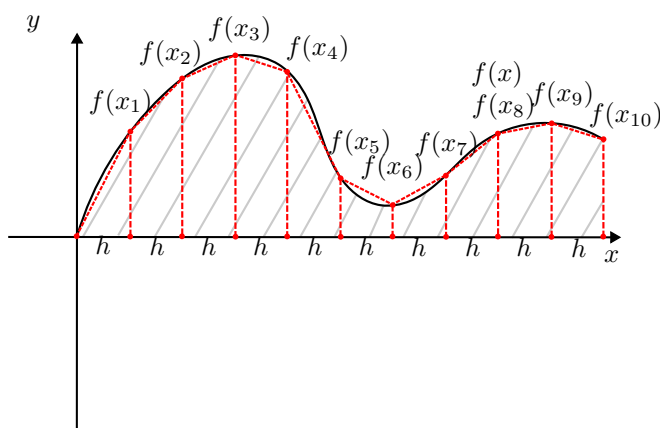


Figure 38: A function shaded below the curve, with trapeziums drawn, equal width.

From this figure, we can notice that each trapezium has a side length of $f(x_i)$ and $f(x_{i+1})$, and a height of h .

Therefore, the area of one of the trapeziums must be of the form:

$$\text{Area of trapezium } i = \frac{h(f(x_i) + f(x_{i+1}))}{2}$$

To approximate the total area of $f(x)$ between two points, we add them all together to get (starting from x_0 and ending at x_n):

$$\begin{aligned} \int_{x_0}^{x_n} f(x)dx &\approx \frac{h(f(x_0) + \textcolor{red}{f(x_1)})}{2} && \text{(first trapezium)} \\ &+ \frac{h(\textcolor{red}{f(x_1)} + \textcolor{blue}{f(x_2)})}{2} && \text{(second trapezium)} \\ &+ \frac{h(\textcolor{blue}{f(x_2)} + f(x_3))}{2} && \text{(third trapezium)} \\ &+ \dots \\ &+ \frac{h(f(x_{n-2}) + \textcolor{brown}{f(x_{n-1})})}{2} && \text{(second-to-last trapezium)} \\ &+ \frac{h(\textcolor{brown}{f(x_{n-1})} + f(x_n))}{2} && \text{(last trapezium)} \end{aligned}$$

Similar terms have been colour-coded. In fact, the right-hand-side can be simplified drastically by collecting the similar terms. You can see that the term at the end of the line appears at the start of the next line, which means that there are repeat terms for every line except the first or the last one. Let's expand and collect the terms together to see what we get:

$$\begin{aligned} \int_{x_0}^{x_n} f(x)dx &\approx \frac{h}{2} \left(f(x_0) + \textcolor{red}{f(x_1)} \right. && \text{(pulling } h/2 \text{ out of the full equation)} \\ &\quad + \textcolor{blue}{f(x_2)} + f(x_3) \\ &\quad + \dots \\ &\quad + f(x_{n-2}) + \textcolor{brown}{f(x_{n-1})} \\ &\quad \left. + \textcolor{brown}{f(x_{n-1})} + f(x_n) \right) \\ &= \frac{h}{2} \left(f(x_0) + 2\textcolor{red}{f(x_1)} + 2\textcolor{blue}{f(x_2)} + \dots + 2\textcolor{brown}{f(x_{n-1})} + f(x_n) \right) && \text{(collecting terms)} \\ &= \frac{h}{2} \left(f(x_0) + f(x_n) + 2(\textcolor{red}{f(x_1)} + \textcolor{blue}{f(x_2)} + \dots + \textcolor{brown}{f(x_{n-1})}) \right) && \text{(rearranging)} \end{aligned}$$

This gives us the trapezium rule from the formula book:

$$\int_{x_0}^{x_n} f(x)dx \approx \frac{h}{2} \left(f(x_0) + f(x_n) + 2(f(x_1) + f(x_2) + \dots + f(x_{n-1})) \right)$$

Now the question is: what is h in terms of n ?

If we know what the start and end points are that we are integrating between, and we know how many points we are going to use, we can figure out what h is.

To split the line-segment between x_0 and x_n into n segments, we need to set h to be

$$h = \frac{x_n - x_0}{n}$$

So that

$$x_{i+1} = x_i + h$$

Note that since we are counting from x_0 to x_n , we actually have $(n + 1)$ points, but n segments between them.

Example 99. By splitting the interval between 0 and 3 with 3 equally spaced segments, use the trapezium rule to estimate the integral:

$$\int_0^3 x^2 dx$$

Then compare this with the answer you get from integrating it directly.

Solution. The 3 equally spaced segments will have $h = \frac{3-0}{3} = \frac{3}{3} = 1$. That means we have:

$$\begin{aligned}x_0 &= 0 \\x_1 &= x_0 + h = 1 \\x_2 &= x_1 + h = 2 \\x_3 &= x_2 + h = 3\end{aligned}$$

Setting $f(x) = x^2$, let's compute each value as a table:

x	0	1	2	3
$f(x) = x^2$	0	1	4	9

Now let's plug this into the trapezium rule:

$$\begin{aligned}\int_{x_0}^{x_n} f(x) dx &\approx \frac{h}{2} \left(f(x_0) + f(x_n) + 2(f(x_1) + f(x_2) + \cdots + f(x_{n-1})) \right) \\ \therefore \int_0^3 x^2 dx &\approx \frac{1}{2} \left(0 + 9 + 2(1 + 4) \right) \\ &= \frac{1}{2} \left(9 + 2(5) \right) \\ &= \frac{19}{2} \\ &= 9.5\end{aligned}$$

Comparing this to computing the integral directly:

$$\begin{aligned}\int_{x_0}^{x_n} f(x) dx &= \int_0^3 x^2 dx \\ &= \left[\frac{1}{3} x^3 \right]_{x=0}^{x=3} \\ &= \frac{1}{3} 3^3 \\ &= 3^2 \\ &= 9\end{aligned}$$

Comparing it with the trapezium rule, we were only out by 0.5 – the trapezium rule with 3 equally spaced segments got 9.5 as the integral whereas the actual answer was 9. Pretty close! $\square$

In the previous question, we found that the trapezium rule overshoot the real answer. If you draw out the function with the trapeziums, you can see this visually – the trapeziums include sections

above the curve since it curves upwards. In fact, you can usually anticipate whether the trapezium rule will over or undershoot the actual integral by looking at the section which does not overlap between the trapeziums and the function itself, look at Figure 36 again to see which parts the trapeziums include more area than the function, and which parts the trapeziums include less area than the function. Notice the patterns in the overlap when the function is curving in the same direction.

Example 100. You are given the following grid of values for a function.

x	-1	0.25	1.5	2.75
$f(x)$	-1	4	6	5

By using the Trapezium rule, estimate the integral of the curve between $x = -1$ and $x = 2.75$.

If you are told that the curve's derivative is always decreasing, what can you say about whether the estimate for the area will be under or over the true area of $f(x)$?

Solution. First we need to determine h . Since h is the distance between two x points, we can deduce that it should be:

$$h = 0.25 - (-1) = 1.25$$

And we can verify that each point is 1.25 apart, meaning that $h = 1.25$ is the distance between each point, and we can use the trapezium rule. Now let's plug this into the trapezium rule:

$$\begin{aligned}
 \int_{x_0}^{x_n} f(x) dx &\approx \frac{h}{2} \left(f(x_0) + f(x_n) + 2(f(x_1) + f(x_2) + \cdots + f(x_{n-1})) \right) \\
 \therefore \int_{-1}^{2.75} f(x) dx &\approx \frac{1.25}{2} \left(-1 + 5 + 2(4 + 6) \right) \\
 &= \frac{1.25}{2} \left(4 + 2(10) \right) \\
 &= \frac{1.25 \times 24}{2} \\
 &= 15
 \end{aligned}$$

Therefore the trapezium rule gives us an estimate of the area as 15.

For the second part of the question, if we draw a sketch of the curve, we know that the slope must be going from steep to less steep since the question told us the derivative is decreasing.

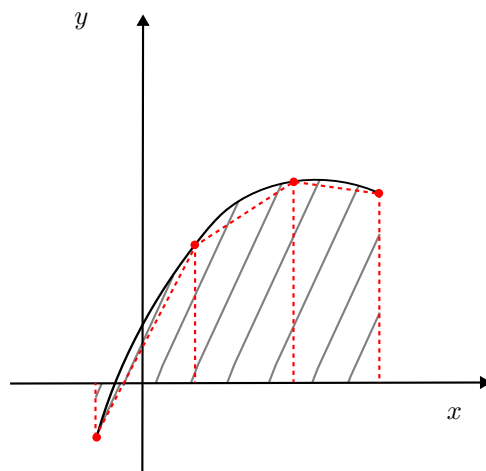


Figure 39: A function shaded below the curve, with trapeziums drawn.

Now it becomes clear that the trapeziums are always under the curve. When the curve is above zero, this means that the trapezium area is less than the curve's area. What about when the function is below zero? In this case, the trapezium area is covering more space than the function is. But since it is negative, this should contribute more “negative area”. However since we have not marked zero itself in the table, the trapezium calculation we are computing has negative side (!), which affects the area in this case.

The actual “trapezium” we are computing for the first trapezium is:

$$\frac{h(-1 + 4)}{2}$$

This makes it unclear whether the area in this case is less than or more than the actual function, at least for the geometrical approach we have been using, but it turns out the trapezium rule still underestimates the true integral in this case too, as long as the curve's derivative is decreasing²⁷. $\square$

12.2 The Newton-Raphson Method

The Newton-Raphson method has the goal of improving an estimate for a root of a function, i.e. finding a point x where some function $f(x) = 0$, given that we know a point close to the root already.

The curve pictured in Figure 40 has three roots. We could try to solve $f(x) = 0$ by rearranging for x , but this is not always possible to do.

Instead, we can try to get *close* to a root. The easiest way to do so is to find two points a and b where the function changes sign between the two points. If the function goes from positive to

²⁷If we instead think of the straight line segment itself between x_0 and x_1 as a function, say $L(x)$, it follows due to the decreasing derivative that $f(x) \geq L(x)$ on this interval, and the “trapezium area” we have been using is given as $\int_{x_0}^{x_1} L(x)dx$. It then follows that $\int_{x_0}^{x_1} f(x)dx \geq \int_{x_0}^{x_1} L(x)dx$, which shows that the trapezium rule still undershoots the true value in this case.

negative, or from negative to positive, we know that it must be zero at some point between. This will give us a point close to the root.

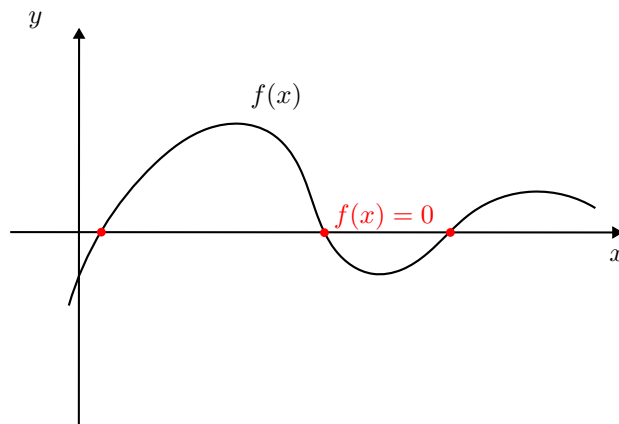


Figure 40: A function with zeros labelled.

Once we have a point close to the root, how can we improve our guess? One method is to notice that near the root, if we zoom in far enough, the function looks like a straight line. Because of this, we might notice that if we were to draw the tangent to the curve at our point, following the tangent until we hit the x axis should land us closer to the root.

This is the Newton-Raphson method:

1. Find a point close to the root
2. Find the tangent to the curve at that point
3. Find the point on the x axis in which the tangent meets the x axis
4. Repeat steps 2-3 until you are satisfied with getting near the root

This is seen pictured in Figure 41, where you can see the tangents leading us closer to the red point we desire to land at.

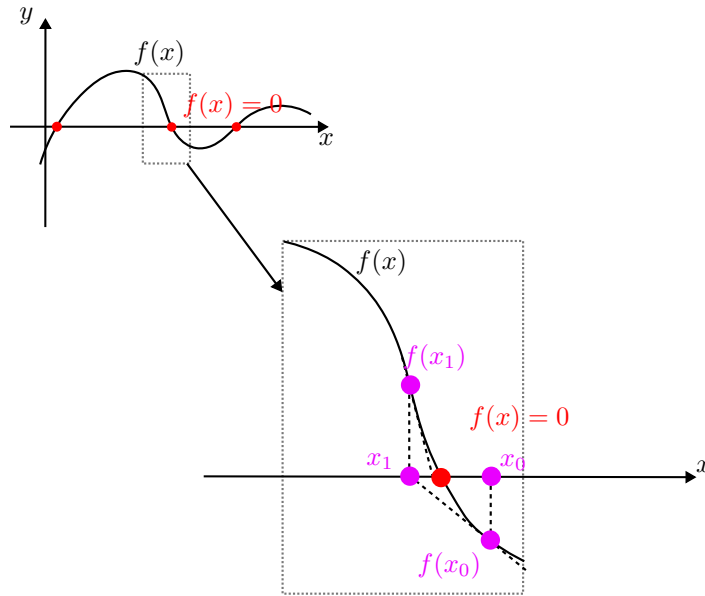


Figure 41: A function with Newton iterations labelled around one root.

So how can we turn the above instructions into maths?

Firstly, we want to find the gradient of our function f at an initial point x_0 . This is given by $f'(x_0)$, as in by differentiating f and plugging in x_0 .

Then given this gradient, we want to find the equation of the straight line which has this gradient, and passes through the point $f(x_0)$, which is exactly the properties we want from the tangent to a curve! It needs to “touch” the curve and have the slope the curve has at that point.

We can write a general straight line as $y = mx + c$, where m is the gradient, and c is the y intercept. But we know that at x_0 , our tangent line passes through $f(x_0)$ with gradient $m = f'(x_0)$.

We can write this tangent observation as an equation:

$$\begin{aligned}
 y &= f'(x_0)x + c && \text{(our equation for the tangent line)} \\
 \therefore f(x_0) &= f'(x_0)x_0 + c && \text{(using our observations above)} \\
 \therefore c &= f(x_0) - f'(x_0)x_0 && \text{(rearranging)} \\
 \therefore y &= f'(x_0)x + f(x_0) - f'(x_0)x_0 && \text{(subbing into original equation)}
 \end{aligned}$$

Now that we have found the equation of a general tangent line to $f(x)$ at the point x_0 , we can find where this crosses the x -axis. To find this, we simply need to set $y = 0$ and solve for x :

$$\begin{aligned}
 0 &= f'(x_0)x + f(x_0) - f'(x_0)x_0 && \text{(setting } y = 0\text{)} \\
 \therefore 0 &= x + \frac{f(x_0)}{f'(x_0)} - x_0 && \text{(dividing by } f'(x_0)\text{)} \\
 \therefore x &= x_0 - \frac{f(x_0)}{f'(x_0)}
 \end{aligned}$$

We can convert this into an iterative process by considering the point on the left side to be the “next” point, and then run through the same process again but swapping x_0 with the next point. This gives the **Newton-Raphson Iteration for solving $f(x) = 0$** :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Notice that we divide by $f'(x_n)$, and therefore this formula does not work if the derivative becomes zero at some point.

Also notice that, since this method assumes that our function “looks like” a straight line when you zoom in, and that we chose an initial point in this general region of straight-ness. If our initial guess of x_0 is not close enough to the actual root, then this method may not approach the root even after many iterations.

Example 101. Consider $f(x) = x^5 + x^3 - 1$. Show that there is a root between 0 and 1. By choosing $x_0 = 1$, compute two Newton-Raphson iterations to get an approximation of a root to this function.

Solution. Firstly, to check that there is a root between 0 and 1, we can plug in 0 and 1 to see if the function crosses zero in-between the two numbers. If one is negative and the other is positive, then we know this must have happened.

First plug in 0:

$$f(0) = 0^5 + 0^3 - 1 = -1$$

Now plug in 1:

$$f(1) = 1^5 + 1^3 - 1 = 1$$

Since at 0 the function is negative and at 1 the function is positive, we know it crossed zero somewhere in-between.

Now for Newton-Raphson, we need to find the derivative of the function.

$$f'(x) = 5x^4 + 3x^2$$

Now plug into the formula starting with $x_0 = 1$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 1 - \frac{f(1)}{f'(1)} \\ &= 1 - \frac{1}{5 + 3} \\ &= 1 - \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$

For the second iteration, we use that $x_1 = 7/8$ to find:

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= \frac{7}{8} - \frac{\left(\frac{7}{8}\right)^5 + \left(\frac{7}{8}\right)^3 - 1}{5\left(\frac{7}{8}\right)^4 + 3\left(\frac{7}{8}\right)^2} \\ &\approx 0.84003 \end{aligned}$$

Therefore, after two Newton-Raphson iterations, the root is approximately

$$x \approx 0.84003.$$

□